

Compare Results

Old File:

USB_PD_R3_2 V1.0 2023-10_Part17.pdf

100 pages (2.47 MB)

15/04/2024 10:51:12

versus

New File:

USB_PD_R3_2 V1.1 2024-10_Chapter 7 Power Supply.pdf

104 pages (2.91 MB)

28/10/2024 11:27:41

Total Changes

1869

Text only comparison

Content

1332 Replacements

264 Insertions

273 Deletions

Styling and Annotations

0 Styling

0 Annotations

Go to First Change (page 1)

7 Power Supply

7.1 Source Requirements

7.1.1 Behavioral Aspects

A **PDUSB Source** exhibits the following behaviors:

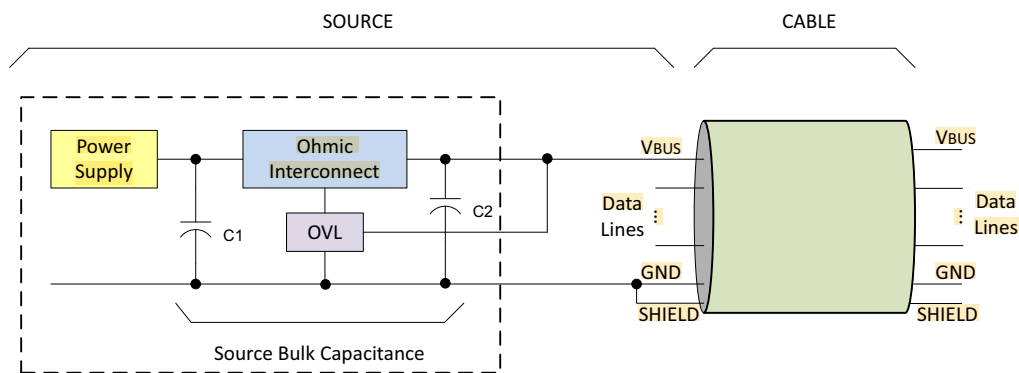
- **Shall** supply **[USB Type-C 2.4]** USB Type-C[®] current to **V_{BUS}** while in a *Default Contract* or *Implicit Contract*.
- **Shall** follow the requirements as specified in **Section 7.1.5, "Response to Hard Resets"** when **Hard Reset Signaling** is received.
- **Shall** control **V_{BUS} voltage** transitions as bound by undershoot, overshoot and transition time requirements.

7.1.2 Source Bulk Capacitance

The **Source** bulk capacitance **Shall Not** be placed between the transceiver isolation impedance and the USB receptacle. The **Source** bulk capacitance consists of C1 and C2 as shown in **Figure 7.1, "Placement of Source Bulk Capacitance"**. The Ohmic Interconnect might consist of PCB traces for power distribution or power switching devices. The Ohmic Interconnect might also be part of the circuit implemented by the **Source** to limit its **V_{BUS} Output Voltage Limit (OVL)** as described in **Section 7.1.7.5, "Output Voltage Limit"**. Though a **Source** **Shall** limit its output voltage, a **Sink** **Shall** implement **Sink OVP** as described in **Section 7.2.9.2, "Input Over Voltage Protection"** to protect against excessive **V_{BUS} input voltage**. The capacitance might be a single capacitor, a capacitor bank or distributed capacitance. If the power supply is shared across multiple ports, the bulk capacitance is defined as **cSrcBulkShared**. If the power supply is dedicated to a single **Port**, the minimum bulk capacitance is defined as **cSrcBulk**.

The **Source** bulk capacitance is allowed to change for a newly **Negotiated** power level. The capacitance change **Shall** occur before the **Source** is ready to operate at the new power level. During a **Power Role Swap**, the **Initial Source** **Shall** transition to **Swap Standby** before operating as the **New Sink**. Any change in bulk capacitance required to complete the **Power Role Swap** **Shall** occur during **Swap Standby**.

Figure 7.1 Placement of Source Bulk Capacitance



7.1.3 Types of Sources

Consistent with the **Power Data Objects** discussed in **Section 6.4.1, "Capabilities Message"**, the power supply types that are available as **Sources** in a USB Power Delivery System are:

- The **Fixed Supply PDO** exposes well-regulated fixed **voltage** power supplies. **Sources** **Shall** support at least one **Fixed Supply** capable of supplying **vSafe5V**. The output **voltage** of a **Fixed Supply** **Shall** remain

within the range defined by the relative tolerance **vSrcNew** and the absolute band **vSrcValid** as listed in [Table 7.23, "Source Electrical Parameters"](#) and described in [Section 7.1.8, "Output Voltage Tolerance and Range"](#).

- The *Variable Supply* (non-Battery) *PDO* exposes less well-regulated *Sources*. The output voltage of a *Variable Supply* (non-Battery) **shall** remain within the absolute maximum output voltage and the absolute minimum output voltage exposed in the *Variable Supply PDO*.
- The *Battery Supply PDO* exposes Batteries than can be connected directly as a *Source* to *VBUS*. The output voltage of a *Battery Supply* **shall** remain within the absolute maximum output voltage and the absolute minimum output exposed in the *Battery Supply PDO*.
- The Programmable Power Supply (PPS) *Augmented Power Data Object* (APDO) exposes a *Source* with an output voltage that can be adjusted programmatically over a defined range. The output voltage of the Programmable Power Supply **shall** remain within a range defined by the relative tolerance **vPpsNew** and the absolute band **vPpsValid**.
- The *Adjustable Voltage Supply* (AVS) *Augmented Power Data Object* (APDO) exposes a *Source* with an output voltage that can be adjusted programmatically over a defined range. The output voltage of the AVS **shall** remain within a range defined by the relative tolerance **vAvsNew** and the absolute band **vAvsValid**.

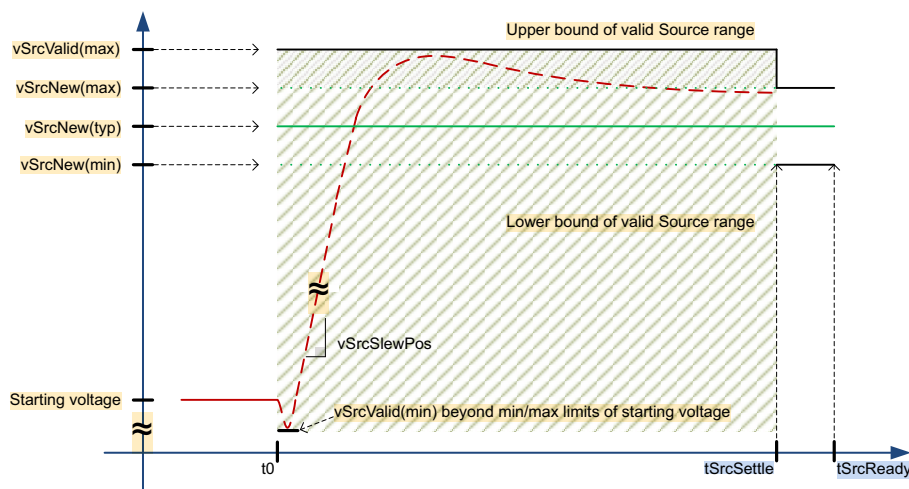
7.1.4 Source Transitions

7.1.4.1 Fixed Supply

7.1.4.1.1 Fixed Supply Positive Voltage Transitions

The *Source* **shall** transition *VBUS* from the starting voltage to the higher new voltage in a controlled manner. The *Negotiated* new voltage (e.g., 5V, 9V, 15V, ...) defines the nominal value for **vSrcNew**. During the positive transition the *Source* **should** be able to supply the *Sink Standby* current and the transient current to charge the total bulk capacitance on *VBUS*. The slew rate of the positive transition **shall not** exceed **vSrcSlewPos**. The transitioning *Source* output voltage **shall** settle within **vSrcNew** by **tSrcSettle**. The *Source* **shall** be able to supply the *Negotiated* power level at the new voltage by **tSrcReady**. The positive voltage transition **shall** remain above **vSrcValid** min of the previous *Explicit Contract* and below **vSrcValid** max of the new *Explicit Contract* ([Figure 7.2, "Transition Envelope for Positive Voltage Transitions"](#)). The voltage **shall** settle to **vSrcNew** within **tSrcSettle**. The starting time, t_0 , in [Figure 7.2, "Transition Envelope for Positive Voltage Transitions"](#) starts **tSrcTransition** after the last bit of the *EOP* of the *GoodCRC* Message has been received by the *Source*.

Figure 7.2 Transition Envelope for Positive Voltage Transitions



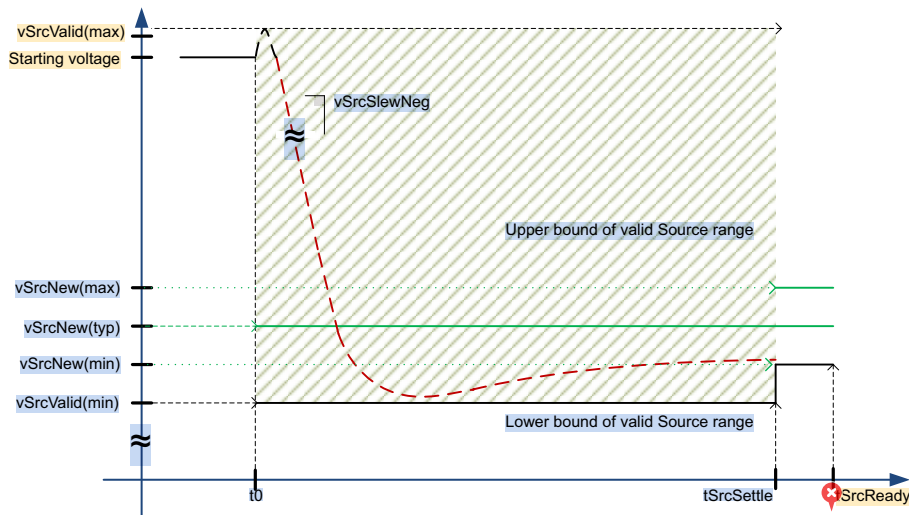
At the start of the positive voltage transition the *VBUS* voltage level **shall not** droop **vSrcValid** min below either **vSrcNew** (i.e., if the starting *VBUS* voltage level is not **vSafe5V**) or **vSafe5V** as applicable.

Section 7.1.14, "Non-application of V_{BUS} Slew Rate Limits" lists transitions that are exempt from the $v_{SrcSlewPos}$ limit.

7.1.4.1.2 Fixed Supply Negative Voltage Transitions

Negative voltage transitions are defined as shown in Figure 7.3, "Transition Envelope for Negative Voltage Transitions" and are specified in a similar manner to positive voltage transitions. Figure 7.3, "Transition Envelope for Negative Voltage Transitions" does not apply to v_{Safe0V} transitions. The slew rate of the negative transition **Shall Not** exceed $v_{SrcSlewNeg}$. The negative voltage transition **Shall** remain below $v_{SrcValid}$ max of the previous *Explicit Contract* and above $v_{SrcValid}$ min of the new *Explicit Contract*, as shown in Figure 7.3, "Transition Envelope for Negative Voltage Transitions". The transitioning Source output voltage **Shall** settle to v_{SrcNew} within $t_{SrcSettle}$. The starting time, t_0 , in Figure 7.3, "Transition Envelope for Negative Voltage Transitions" starts $t_{SrcTransition}$ after the last bit of the *EOP* of the *GoodCRC* Message has been received by the Source.

Figure 7.3 Transition Envelope for Negative Voltage Transitions



If the newly *Negotiated* voltage is v_{Safe5V} , then the $v_{SrcValid}$ limits **Shall** determine the transition window and the transitioning Source **Shall** settle within the v_{Safe5V} limits by $t_{SrcSettle}$.

Section 7.1.14, "Non-application of V_{BUS} Slew Rate Limits" lists transitions that are exempt from the $v_{SrcSlewNeg}$ limit.

7.1.4.2 SPR Programmable Power Supply (PPS)

7.1.4.2.1 SPR Programmable Power Supply Voltage Transitions

The Programmable Power Supply (PPS) **Shall** transition V_{BUS} over the defined voltage range in a controlled manner. The Output Voltage value in the Programmable *RDO* defines the nominal value of the PPS output voltage after completing a voltage change and **Shall** settle within the limits defined by v_{PpsNew} by $t_{PpsSrcTransSmall}$ for steps smaller than or equal to $v_{PpsSmallStep}$, or else, within the limits defined by v_{PpsNew} by $t_{PpsSrcTransLarge}$, but only in case the Programmable Power Supply is not in *CL* mode. Any overshoot beyond v_{PpsNew} **Shall Not** exceed $v_{PpsValid}$ at any time. Any undershoot beyond v_{PpsNew} **Shall Not** exceed $v_{PpsValid}$ for currents not resulting in *CL* mode. The PPS output voltage **May** change in a step-wise or linear manner and the slew rate of either type of change **Shall Not** exceed $v_{PpsSlewPos}$ for voltage increases or $v_{PpsSlewNeg}$ for voltage decreases. The nominal requested voltage of all linear voltage changes **Shall** equate to an integer number of *LSB* changes. An *LSB* change of the PPS output voltage is defined as $v_{PpsStep}$. A PPS **Shall** be able to supply the *Negotiated* current level as it changes its output voltage to the requested level. All PPS voltage increases **Shall**

result in a **voltage** that is greater than or equal to the previous PPS output **voltage**. Likewise, all PPS **voltage** decreases **Shall** result in a **voltage** that is less than or equal to the previous PPS output **voltage**.

Since a *Sink* can draw current up to the *Negotiated APDO* current level in case of a **voltage** step, the **voltage** might not increase to the requested level due to the power supply operating in *CL* mode. Likewise, since a *Sink* can have a *Battery* connected to *VBUS*, the **voltage** might not decrease to the requested level due to the *Battery* voltage being higher than the output **voltage** set point the *Source* is transitioning to. Were the *Source* to rely on checking the **voltage** on *VBUS*, in either case, to determine when its power supply is ready a *PS_RDY* Message would never be sent.

When the PPS **voltage** steps up or down, a *PS_RDY* Message **Shall** be sent within:

- *tPpsSrcTransLarge* after the last bit of the *GoodCRC* Message following the *Accept* Message for steps larger than *vPpsSmallStep*.
- *tPpsSrcTransSmall* after the last bit of the *GoodCRC* Message following the *Accept* Message for steps less than or equal to *vPpsSmallStep* provided that either the **voltage** on *VBUS* has reached *vPpsNew* or the power supply is in *CL* mode.

When *vPpsNew* is lower than the *Battery* voltage, or the *Source*'s primary power is cut off the *Sink* **Shall** immediately disconnect its *Battery* from *VBUS*. In these situations, the output current could reverse polarity and the *Sink* is not allowed to source current (see [Section 7.2.1, "Behavioral Aspects"](#) and [Section 7.2.9, "Robust Sink Operation"](#)).

[Figure 7.4, "PPS Positive Voltage Transitions"](#) and [Figure 7.5, "PPS Negative Voltage Transitions"](#) below show the output **voltage** behavior of a Programmable Power Supply in response to positive and negative **voltage** change requests. The parameters *vPpsMinVoltage* and *vPpsMaxVoltage* define the lower and upper limits of the PPS range respectively (see [Table 10.11, "SPR Programmable Power Supply Voltage Ranges"](#) for required ranges). *vPpsMinVoltage* corresponds to the *Minimum Voltage* field in the PPS *APDO* and *vPpsMaxVoltage* corresponds to *Maximum Voltage* field in the PPS *APDO*. If the *Sink* negotiates for a new PPS *APDO*, then the transition between the two PPS *APDO*s **Shall** occur as described in [Section 7.3.1, "Transitions caused by a Request Message"](#).

Figure 7.4 PPS Positive Voltage Transitions

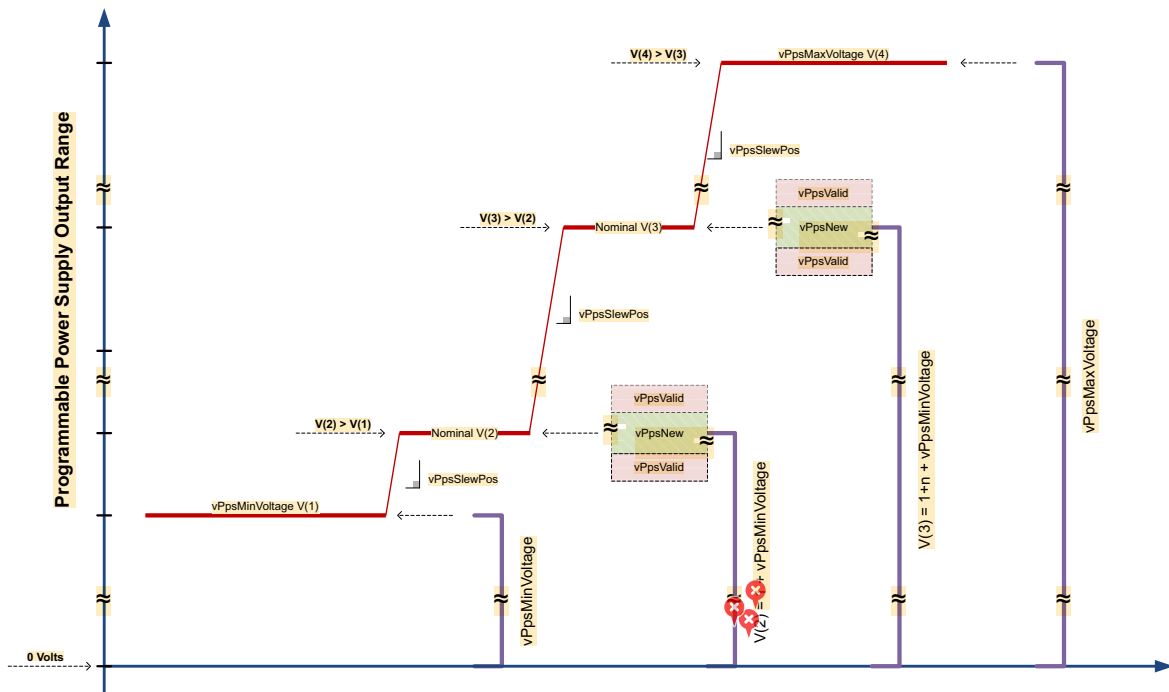
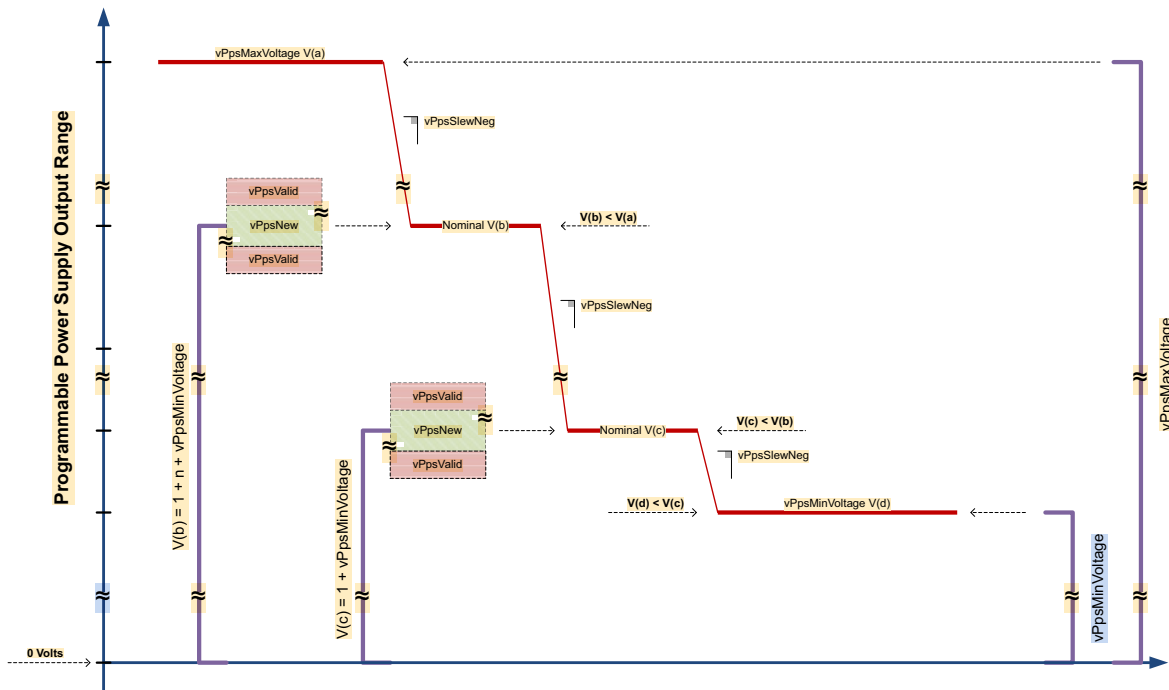


Figure 7.5 PPS Negative Voltage Transitions



Section 7.1.14, "Non-application of VBUS Slew Rate Limits" lists transitions that are exempt from the *vPpsSlewNeg* and *vPpsSlewPos* limits.

See Section 7.1.8.1, "AVS/PPS Output Voltage Ripple" for output voltage ripple limits.

See Section 7.1.8.2, "AVS/PPS DNL Errors and Output Voltage/Current Tolerance" for output voltage and current DNL step adjustments.

7.1.4.2.2 SPR Programmable Power Supply Current Limit

The Programmable Power Supply operating in *SPR PPS Mode* **shall** limit its output current to the *Operating Current* field value in the *APDO* when the Sink attempts to draw more current than the *Operating Current* field value level. The programming step size for the *Operating Current* is *iPpsCLStep*. All programming changes of the *Operating Current* **shall** settle to the new *Operating Current* field value within *tPpsCLProgramSettle*. The *SPR PPS* *Operating Current* regulation accuracy during *Current Limit* is defined as *iPpsCLNew*. The minimum programmable *Current Limit* level is *iPpsCLMin*. A Source that supports *SPR PPS Mode* **shall** support *Current Limit* programmability between *iPpsCLMin* and the Maximum Current value in the *SPR PPS APDO*. A Source which receives a request for current below *iPpsCLMin* **should** reject the request. A Source that accepts a request for current below *iPpsCLMin* **shall** set its current limit at 1A.

The response of an *SPR PPS* to a load change depends on the Operating mode of the *SPR PPS* and the magnitude of the load change. These dependencies lead to one of four possible responses of an *SPR PPS* to any load change. They are differentiated by the value of the PPS Status OMF before and after the load change:

- If the PPS Status OMF is cleared both before and after the load change, the *SPR PPS* responds solely by maintaining the output voltage. The *SPR PPS* output voltage **shall** remain within *vPpsValid* range. The *SPR PPS* response to the load change **shall** settle within the *vPpsNew* tolerance band by the time *tPpsTransient*. The Operating Mode Flag **shall** remain cleared during the load change response of the *SPR PPS*.
- If the PPS Status OMF is cleared before the load change and set after the load change, the *SPR PPS* responds by reducing its output voltage to limit the *SPR PPS* output current. The *SPR PPS* output current **shall** stay within the *iPpsCVCLTransient* range once it reaches the *iPpsCVCLTransient* range. The *SPR*

PPS response to the load change **Shall** settle within the *iPpsCLNew* tolerance band by the time *tPpsCVCLTransient*. The Operating Mode Flag **Shall** be set when the SPR PPS load change response settles.

- If the PPS Status OMF is set both before and after the load change, the SPR PPS responds by adjusting its output voltage to maintain the output current. The SPR PPS output current **Shall** stay within the *iPpsCLTransient* range. The SPR PPS response to the load change **Shall** settle within the *iPpsCLNew* tolerance band by the time *tPpsCLSettle*. The Operating Mode Flag **Shall** remain set during the load change response of the SPR PPS.
- If the PPS Status OMF is set before the load change and cleared after the load change, the PPS responds to the load change by increasing its output voltage to *vPpsNew* and then maintaining it. The SPR PPS output voltage **Shall** stay within the *vPpsCLCVTransient* range. The SPR PPS response to the load change **Shall** settle within the *vPpsNew* tolerance band by the time *tPpsCLCVTransient*. The Operating Mode Flag **Shall** be cleared when the PPS load change response settles.

The SPR PPS Source **Shall** maintain its output voltage at the value requested in the PPS RDO for all static and dynamic load conditions except when in *Current Limit* operation. In response to any static or dynamic load condition during *Current Limit* operation that causes the SPR PPS output voltage to drop below *vPpsShutdown* the Source **May** send *Hard Reset* Signaling and **Shall** discharge *V_{BUS}* to *vSafe0V* then resumes USB Default Operation at *vSafe5V*.

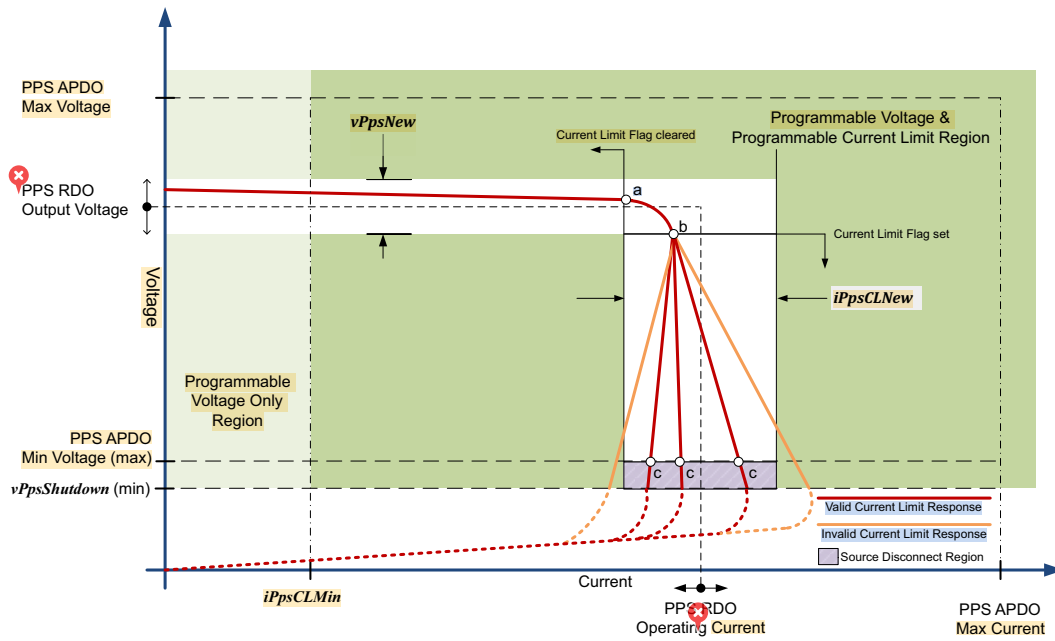
When the Sink attempts to draw more current than the Operating Current in the RDO, the Source **Shall** limit its output current. The current available from the Source during *Current Limit* mode **Shall** meet *iPpsCLNew*. The Sink **May Not** reduce its Operating Current request in the RDO when the PPS Status OMF is set.

Current limiting **Shall** be performed by the SPR PPS Source. Sinks that rely on PPS Current Limiting **Shall** meet the requirements of [Section 7.2.9, "Robust Sink Operation"](#). The Source **Shall Not** shutdown or otherwise disrupt the available output power while in *Current Limit* mode unless another protection mechanism as outlined in [Section 7.1.7, "Robust Source Operation"](#) is engaged to protect the Source from damage.

An SPR PPS Source that is operating in *Current Limit* **Shall Not** change its set-point in a manner that exceeds *iPpsCLLoadStepRate* or *iPpsCLLoadReleaseRate*.

The relationship between SPR PPS programmable output voltage and SPR PPS programmable *Current Limit* **Shall** be as shown in [Figure 7.6, "SPR PPS Programmable Voltage and Current Limit"](#). The transition between the *Constant Voltage* mode and the *Current Limit* mode occurs between points a and b. The PPS Status OMF **Shall** be set or cleared within this region. In *Current Limit* mode when the load resistance changes, the output current of the Source **Shall** stay within *iPpsCLNew*. The proper behavior is represented by point c.

Figure 7.6 SPR PPS Programmable Voltage and Current Limit



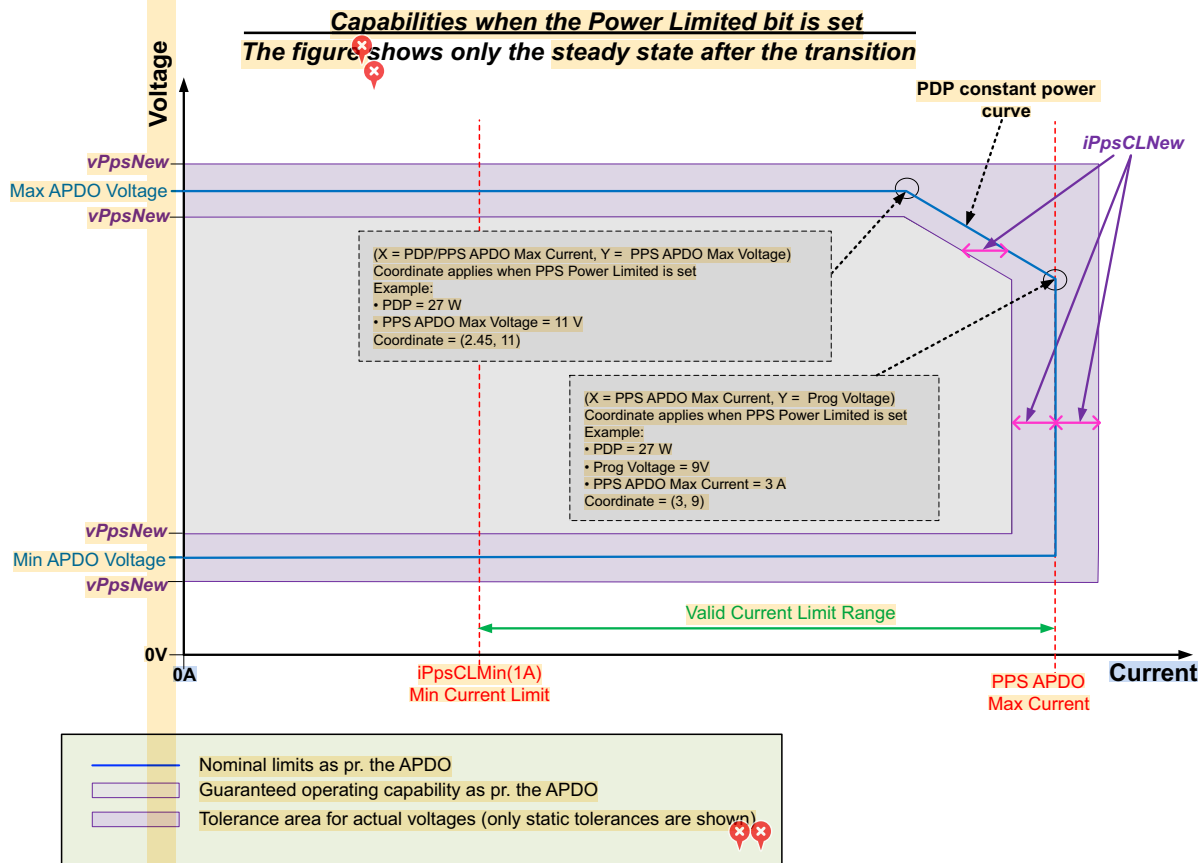
Point *a* represents entry into the transition region between Constant Voltage mode and Current Limit mode.
 Point *b* represents exit from the transition region between Constant Voltage mode and Current Limit mode.
 Point *c* represents the exit from the *iPpsCLNew* region as the voltage drops below the PPS APDO Min Voltage.
 The Source **May** disconnect at any point inside the tolerance range of the minimum voltage defined in the PPS APDO.

7.1.4.2.3 SPR PPS Constant Power Mode

In Constant Power mode (when the PPS Power Limited bit is set) the Source **May** supply power that exceeds the Source's *PDP Rating*. Sinks **May** limit their Operating Current request in the *RDO* and **Shall** meet the requirements of [Section 7.2.9, "Robust Sink Operation"](#).

The tolerances along the Constant Power Curve **Shall Not** extend into the Guaranteed Capability Area of [Figure 7.7, "SPR PPS Constant Power"](#).

Figure 7.7 SPR PPS Constant Power



7.1.4.3

Adjustable Voltage Supply (AVS)

7.1.4.3.1

Adjustable Voltage Supply Voltage Transitions

The *Adjustable Voltage Supply (AVS)* **Shall** transition *V_{BUS}* over the defined voltage range in a controlled manner. The Output Voltage value in the AVS RDO defines the nominal value of the AVS output voltage after completing a voltage change and **Shall** settle within the limits defined by *vAvsNew* by *tAvsSrcTransSmall* for steps smaller than or equal to *vAvsSmallStep*, or else, within the limits defined by *vAvsNew* by *tAvsSrcTransLarge* for steps larger than *vAvsSmallStep*. Any overshoot beyond *vAvsNew* **Shall Not** exceed *vAvsValid* at any time. Any undershoot beyond *vAvsNew* **Shall Not** exceed *vAvsValid* at any time. The AVS output voltage **May** change in a stepwise or linear manner and the slew rate of either type of change **Shall Not** exceed *vAvsSlewPos* for voltage increases or *vAvsSlewNeg* for voltage decreases. The nominal requested voltage of all linear voltage changes **Shall** equate to an integer number of *LSB* changes. An *LSB* change of the AVS output voltage is defined as *vAvsStep*. An AVS **Shall** be able to supply the *Negotiated* current level as it changes its output voltage to the requested level if the change of output voltage is less than or equal to *vAvsSmallStep* relative to *vAvsNew*. All AVS voltage increases **Shall** result in a voltage that is greater than or equal to the previous AVS output voltage. Likewise, all AVS voltage decreases **Shall** result in a voltage that is less than or equal to the previous AVS output voltage. Any time the Source enters the AVS range of operation that voltage transition is considered a voltage step larger than *vAvsSmallStep*.

When the AVS voltage steps up or down, a *PS_RDY* Message **Shall** be sent within:

- *tAvsSrcTransLarge* after the last bit of the *GoodCRC* Message following the *Accept* Message for steps larger than *vAvsSmallStep*.
- *tAvsSrcTransSmall* after the last bit of the *GoodCRC* Message following the *Accept* Message for steps less than or equal to *vAvsSmallStep* provided the voltage on *V_{BUS}* has reached *vAvsNew*.

[Figure 7.8, "AVS Positive Voltage Transitions"](#) and [Figure 7.9, "AVS Negative Voltage Transitions"](#) below show the output voltage behavior of an AVS in response to positive and negative voltage change requests. The parameters *vAvsMinVoltage* and *vAvsMaxVoltage* define the lower and upper limits of the AVS range respectively:

- For *SPR* AVS Sources there are two possible voltage ranges where the *vAvsMinVoltage* is always 9V and *vAvsMaxVoltage* is either 15V or 20V depending on the Source's PDP. See [Table 10.9, "SPR Adjustable Voltage Supply \(AVS\) Voltage Ranges"](#).
- For *EPR* AVS Sources *vAvsMinVoltage* corresponds to *Minimum Voltage* field (always 15V) in the *EPR* AVS APDO and *vAvsMaxVoltage* corresponds to *Maximum Voltage* field in the *EPR* AVS APDO. See [Table 10.15, "EPR Adjustable Voltage Supply \(AVS\) Voltage Ranges"](#) for required ranges.

Figure 7.8 AVS Positive Voltage Transitions

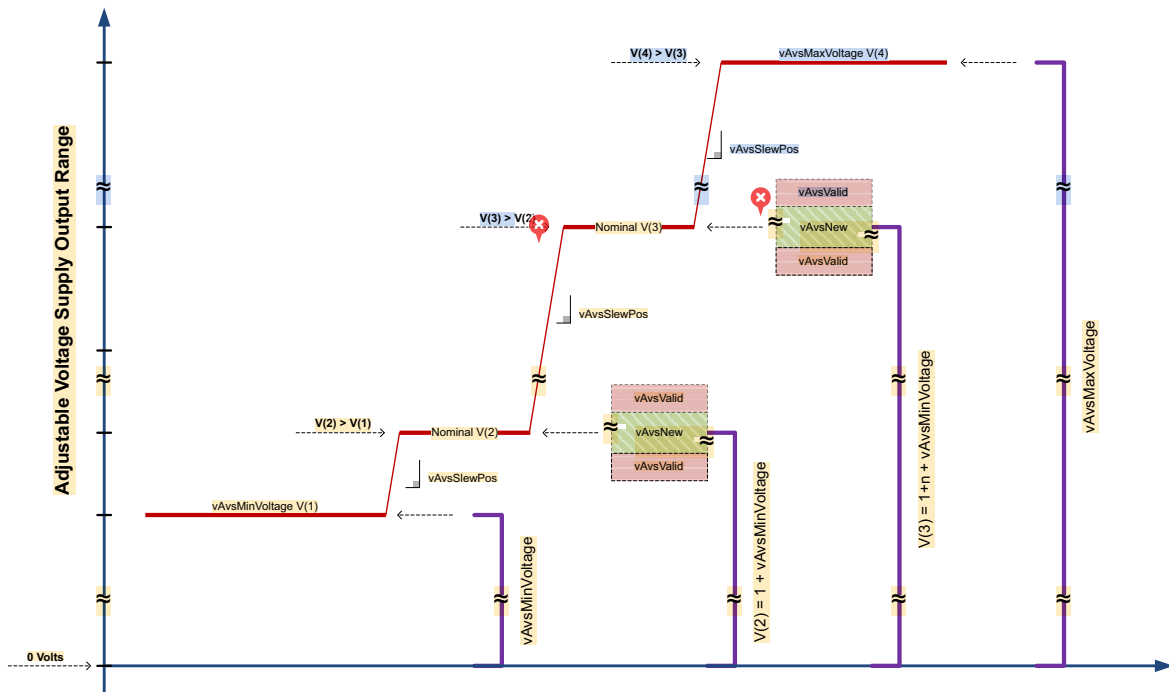
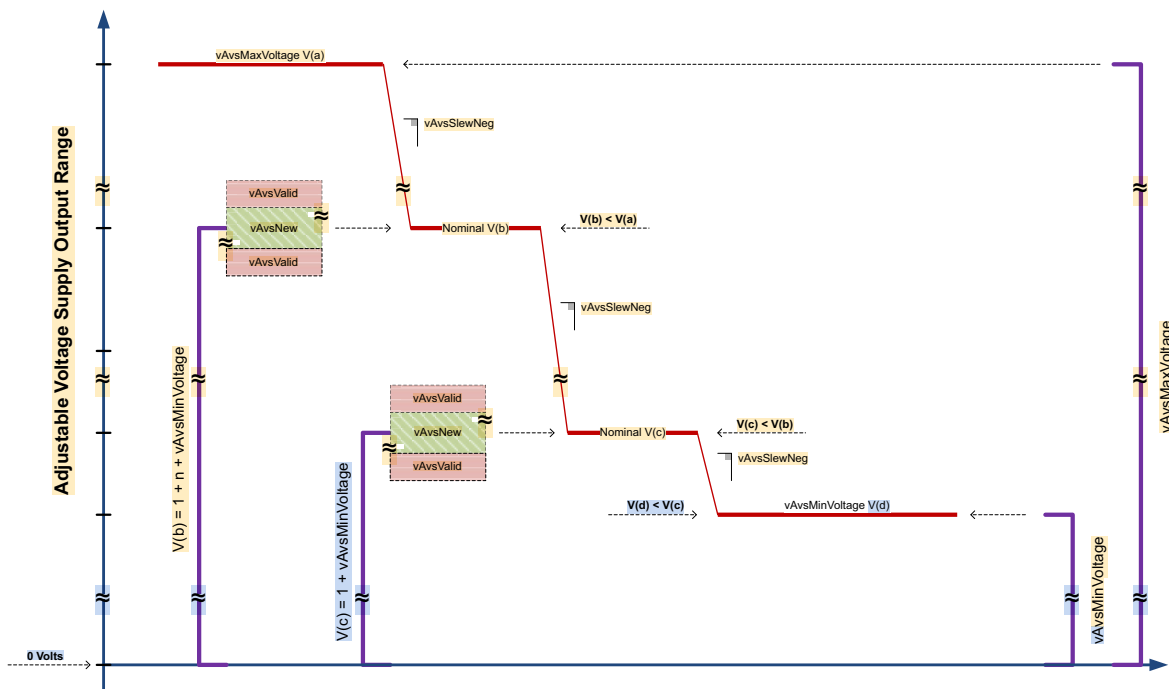


Figure 7.9 AVS Negative Voltage Transitions



See [Section 7.1.8.1, "AVS/PPS Output Voltage Ripple"](#) for output voltage ripple limits.

See [Section 7.1.8.2, "AVS/PPS DNL Errors and Output Voltage/Current Tolerance"](#) for output voltage **DNL** step adjustments.

7.1.4.3.2 Adjustable Voltage Supply Current

The AVS **Shall** maintain its output voltage at the value requested in the AVS RDO for all static and dynamic load conditions that do not exceed the Operating Current in the RDO. Unlike the SPR PPS programmable current, the AVS programmable power **May** range from zero to the PDP.

The maximum operating current:

- For SPR Sources, the maximum operating current is defined in the SPR **Source_Capabilities** Message **Maximum Current 15V/Maximum Current 20V** fields.
- For EPR Sources, the maximum operating current has to be calculated as the lower of the **PDP** field value/**Output Voltage** or 5A whichever is lower. See [Table 10.12, "EPR Source Capabilities based on the Port Maximum PDP and using an EPR Capable Cable"](#).

7.1.5 Response to Hard Resets

Hard Reset Signaling indicates a communication failure has occurred and the Source **Shall** stop driving V_{CONN} , **Shall** remove R_p from the V_{CONN} pin and **Shall** drive V_{BUS} to **vSafe0V** as shown in [Figure 7.10, "Source \$V_{BUS}\$ and \$V_{CONN}\$ Response to Hard Reset"](#). The USB connection **May** reset during a Hard Reset since the V_{BUS} voltage will be less than **vSafe5V** for an extended period of time. After establishing the **vSafe0V** voltage condition on V_{BUS} , the Source **Shall** wait **tSrcRecover** before re-applying V_{CONN} and restoring V_{BUS} to **vSafe5V**. A Source **Shall** conform to the V_{CONN} timing as specified in [\[USB Type-C 2.4\]](#).

A Sink that enters Hard Reset can have **cSnkBulkPd** present until V_{BUS} drops below **vSafe0V**. The Source **Shall** take this into consideration.

Device operation during and after a Hard Reset is defined as follows:

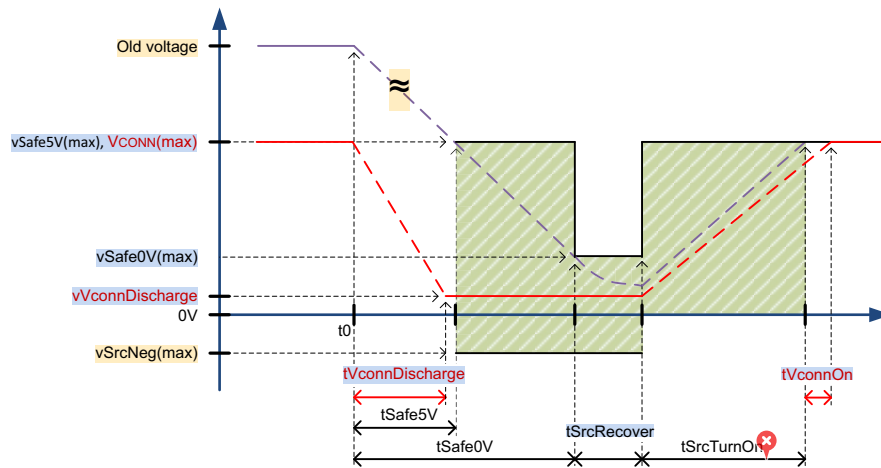
- ❗❗ Self-powered devices **Should Not** disconnect from USB during a Hard Reset (see [Section 9.1.2, "Mapping to USB Device States"](#)).
- Self-powered devices operating at more than **vSafe5V** **May Not** maintain full functionality after a Hard Reset.
- Bus powered devices will disconnect from USB during a Hard Reset due to the loss of their power source.

When a Hard Reset occurs the Source **Shall** stop driving V_{CONN} , **Shall** remove R_p from the V_{CONN} pin and **Shall** start to transition the V_{BUS} voltage to **vSafe0V** either:

- **tPSHardReset** after the last bit of the **Hard Reset** Signaling has been received from the Sink or
- **tPSHardReset** after the last bit of the **Hard Reset** Signaling has been sent by the Source.

The Source **Shall** meet both **tSafe5V** and **tSafe0V** relative to the start of the voltage transition as shown in [Figure 7.10, "Source \$V_{BUS}\$ and \$V_{CONN}\$ Response to Hard Reset"](#).

Figure 7.10 Source V_{BUS} and V_{CONN} Response to Hard Reset



V_{CONN} will meet $t_{VCONNDischarge}$ relative to the start of the voltage transition as shown in [Figure 7.10, "Source \$V_{BUS}\$ and \$V_{CONN}\$ Response to Hard Reset"](#) due to the discharge circuitry in the Cable Plug. V_{CONN} **Shall** meet $t_{VCONNOn}$ relative to V_{BUS} reaching v_{Safe5V} .

Note: $t_{VCONNOn}$ and $t_{VCONNDischarge}$ are defined in [\[USB Type-C 2.4\]](#).

7.1.6 Changing the Output Power Capability

Some USB Power Delivery *Negotiations* will require the *Source* to adjust its output power capability without changing the output voltage. In this case the *Source* **Shall** be able to supply a higher or lower load current within $t_{SrcReady}$.

7.1.7 Robust Source Operation

7.1.7.1 Output Over Current Protection

Sources **Shall** implement over current protection to prevent damage from output current that exceeds the current handling capability of the *Source*. The definition of current handling capability is left to the discretion of the *Source* implementation and **Shall** take into consideration the current handling capability of the connector contacts. If the over current protection implementation does not use a *Hard Reset* or *Error Recovery*, it **Shall Not** interfere with the *Negotiated V_{BUS}* current level.

After three consecutive over current events *Source* **Shall** go to *ErrorRecovery*.

Sources **Should** attempt to send *Hard Reset Signaling* when over current protection engages followed by an *Alert Message* indicating an *OCP* event once an *Explicit Contract* has been established. The over current protection response **May** engage at either the *Port* or system level. Systems or ports that have engaged over current protection **Should** attempt to resume *USB Default Operation* after determining that the cause of over current is no longer present and **May** latch off to protect the *Port* or system. The definition of how to detect if the cause of over current is still present is left to the discretion of the *Source* implementation.

The *Source* **Shall** *Re-negotiate* with the *Sink* after choosing to resume *USB Default Operation*. The decision of how to *Re-negotiate* after an over current event is left to the discretion of the *Source* implementation.

The *Source* **Shall** prevent continual system or *Port* cycling if over current protection continues to engage after initially resuming either *USB Default Operation* or *Re-negotiation*. Latching off the *Port* or system is an acceptable response to recurring over current.

During the over current response and subsequent system or *Port* shutdown, all affected *Source* ports operating with V_{BUS} greater than v_{Safe5V} **Shall** discharge V_{BUS} to v_{Safe5V} by the time t_{Safe5V} and v_{Safe0V} by the time t_{Safe0V} .

7.1.7.2 Over Temperature Protection

Sources **Shall** implement Over Temperature Protection (OTP) to prevent damage from temperature that exceeds the thermal capability of the Source. The definition of thermal capability and the monitoring locations used to trigger the over temperature protection are left to the discretion of the Source implementation.

In order to avoid reaching an OTP event, Sources **May** proactively reduce the available power being offered to the Sink, even though these offers might be lower than the Source would be expected to offer during normal thermal operating conditions. Prior to reducing power, the Source **Should** generate **Alert Message** indicating an Operating Condition Change and set the Temperature Status bit in the SOP **Status Message** to Warning (10b).

Sources **Should** attempt to send **Hard Reset Signaling** when OTP engages followed by an **Alert Message** indicating an OTP event once an **Explicit Contract** has been established. The OTP response **May** engage at either the **Port** or system level. Systems or ports that have engaged OTP **Should** attempt to resume **USB Default Operation** and **May** latch off to protect the **Port** or system.

The Source **Shall** **Re-negotiate** with the Sink **After** choosing to resume **USB Default Operation**. The decision of how to **Re-negotiate** after an over temperature event is left to the discretion of the Source implementation.

The Source **Shall** prevent continual system or **Port** cycling if over temperature protection continues to engage after initially resuming either **USB Default Operation** or **Re-negotiation**. Latching off the **Port** or system is an acceptable response to recurring over temperature.

During the OTP and subsequent system or **Port** shutdown, all affected Source ports operating with **VBUS** greater than **vSafe5V** **Shall** discharge **VBUS** to **vSafe5V** by the time **tSafe5V** and **vSafe0V** by the time **tSafe0V**.

7.1.7.3 vSafe5V Externally Applied to Ports Supplying vSafe5V

Safe operation mandates that Power Delivery Sources **Shall** be tolerant of **vSafe5V** being present on **VBUS** when simultaneously applying power to **VBUS**. Normal USB PD communication **Shall** be supported when this **vSafe5V** to **vSafe5V** connection exists.

7.1.7.4 Detach

A USB **Detach** is detected electrically using CC detection on the **USB Type-C** connector. When the Source is **Detached** the Source **Shall** transition to **vSafe0V** by **tSafe0V** relative to when the **Detach** event occurred. During the transition to **vSafe0V** the **VBUS** voltage **Shall** be below **vSafe5V** max by **tSafe5V** relative to when the **Detach** event occurred and **Shall Not** exceed **vSafe5V** max after this time.

Sources operating in **EPR Mode** need to avoid creating large differential voltages at the connector. See Appendix H in the **[USB Type-C 2.4]** specification for background information. To achieve this, Sources operating in **EPR Mode**, upon detecting a disconnect, **Shall** stop sourcing current and minimize **VBUS** capacitance. There **May** continue to be current sourced from the Source bulk capacitance, but that **Should** also be minimized by disconnecting as much of the Source bulk capacitance as possible. For example, the Source can stop sourcing from the Power Supply and the C1 portion of the Source bulk capacitance in **Figure 7.1, "Placement of Source Bulk Capacitance"** by disabling the Ohmic Interconnect switch.

The Source **Should** detect the disconnect, stop sourcing current, and minimize the **VBUS** capacitance as quickly as practical. If this is done after the CC contacts disconnect and before the **VBUS** contacts disconnect there is less risk of large differential voltages at the connector.

Note: A USB-PD transmission by the Source during a disconnect event will delay disconnect detection by the Source.

7.1.7.5 Output Voltage Limit

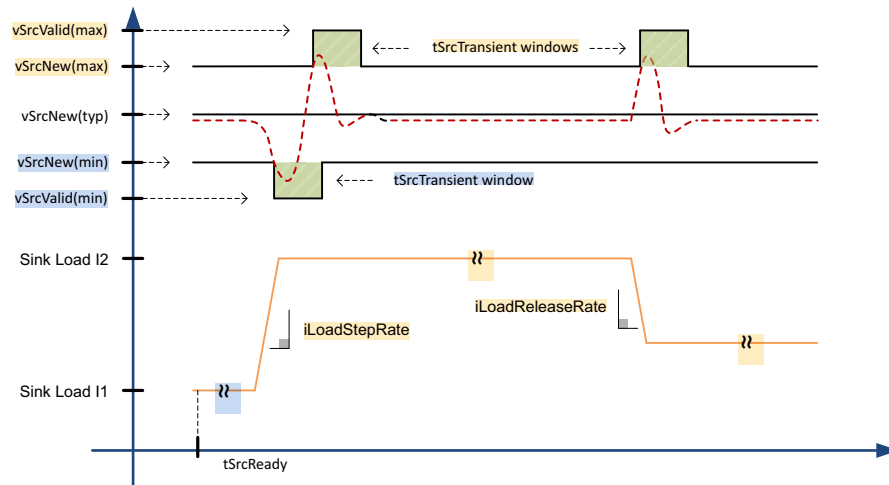
The output voltage of Sources **Shall** account for **vSrcNew**, **vSrcValid** or **vPpsNew**, **vPpsValid** or **vAvsNew**, **vAvsValid** as determined by the **Negotiated VBUS** value. Sources **Shall** meet applicable safety and regulatory requirements.

7.1.8 Output Voltage Tolerance and Range

After a voltage transition is complete (i.e., after *tSrcReady*) and during static load conditions the *Source* output voltage **Shall** remain within the *vSrcNew* or *vSafe5V* limits as applicable. The ranges defined by *vSrcNew* and *vSafe5V* account for DC regulation accuracy, line regulation, load regulation and output ripple. After a voltage transition is complete (i.e., after *tSrcReady*) and during transient load conditions the *Source* output voltage **Shall Not** go beyond the range specified by *vSrcValid*. The amount of time the *Source* output voltage can be in the band between either *vSrcNew* or *vSafe5V* and *vSrcValid* **Shall Not** exceed *tSrcTransient*. Refer to [Table 7.23, "Source Electrical Parameters"](#) for the output voltage tolerance specifications. [Figure 7.11, "Application of vSrcNew and vSrcValid limits after tSrcReady"](#) illustrates the application of *vSrcNew* and *vSrcValid* after the voltage transition is complete.

The *vSrcNew* and *vSrcValid* limits **Shall Not** apply to *VBUS* during the *VBUS* discharge and switchover that occurs during a *Fast Role Swap* as described in [Section 7.1.13, "Fast Role Swap"](#).

Figure 7.11 Application of vSrcNew and vSrcValid limits after tSrcReady

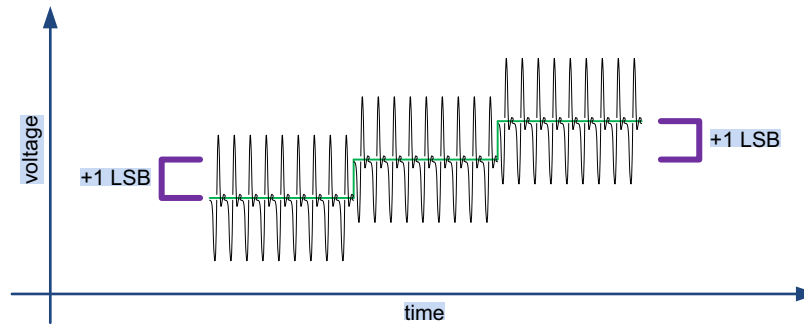


The *Source* output voltage **Shall** be measured at the connector receptacle. The stability of the *Source* **Shall** be tested in 25% load step increments from minimum load to maximum load and also from maximum load to minimum load. The transient behavior of the load current is defined in [Section 7.2.6, "Transient Load Behavior"](#). The time between each step **Shall** be sufficient to allow for the output voltage to settle between load steps. In some systems it might be necessary to design the *Source* to compensate for the voltage drop between the output stage of the power supply electronics and the receptacle contact. The determination of whether compensation is necessary is left to the discretion of the *Source* implementation.

7.1.8.1 AVS/PPS Output Voltage Ripple

The AVS/PPS output voltage ripple is expected to exceed the magnitude of one or more *LSB* as show in the [Figure 7.12, "Expected AVS/PPS Ripple Relative to an LSB"](#).

Figure 7.12 Expected AVS/PPS Ripple Relative to an LSB



7.1.8.2 AVS/PPS DNL Errors and Output Voltage/Current Tolerance

The PPS voltage and current discrete *LSB* steps have a *DNL* tolerance as shown in [Figure 7.13, "Allowed DNL errors and tolerance of Voltage and Current in AVS/PPS mode"](#) below. In absolute terms the step size of the *LSB* for both voltage and current is defined by *vPpsStep*/*vAvsStep* for voltage and *iPpsCLStep* for current. Several examples of *Valid* *LSB* steps are shown in [Figure 7.13, "Allowed DNL errors and tolerance of Voltage and Current in AVS/PPS mode"](#):

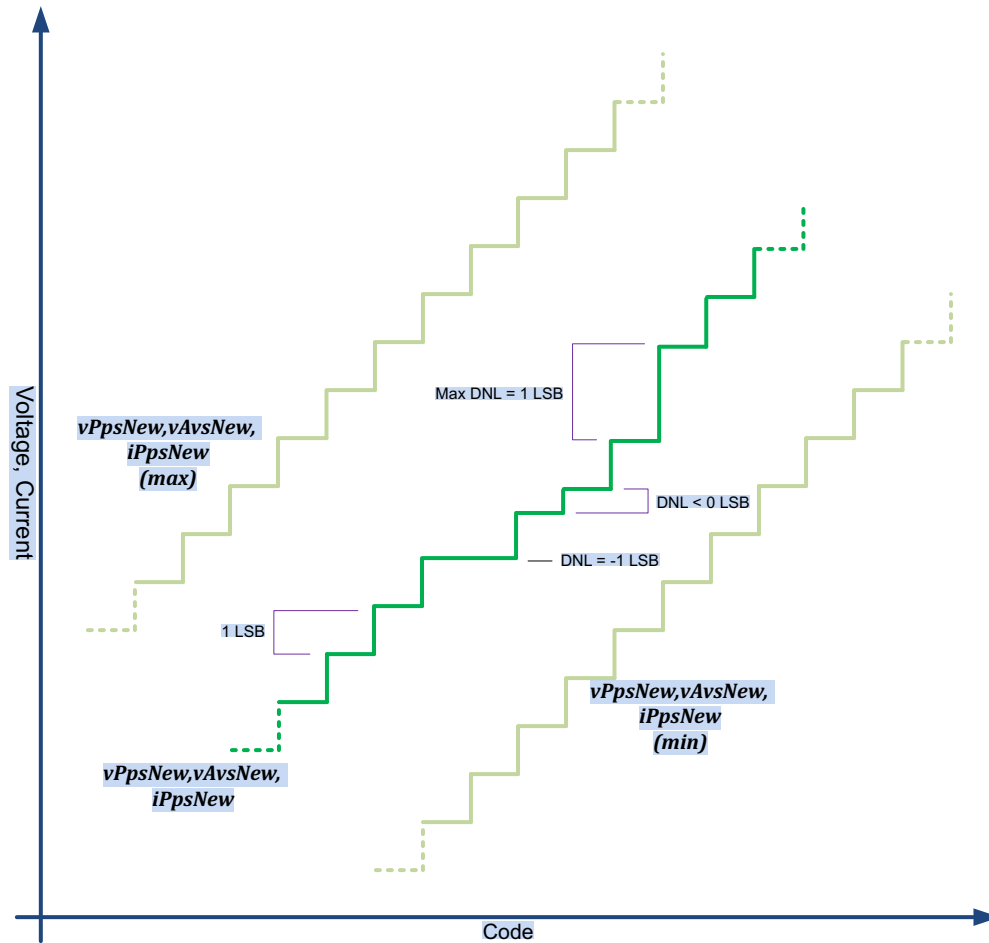
- The upper end of the *DNL* error (+1 *LSB*) shows the case where one step is effectively skipped.
- The lower end of the *DNL* error (-1 *LSB*) shows the case where the voltage or current set-point remained the same.

The ideal scenario for the *DNL* error (=0) matches the typical step size for the voltage or current.

The intent of *DNL* is to guarantee that changes to the voltage/current have the correct directionality, and that the maximum step size is clearly defined.

Note: The Source *Should* avoid scenarios where multiple consecutive steps have errors close to the Maximum and Minimum *DNL*.

Figure 7.13 Allowed DNL errors and tolerance of Voltage and Current in AVS/PPS mode



7.1.8.3 Programmable Power Supply Output Voltage Tolerance and Range

After a voltage transition of a Programmable Power Supply is complete (i.e. after *tPpsSrcTransSmall* or *tPpsSrcTransLarge*) and during static load conditions the Source output voltage **Shall** remain within the *vPpsNew* limits. The range defined by *vPpsNew* accounts for DC regulation accuracy, line regulation, load regulation and output ripple. After a voltage transition is complete (i.e. after *tPpsSrcTransSmall* or *tPpsSrcTransLarge*) and during transient load conditions the Source output voltage **Shall Not** go beyond the range specified by *vPpsValid*. The amount of time the Source output voltage can be in the band between *vPpsNew* and *vPpsValid* **Shall Not** exceed *tPpsTransient*.

7.1.8.4 Adjustable Voltage Supply Output Voltage tolerance and Range

After a voltage transition of an AVS is complete (i.e. after *tAvsSrcTransSmall* or *tAvsSrcTransLarge*) and during static load conditions the Source output voltage **Shall** remain within the *vAvsNew* limits. The range defined by *vAvsNew* accounts for DC regulation accuracy, line regulation, load regulation and output ripple. After a voltage transition is complete (i.e. after *tAvsSrcTransSmall* or *tAvsSrcTransLarge*) and during transient load conditions the Source output voltage **Shall Not** go beyond the range specified by *vAvsValid*. The amount of time the Source output voltage can be in the band between *vAvsNew* and *vAvsValid* **Shall Not** exceed *tAvsTransient*.

7.1.9 Charging and Discharging the Bulk Capacitance on **VBUS**

The *Source* **Shall** charge and discharge the bulk capacitance on **VBUS** whenever the *Source* voltage is *Negotiated* to a different value. The charging or discharging occurs during the *voltage* transition and **Shall Not** interfere with the *Source's* ability to meet *tSrcReady*.

7.1.10 Swap Standby for Sources

Sources and *Sinks* of a *Dual-Role Power Port* **Shall** support *Swap Standby*. *Swap Standby* occurs for the *Source* after the *Source* power supply has discharged the bulk capacitance on **VBUS** to *vSafe0V* as part of the *Power Role Swap* transition.

While in *Swap Standby*:

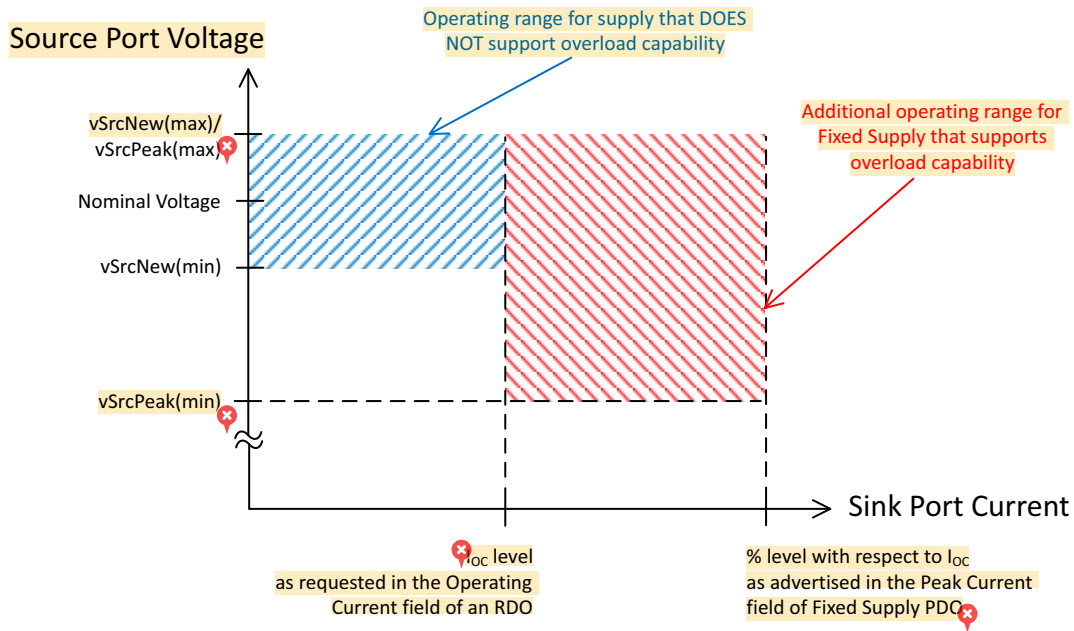
- The *Source* **Shall Not** drive **VBUS** that is therefore expected to remain at *vSafe0V*.
- Any discharge circuitry that was used to achieve *vSafe0V* **Shall** be removed from **VBUS**.
- The *Dual-Role Power Port* **Shall** be configured as a *Sink*.
- The USB connection **Shall Not** reset even though *vSafe5V* is no longer present on **VBUS** (see [Section 9.1.2, "Mapping to USB Device States"](#)).

The *PS_RDY* Message associated with the *Source* being in *Swap Standby* **Shall** be sent after the **VBUS** drive is removed. The time for the *Source* to transition to *Swap Standby* **Shall Not** exceed *tSrcSwapStdby*. Upon entering *Swap Standby*, the *Source* has relinquished its *Power Role* as *Source* and is ready to become the *New Sink*. The transition time from *Swap Standby* to being the *New Sink* **Shall** be no more than *tNewSnk*. The *New Sink* **May** start using power after the new *Source* sends the *PS_RDY* Message.

7.1.11 Source Peak Current Operation

A *Source* that has the *Fixed Supply PDO* or *AVS APDO* Peak Current bits set to 01b, 10b and 11b **Shall** be designed to support one of the overload *Capabilities* defined in [Table 6.10, "Fixed Power Source Peak Current Capability"](#) or [Table 6.16, "EPR AVS Power Source Peak Current Capability"](#) respectively. The overload conditions are bound in magnitude, duration and duty cycle as listed in [Table 6.10, "Fixed Power Source Peak Current Capability"](#) or [Table 6.16, "EPR AVS Power Source Peak Current Capability"](#). *Sources* are not required to support continuous overload operation. When overload conditions occur, the *Source* is allowed the range of *vSrcPeak* (instead of *vSrcNew*) relative to the nominal value (see [Figure 7.14, "Source Peak Current Overload"](#)). When the overload capability is exceeded, the *Source* is expected take whatever action is necessary to prevent electrical or thermal damage to the *Source*. The *Source* **May** send a new *Source_Capabilities* Message with the *Fixed Supply PDO* or *AVS APDO* Peak Current bits set to 00b to prohibit overload operation even if an overload capability was previously *Negotiated* with the *Sink*.

Figure 7.14 Source Peak Current Overload



7.1.12 Source Capabilities Extended Parameters

Implementers can choose to make available certain characteristics of a *PDUSB Source* as a set of **Static** and/or dynamic parameters to improve interoperability between external power sources and portable computing devices. The complete list of reportable **Static** parameters is described in full in [Section 6.5.1](#), "[Source Capabilities Extended Message](#)" and listed in [Figure 6.37](#), "[Source Capabilities Extended Message](#)". The subset of parameters listed below directly represent *Source Capabilities* and are described in the rest of this section.

- Voltage Regulation.
- Holdup Time.
- Compliance.
- Peak Current.
- Source Inputs.
- Batteries.

7.1.12.1 Voltage Regulation Field

The power consumption of a device can change dynamically. The ability of the *Source* to regulate its voltage output might be important if the device is sensitive to fluctuations in voltage. The **Voltage Regulation** bit field is used to convey information about the *Sources* output regulation and tolerance to various load steps.

7.1.12.1.1 Load Step Slew Rate

The default load step slew rate is established at 150mA/μs. A *Source* **Shall** meet the following requirements under the load step reported in the [Source Capabilities Extended Message](#):

- The *Source* **Shall** maintain *V_{BUS}* regulation within the *vSrcValid* range.
- The noise on the *CC* line **Shall** remain below *vNoiseIdle* and *vNoiseActive*.

Test conditions require a change in both positive and negative load steps from 1Hz to 5000Hz, up to the *Advertised* Load Step Magnitude of the full load output including from both 10 mA and 10% initial load. The *Source* **Shall** ensure that PD Communications meet the transmit and receive masks as specified in [Section 5.8.2, "Transmit and Receive Masks"](#) under all load conditions.

7.1.12.1.2 Load Step Magnitude

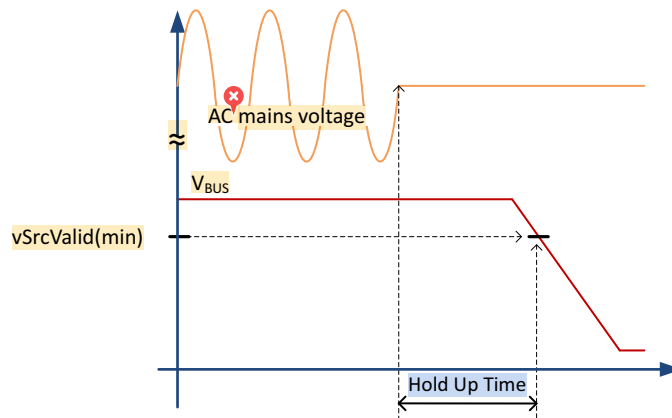
The default load step magnitude rate **Shall** be 25% of I_{OC} . The *Source* **May** report higher capability tolerating a load step of 90% of I_{OC} .

7.1.12.2 Holdup Time Field

The *Holdup Time* field **Shall** return a numeric value of the number of milliseconds the output voltage stays in regulation upon a short interruption of the *AC Supply*.

An *AC Supplied Source* **Shall** report its holdup time in this field. The holdup time is measured with the load at rated maximum, with the *AC Supply* at 115VAC rms and 60Hz (or at 230VAC rms and 50Hz for a *Source* that does not support 115VAC *AC Supply*). The reported time describes the minimum length of time from the last completed *AC Supply* input cycle (zero-degree phase angle) until when the output voltage decays below $v_{SrcValid}$ (min). *Sources* are recommended to support a minimum of 3ms and are preferred to support over 10 milliseconds holdup time (equivalent to a half cycle drop from the *AC Supply*). See [Figure 7.15, "Holdup Time Measurement"](#).

Figure 7.15 Holdup Time Measurement



7.1.12.3 Compliance Field

An *SPR Source* claiming *LPS*, PS1 or PS2 compliance (see [\[IEC 62368-1\]](#)) **Shall** report its *Capabilities* in the *Compliance* field. Since the *SPR Source* **May** have several potential output voltage and current settings, every *SPR Source* supply (each indicated by a *PDO*) **Shall** be compliant to *LPS* requirements.

Note: According to the requirements of [\[IEC 60950-1\]](#) and/or [\[IEC 62368-3\]](#), a device tested and certified with an *LPS Source* (*SPR Source* or *EPR Source* operating in *SPR Mode*) is prohibited from using a non-*LPS Source* (*EPR Source* operating in *EPR Mode*). Alternatively, [\[IEC 62368-1\]](#), classifies power sources according to their maximum, constrained power output (15watts or 100watts).

7.1.12.4 Peak Current

The *Source* reports its ability to source peak current delivery in excess of the *Negotiated* amount in the *Peak Current* field. The duration of peak current **Shall** be followed by a current consumption below the Operating Current (I_{OC}) in order to maintain average power delivery below the I_{OC} current.

A Source **May** have greater capability to source peak current than can be reported using the **Peak Current** field in the **Fixed Supply PDO** or **AVS APDO**. In this case the Source **Shall** report its additional capability in the **Peak Current1/Peak Current2/Peak Current3** fields in the **Source_Capabilities_Extended** Message.

Each overload period **Shall** be followed by a period of reduced current draw such that the rolling average current over the Overload Period field value with the specified Duty Cycle field value (see [Section 6.5.1.10, "Peak Current Field"](#)) **Shall Not** exceed the **Negotiated** current. This is calculated as:

Period of reduced current = (1 - value in Duty Cycle field/100) * value in Overload Period field

7.1.12.5 Source Inputs

The **Source Inputs** field identifies the possible inputs that provide power to the Source.

Note: Some Sources are only powered by a Battery (e.g., an automobile) rather than the more common AC Supply.

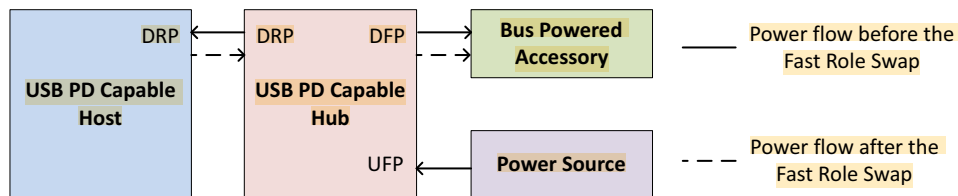
7.1.12.6 Batteries

The **Number of Batteries/Battery Slots** field **Shall** report the number of Batteries the Source supports. The Source **Shall** independently report the number of Hot Swappable Batteries and the number of **Fixed Batteries**.

7.1.13 Fast Role Swap

A **Fast Role Swap** limits the interruption of **VBUS** power to a bus powered accessory connected to a **Hub DFP** that has a **UFP Attached** to a power source and a **DRP Attached** to a **Host Port** supporting **DRP** as shown in [Figure 7.16, "VBUS Power during Fast Role Swap"](#).

Figure 7.16 VBUS Power during Fast Role Swap



When the power source connected to the **Hub UFP** stops sourcing power and **VBUS** at the **Hub DRP** connector discharges below **vSrcValid**(min), if **VBUS** has been **Negotiated** to a higher voltage than **vSafe5V**, or **vSafe5V** (min) the **Fast Role Swap Request** **Shall** be sent from the **Hub DRP** to the **Host DRP** and the **Hub DRP** **Shall** sink power. In the **Fast Role Swap** use case, the **Hub DRP** behaves like a bidirectional power path. The **Hub DRP** **Shall Not** enable **VBUS** discharge circuitry when changing operation from **Initial Source** to **New Sink**. The **Hub DFP Port(s)** **Shall** support default **USB Type-C Current** (see [\[USB Type-C 2.4\]](#)) until a new **Explicit Contract** is **Negotiated**.

After sending the **Fast Role Swap Request** and while **VBUS > vSafe5V** (min), the **New Sink** **Shall Not** draw more than **iNewFrsSink** until the **New Source** has applied its **R_p**. The **New Sink** **Shall Not** draw more than **iSnkStdby** from **VBUS** until **tSnkFRSwap** after it has started sending the **Fast Role Swap Request** or **VBUS** has fallen below **vSafe5V** (min). The **tSnkFRSwap** time **Shall** start at the beginning of the **Fast Role Swap Request** or when **VBUS** falls below **vSafe5V** (min), whichever comes later. After waiting for **tSnkFRSwap**, the **New Sink** **Shall Not** draw more than **iNewFrsSink** until the **New Source** has applied its **R_p**. After the **New Source** has applied its **R_p**, the **New Sink** **Shall** be limited to **USB Type-C Current** (see [\[USB Type-C 2.4\]](#)) in an **Implicit Contract** until a new **Explicit Contract** is **Negotiated**. All Sink requirements **Shall** apply to the **New Sink** after the **Fast Role Swap** is complete. The **Fast Role Swap** response of the **Host DRP** is described in [Section 7.2.10, "Fast Role Swap"](#) since the **Host DRP** is operating as the **Initial Sink** prior to the **Fast Role Swap**.

After the **VBUS voltage** level at the **Hub DRP** connector drops below **vSafe5V** a **PS_RDY** Message **Shall** be sent to the **Host DRP** as shown in the **Fast Role Swap** transition diagram of [Section 7.3.4, "Transitions Caused by Fast Role Swap"](#).

Figure 7.17, "*VBUS detection and timing during Fast Role Swap, initial VBUS (at new source) > vSafe5V(min)*" and Figure 7.18, "*VBUS detection and timing during Fast Role Swap, initial VBUS (at new source) < vSafe5V(min)*" show the VBUS detection and timing for the New Source during a Fast Role Swap after the Fast Role Swap Request has been received. The New Source **May** turn on the VBUS output switch once VBUS is below vSafe5V (max). In this case, the New Source prevents VBUS from falling below vSafe5V (min). The new source **Shall** turn on the VBUS output switch within tSrcFRSwap of falling below vSafe5V (min).

VBUS might have started at vSafe5V or at higher voltage. When the Fast Role Swap Request is detected, VBUS could therefore be either above vSafe5V (max), within the vSafe5V range, or below vSafe5V (min). If the Fast Role Swap Request is detected when VBUS is below vSafe5V (min), then the new source **Shall** turn on the VBUS output switch within tSrcFRSwap of detecting the Fast Role Swap Request. In this case, the maximum time from the beginning of the Fast Role Swap Request to VBUS being sourced **May** be tSrcFRSwap (max) + tFRSwapRx (max).

Figure 7.17 VBUS detection and timing during Fast Role Swap, initial VBUS (at new source) > vSafe5V(min)

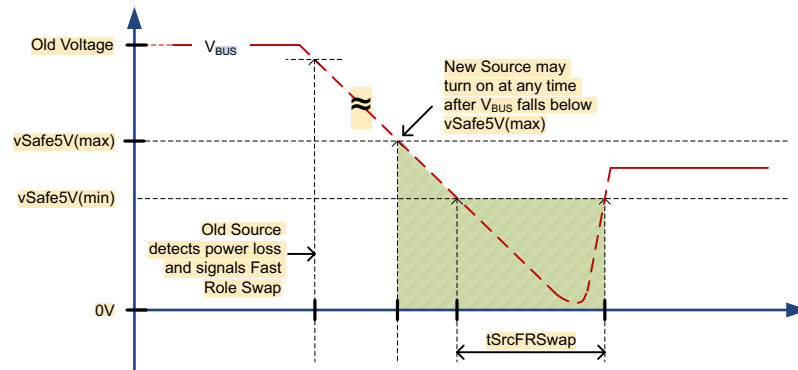
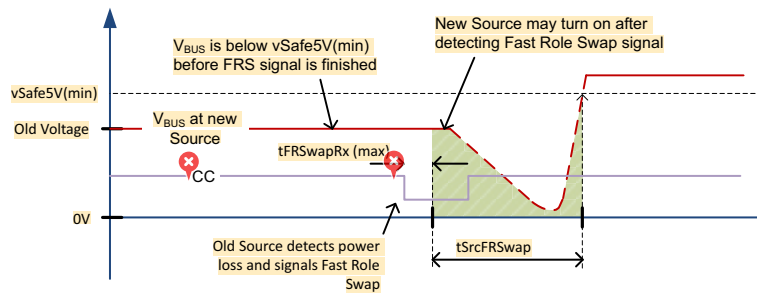


Figure 7.18 VBUS detection and timing during Fast Role Swap, initial VBUS (at new source) < vSafe5V(min)



7.1.14 Non-application of VBUS Slew Rate Limits

Scenarios where vSrcSlewPos and vPpsSlewPos VBUS slew rate limits do not apply and VBUS **May** transition faster than specified are as follows:

- When first applying VBUS after an Attach.
- When applying VBUS as part of a Power Role Swap to Source Power Role.
- When increasing VBUS from vSafe0V to vSafe5V during a Hard Reset.
- During a Fast Role Swap when the Initial Sink applies VBUS.

Scenarios where $v_{SrcSlewNeg}$ and $v_{PpsSlewNeg}$ V_{BUS} slew rate limits do not apply and V_{BUS} **May** transition faster than specified are as follows:

- When discharging V_{BUS} to v_{Safe0V} during a *Hard Reset*.
- When discharging V_{BUS} to v_{Safe0V} as part of a *Power Role Swap* to Sink *Power Role*.
- When discharging V_{BUS} to v_{Safe0V} after a *Detach*.
- During a *Fast Role Swap* when the V_{BUS} power source connected to the *Hub UFP* stops sourcing power.

7.1.15 VCONN Power Cycle

7.1.15.1 UFP VCONN Power Cycle

The *Data Reset* process requires the *DFP* to be the *VCONN Source* by the end of the process. In the case where the *UFP* is the *VCONN Source*, the following steps **Shall** be followed:

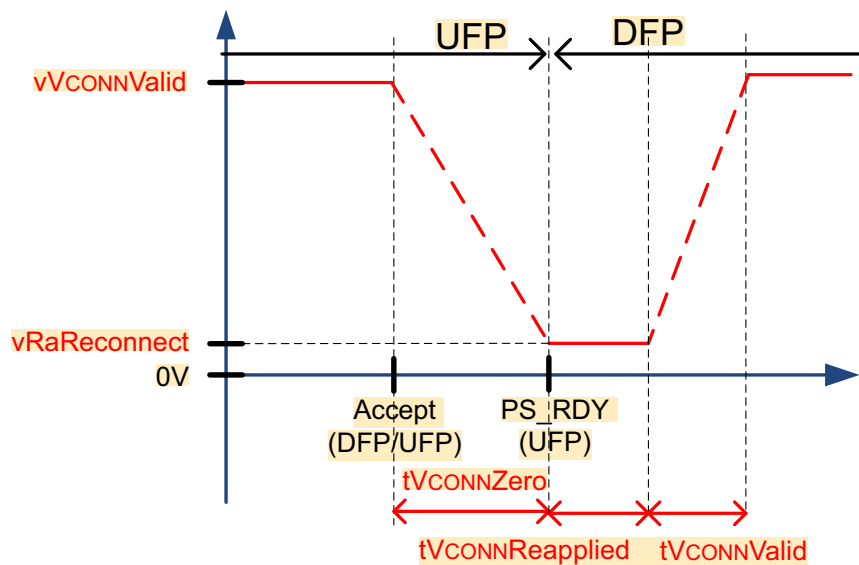
- Following the last bit of the *GoodCRC Message* acknowledging the *Accept Message* in response to the *Data_Reset Message*, the *UFP Shall* turn off *VCONN* and ensure it is below $v_{RaReconnect}$ (see [USB Type-C 2.4]) within $t_{VCONNZero}$.
- When *VCONN* is below $v_{RaReconnect}$, the *UFP Shall* send a *PS_RDY Message*.

Note: If the *UFP* was not sourcing *VCONN*, it still sends the *PS_RDY Message*.

- The *DFP Shall* wait $t_{VCONNReapplied}$ following the last bit of the *GoodCRC Message* acknowledging the *PS_RDY Message* before sourcing *VCONN*. The *DFP Shall* ensure *VCONN* is within $v_{VCONNValid}$ (see [USB Type-C 2.4]) within $t_{VCONNValid}$.

Figure 7.19, "Data Reset UFP VCONN Power Cycle" below illustrates the *UFP VCONN Power Cycle* process.

Figure 7.19 Data Reset UFP VCONN Power Cycle



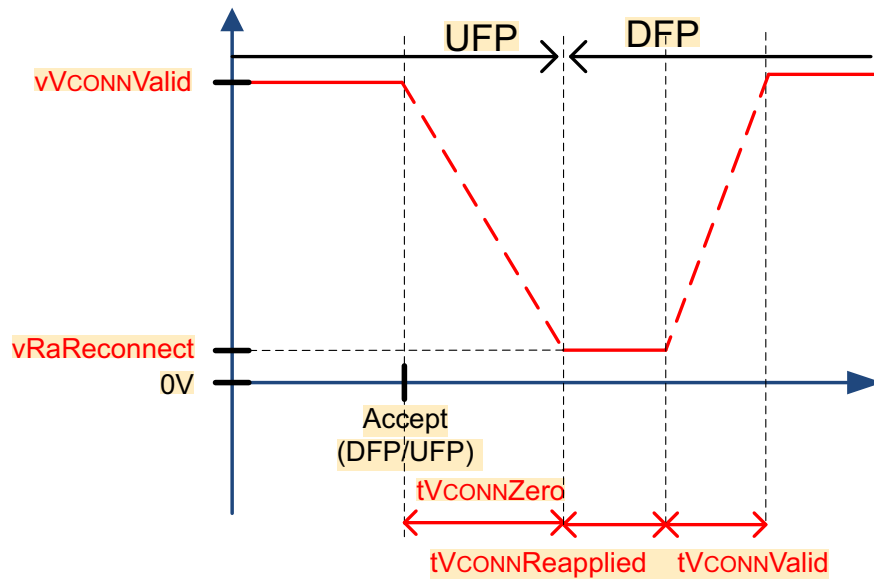
7.1.15.2 DFP VCONN Power Cycle

* The *Data Reset* process requires the *DFP* to be the *VCONN Source* by the end of the process. In the case where the *DFP* is the *VCONN Source*, the following steps **Shall** be followed:

- 1) If the *DFP* sent the *Data_Reset* Message and is sourcing *VCONN* then it **Shall** turn off *VCONN* and ensure it is below *vRaReconnect* (see [USB Type-C 2.4]) within *tVCONNZero* of the last bit of the *GoodCRC* Message acknowledging the *Accept* Message in response to the *Data_Reset* Message.
- 2) If the *UFP* sent the *Data_Reset* Message then the *DFP* **Shall** turn off *VCONN* and ensure it is below *vRaReconnect* (see [USB Type-C 2.4]) within *tVCONNZero* following the last bit of the *GoodCRC* Message acknowledging the *Accept* Message in response to the *Data_Reset* Message.
- 3) When *VCONN* is below *vRaReconnect*, the *DFP* **Shall** wait *tVCONNReapplied* before sourcing *VCONN*.
- 4) The *DFP* **Shall** ensure *VCONN* is within *vVCONNValid* (see [USB Type-C 2.4]) within *tVCONNValid*.

Figure 7.20, "Data Reset DFP VCONN Power Cycle" below illustrates the *DFP VCONN* Power Cycle process.

Figure 7.20 Data Reset DFP VCONN Power Cycle



7.2 Sink Requirements

7.2.1 Behavioral Aspects

A **PDUSB Sink** exhibits the following behaviors:

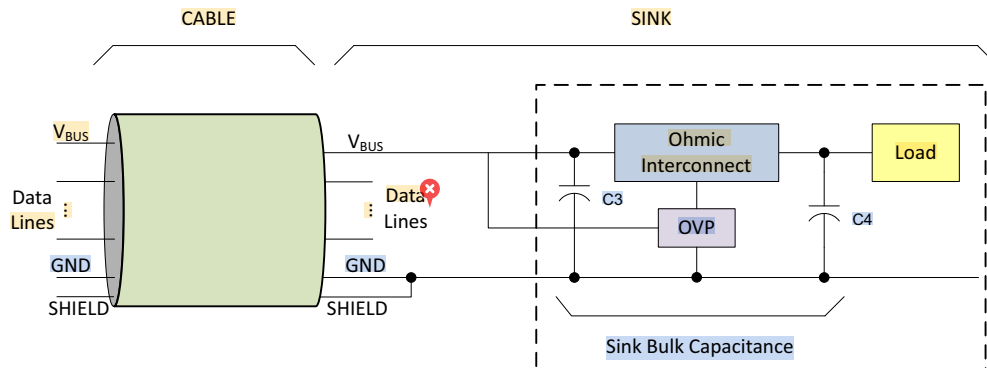
- **Shall Not** draw more than **[USB Type-C 2.4]** USB Type-C Current from **V_{BUS}** while in a **Default Contract** or **Implicit Contract**.
- **Shall** follow the requirements as specified in **Section 7.1.5, "Response to Hard Resets"** when **Hard Reset Signaling** is received.
- **Shall** control **V_{BUS}** in-rush current when increasing current consumption according to **[USB 2.0]** or **[USB 3.2]** as appropriate.

7.2.2 Sink Bulk Capacitance

The **Sink** bulk capacitance consists of **C3** and **C4** as shown in **Figure 7.21, "Placement of Sink Bulk Capacitance"**. The Ohmic Interconnect might consist of PCB traces for power distribution or power switching devices. The Ohmic Interconnect is expected to be part of an input Over Voltage Protection (**Sink OVP**) circuit implemented by the **Sink** as described in **Section 7.2.9.2, "Input Over Voltage Protection"** to protect against excessive **V_{BUS}** input voltage. A **Sink** **Shall** implement **OVP**. The **Sink** **Shall Not** rely on the **Source** output voltage limit for its input **OVP**. The capacitance might be a single capacitor, a capacitor bank or distributed capacitance. An upper bound of **cSnkBulkPd** **Shall Not** be exceeded so that the transient charging, or discharging, of the total bulk capacitance on **V_{BUS}** can be accounted for during voltage transitions.

The **Sink** bulk capacitance that is within the **cSnkBulk** max or **cSnkBulkPd** max limits is allowed to change to support a newly **Negotiated** power level. The capacitance can be changed when the **Sink** enters **Sink Standby** or during a voltage transition or when the **Sink** begins to operate at the new power level. Changing the **Sink** bulk capacitance **Shall Not** cause a transient current on **V_{BUS}** that violates the present **Contract**. During a **Power Role Swap** the **Default Sink** **Shall** transition to **Swap Standby** before operating as the **New Source**. Any change in bulk capacitance required to complete the **Power Role Swap** **Shall** occur during **Swap Standby**.

Figure 7.21 Placement of Sink Bulk Capacitance



7.2.3 Sink Standby

The **Sink** **Shall** transition to **Sink Standby** before a positive voltage transition of **V_{BUS}**. During **Sink Standby** the **Sink** **Shall** reduce the current drawn to **iSnkStdby**. This allows the **Source** to manage the voltage transition as well as supply sufficient operating current to the **Sink** to maintain PD operation during the transition. The **Sink** **Shall** complete this transition to **Sink Standby** within **tSnkStdby** after evaluating the **Accept Message** from the **Source**. The transition when returning to **Sink** operation from **Sink Standby** **Shall** be completed within **tSnkNewPower**. The **iSnkStdby** requirement **Shall** only apply if the **Sink** current draw is higher than this level.

See **Section 7.3, "Transitions"** for details.

7.2.3.1 Programmable Power Supply Sink Standby

A Sink is not required to transition to Sink Standby when operating within the *Negotiated* PPS APDO. A Sink **May** consume the Operating Current value in the PPS RDO during PPS output voltage changes. However, prior to operating the SPR PPS in Current Limit, the Sink **Shall** program the PPS Operating Voltage to the lowest practical level that satisfies the Sink load requirement. Doing so will minimize the inrush current that occurs when the transition to Current Limit occurs. When operating with an SPR PPS Source that is in Current Limit, the Sink **Shall Not** change its load in a manner that exceeds *iPpsCLLoadStepRate* or *iPpsCLLoadReleaseRate*. The load change magnitude **Shall Not** request a change to the Current Limit set-point that exceeds *iPpsCLLoadStep*.

If the Sink *Negotiates* for a new PPS APDO, that is expected to increase *V_{BUS}* voltage, then the Sink **Shall** transition to Sink Standby while changing between PPS APDOs as described in [Section 7.3.1, "Transitions caused by a Request Message"](#).

7.2.4 Suspend Power Consumption

When Source has set its *USB Suspend Supported* flag (see [Section 6.4.1.2.1.2, "USB Suspend Supported"](#)), a Sink **Shall** go to the lowest power state during USB suspend. The lowest power state **Shall** be *pSnkSusp* or lower for a PDUSB Peripheral and *pHubSusp* or lower for a PDUSB Hub. There is no requirement for the Source voltage to be changed during USB suspend.

7.2.5 Zero Negotiated Current

When a Sink Requests zero current as part of a power *Negotiation* with a Source, the Sink **Shall** go to the lowest power state, *pSnkSusp* or lower, where it can still communicate using PD signaling.

7.2.6 Transient Load Behavior

When a Sink's operating current changes due to a load step, load release or any other change in load level, the positive or negative overshoot of the new load current **Shall Not** exceed the range defined by *iOvershoot*. For the purposes of measuring *iOvershoot* the new load current value is defined as the average steady state value of the load current after the load step has settled. The rate of change of any shift in Sink load current during normal operation **Shall Not** exceed *iLoadStepRate* (for load steps) and *iLoadReleaseRate* (for load releases) as measured at the Sink receptacle.

The Sink's operating current **Shall Not** change faster than the value reported in the Source's Load Step Slew Rate in its *Voltage Regulation* bit field and **Shall** ensure that PD Communications meet the transmit and receive masks as specified in [Section 5.8.2, "Transmit and Receive Masks"](#).

7.2.7 Swap Standby for Sinks

The Sink functionality in a Dual-Role Power Port **Shall** support Swap Standby. Swap Standby occurs for the Sink after evaluating the *Accept* Message from the Source during a Power Role Swap. While in Swap Standby the Sink's current draw **Shall Not** exceed *iSnkSwapStdby* from *V_{BUS}* and the Dual-Role Power Port **Shall** be configured as a Source after *V_{BUS}* has been discharged to *vSafe0V* by the existing Initial Source. The Sink's USB connection **Should Not** be reset even though *vSafe5V* is not present on the *V_{BUS}* conductor (see [Section 9.1.2, "Mapping to USB Device States"](#)). The time for the Sink to transition to Swap Standby **Shall** be no more than *tSnkSwapStdby*. When in Swap Standby the Sink has relinquished its Power Role as Sink and will prepare to become the New Source. The transition time from Swap Standby to New Source **Shall** be no more than *tNewSrc*.

7.2.8 Sink Peak Current Operation

Sinks **Shall** only make use of a Source overload capability when the corresponding Fixed Supply PDO Peak Current (see [Section 6.4.1.2.1.8, "Peak Current"](#)) or AVS APDO Peak Current (see [Section 6.4.1.2.4.3.2, "Peak Current"](#)) bits are set to 01b, 10b and 11b. Sinks **Shall** manage thermal aspects of the overload event by not exceeding the average *Negotiated* output of a Fixed Supply or AVS that supports Peak Current operation.

Sinks that depend on the Peak Current capability for enhanced system performance **Shall** also function correctly when Attached to a Source that does not offer the Peak Current capability or when the Peak Current capability has been inhibited by the Source.

7.2.9 Robust Sink Operation

7.2.9.1 Sink Bulk Capacitance Discharge at Detach

When a *Source* is *Detached* from a *Sink*, the *Sink* **shall** continue to draw power from its input bulk capacitance until *VBUS* is discharged to **vSafe5V** or lower by no longer than **tSafe5V** from the *Detach* event. This safe *Sink* requirement **shall** apply to all *Sinks* operating with a *Negotiated VBUS* level greater than **vSafe5V** and **shall** apply during all low power and high-power operating modes of the *Sink*.

If the *Detach* is detected during a *Sink* low power state, such as USB Suspend, the *Sink* can then draw as much power as needed from its bulk capacitance since a *Source* is no longer *Attached*. In order to achieve a successful *Detach* detect based on *VBUS* voltage level droop, the *Sink* power consumption **shall** be high enough so that *VBUS* will decay below **vSrcValid**(min) well within **tSafe5V** after the *Source* bulk capacitance is removed due to the *Detach*. Once adequate *VBUS* droop has been achieved, a discharge circuit can be enabled to meet the safe *Sink* requirement.

To illustrate the point, the following set of *Sink* conditions will not meet the safe *Sink* requirement without additional discharge circuitry:

- *Negotiated VBUS* = 20V.
- Maximum allowable supplied *VBUS* voltage = 21.55V.
- Maximum bulk capacitance = 30μF.
- Power consumption at *Detach* = 12.5mW.

When the *Detach* occurs (hence removal of the *Source* bulk capacitance) the 12.5mW power consumption will draw down the *VBUS* voltage from the worst-case maximum level of 21.55V to 17V in approximately 205ms. At this point, with *VBUS* well below **vSrcValid** (min) an approximate 100mW discharge circuit can be enabled to increase the rate of *Sink* bulk capacitance discharge and meet the safe *Sink* requirement. The power level of the discharge circuit is dependent on how much time is left to discharge the remaining voltage on the *Sink* bulk capacitance. If a *Sink* has the ability to detect the *Detach* in a different manner and in much less time than **tSafe5V**, then this different manner of detection can be used to enable a discharge circuit, allowing even lower power dissipation during low power modes such as USB Suspend.

In most applications, the safe *Sink* requirement will limit the maximum *Sink* bulk capacitance well below the **cSinkBulkPd** limit. A *Detach* occurring during *Sink* high power operating modes must quickly discharge the *Sink* bulk capacitance to **vSafe5V** or lower as long as the *Sink* continues to draw adequate power until *VBUS* has decayed to **vSafe5V** or lower.

7.2.9.2 Input Over Voltage Protection

Sinks **shall** implement input Over-Voltage Protection (OVP) to prevent damage from input voltage that exceeds the voltage handling capability of the *Sink*. The definition of voltage handling capability is left to the discretion of the *Sink* implementation. The over voltage response of *Sinks* **shall not** interfere with normal PD operation and **shall** account for **vSrcNew**, **vSrcValid** or **vPpsNew**, **vPpsValid** as determined by the *Negotiated VBUS* value. *SPR Sinks* **should** tolerate input voltages as high as **vSprMax** and **shall** meet applicable safety requirements if **vSprMax** is exceeded. Likewise, *EPR Sinks* **should** tolerate input voltages as high as **vEprMax** and **shall** meet applicable safety requirements if **vEprMax** is exceeded.

Sinks **should** attempt to send **Hard Reset Signaling** when OVP engages followed by an **Alert Message** indicating an OVP event once an *Explicit Contract* has been established. The OVP response **may** engage at either the *Port* or system level. Systems or ports that have engaged OVP **shall** resume *USB Default Operation* when the *Source* has re-established **vSafe5V** on *VBUS*.

The *Sink* **shall** be able to *Re-negotiate* with the *Source* after resuming *USB Default Operation*. The decision of how to respond to *Re-negotiation* after an OVP event is left to the discretion of the *Sink* implementation.

The *Sink* **shall** prevent continual system or *Port* cycling if OVP continues to engage after initially resuming either *USB Default Operation* or *Re-negotiation*. Latching off the *Port* or system is an acceptable response to recurring over voltage.

7.2.9.3 Over Temperature Protection

Sinks **Shall** implement over temperature protection (OTP) to prevent damage from temperature that exceeds the thermal capability of the Sink. The definition of thermal capability and the monitoring locations used to trigger the over temperature protection are left to the discretion of the Sink implementation.

Sinks **Shall** attempt to send **Hard Reset Signaling** when over temperature protection engages followed by an **Alert Message** indicating an OTP event once an **Explicit Contract** has been established. The over temperature protection response **May** engage at either the **Port** or system level. Systems or ports that have engaged over temperature protection **Should** attempt to resume **USB Default Operation** after sufficient cooling is achieved and **May** latch off to protect the **Port** or system. The definition of sufficient cooling is left to the discretion of the Sink implementation.

The Sink **Shall** be able to **Re-negotiate** with the Source after resuming **USB Default Operation**. The decision of how to respond to **Re-negotiation** after an over temperature event is left to the discretion of the Sink implementation.

The Sink **Shall** prevent continual system or **Port** cycling if over temperature protection continues to engage after initially resuming either **USB Default Operation** or **Re-negotiation**. Latching off the **Port** or system is an acceptable response to recurring over temperature.

7.2.9.4 Over Current Protection

Sinks that operate with a Programmable Power Supply **Shall** implement their own internal current protection mechanism to protect against internal **V_{BUS}** current faults as well as erratic **Source** current regulation. The Sink **Shall** never draw higher current than the Maximum Current value in the PPS **APDO**.

7.2.10 Fast Role Swap

As described in [Section 7.1.13, "Fast Role Swap"](#) a Fast Role Swap limits the interruption of **V_{BUS}** power to a bus powered accessory connected to a Hub DFP that has a **UFP Attached** to a power source and a **DRP Attached** to a Host **Port** that supports **DRP**. This configuration is shown in [Figure 7.16, "V_{BUS} Power during Fast Role Swap"](#).

The Host **DRP**, upon establishing an **Explicit Contract**, **Shall** query the **Initial Source's Sink Capabilities** to determine whether the **Initial Source** supports Fast Role Swap, and what level of current it requires. If the **Sink Capabilities Message** received from the **Initial Source** has at least one of the **Fast Role Swap required USB Type-C Current** bits set, and the Host **DRP** is able to source the requested current at 5V, the Host **DRP May** arm itself for Fast Role Swap. If the Host **DRP** has not queried the **Sink Capabilities** from the **Initial Source**, or if the **Sink Capabilities Message** reports no Fast Role Swap support or a current that is beyond what the Host **DRP** is able or willing to source in the event of a Fast Role Swap, the Host **DRP Shall Not** arm itself for Fast Role Swap and **Shall Ignore** any Fast Role Swap **Requests** that are detected.

When the Host **DRP** that supports Fast Role Swap detects the **Fast Role Swap Request**, the Host **DRP Shall** stop sinking current and **Shall** be ready and able to source **vSafe5V** if the residual **V_{BUS} voltage** level at the Host **DRP** connector is greater than **vSafe5V**. When the residual **V_{BUS} voltage** level at the Host **DRP** connector discharges below **vSafe5V(min)** the Host **DRP** as the **New Source Shall** supply **vSafe5V** to the Hub **DRP** within **tSrcFRSwap**. The Host **DRP Shall Not** enable **V_{BUS}** discharge circuitry when changing **Power Roles** from **Initial Sink** to **New Source**.

The **New Source Shall** supply **vSafe5V** at **USB Type-C Current** (see [\[USB Type-C 2.4\]](#)) at the value Advertised in the **Fast Role Swap required USB Type-C Current** field (see [Section 6.4.1.3.1.6, "Fast Role Swap USB Type-C Current"](#)). All **Source** requirements **Shall** apply to the **New Source** after the Fast Role Swap is complete. The Fast Role Swap response of the Hub **DRP** is described in [Section 7.1.13, "Fast Role Swap"](#) since the Hub **DRP** is operating as the **Initial Source** prior to the Fast Role Swap.

After the Host **DRP** is providing **V_{BUS}** power to the Hub **DRP**, a **PS_RDY Message Shall** be sent to the Hub **DRP** as defined by the Fast Role Swap **Request** and the **AMS** detailed in [Section 7.3.4, "Transitions Caused by Fast Role Swap"](#).

7.3 Transitions

The following sections illustrate the power supply's response to various types of *Negotiations*. The *Negotiations* are triggered by certain *Messages* or *Signaling*. It provides examples of the transitions and is organized around each of the *Messages* and *Signals* that result in a response from the power supply. The response to a *Message* or *Signal* can result in different transitions depending upon the power supply's starting conditions and the requested change.

- Transitions caused by a *Request Message*:
 - Generic transition between *(A)PDO s*:
 - Increase the current.
 - Increase the *voltage*.
 - Increase the *voltage* and the current.
 - Increase the *voltage* and decrease the current.
 - Decrease the *voltage* and increase the current.
 - Decrease the *voltage* and the current.
 - No change in Current or *voltage*.
 - Transitions within the same *PDO (Fixed Supply, Battery Supply, Variable Supply)*:
 - Increase the current.
 - Decrease the current.
 - No change in current.
 - Transitions within the same PPS *APDO*:
 - Increasing the Programmable Power Supply (PPS) *voltage*.
 - Decreasing the Programmable Power Supply (PPS) *voltage*.
 - Increasing the Programmable Power Supply (PPS) Current.
 - Decreasing the Programmable Power Supply (PPS) Current.
 - Same Request Programmable Power Supply (PPS).
 - Transitions within the same *AVS APDO*:
 - Increasing the *Adjustable Voltage Supply (AVS) voltage*
 - Decreasing the *Adjustable Voltage Supply (AVS) voltage*
 - Same Request *Adjustable Voltage Supply (AVS)*
- Transitions caused by the *PR_Swap Message*:
 - *Source* requests a *Power Role Swap*
 - *Sink* requests a *Power Role Swap*
- Transitions caused by *Hard Reset Signaling*:
 - *Source* issues *Hard Reset Signaling*.
 - *Sink* issues *Hard Reset Signaling*.
- Transitions caused by the *Fast Role Swap Request*:
 - *Source* asserts R_d at its preferred *[USB Type-C 2.4]* current.

7.3.1 Transitions caused by a Request Message

This section describes transitions that are caused by a *Request Message*.

7.3.1.1 Changing the Source between Different (A)PDOs

In these transition descriptions the term (A)PDO is used to describe any *Power Data Object*, regardless of whether it is a *PDO* or an *APDO* in the *Capabilities Message*.

This section describes transitions in response to a *Request Message*:

- From one (A)PDO to another (A)PDO
- From an *Implicit Contract* to an *Explicit Contract*
- From *[USB Type-C 2.4]operation* to the *First Explicit Contract*

These transitions usually result in a voltage change but is not required.

The interaction of the *Device Policy Manager*, the *Port Policy Engine* and the *Power Supply* that **shall** be followed when increasing the current is shown in [Figure 7.23, "Transition Diagram for Increasing the Voltage and Current"](#) and [Figure 7.25, "Transition Diagram for Decreasing the Voltage and Increasing the Current"](#).

The *Source* voltage as the transition starts **shall** be any voltage within the **Valid** *V_{BUS}* range of the previous *Source* *PDO* or *APDO*. The *Source* voltage after the transition is complete **shall** be any voltage within the **Valid** *V_{BUS}* range of the *New Source* *PDO* or *APDO*. The sequence that **shall** be followed is described in [Table 7.2, "Sequence Diagram for Increasing the Voltage and Current"](#) and [Table 7.4, "Sequence Description for Decreasing the Voltage and Increasing the Current"](#). The timing parameters that **shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Note: In this figure, the *Sink* has previously sent a *Request Message* to the *Source*.

The voltage is considered to increase if the change from V_{OLD} to V_{NEW} is greater than **vSmallStep**. The determination **shall** be based on the nominal (A)PDO voltage before and after, unless either (A)PDO is *Battery Supply* or *Variable Supply* when the worst case of the following is assumed in making this determination.

- Minimum voltage to voltage.
- Minimum voltage to Maximum voltage.
- Voltage to Maximum voltage.

The following sections begin with a description of the generic process followed by more specific examples of the most common transitions.

7.3.1.1.1



Examples of changes from one (A)PDO to another (A)PDO

The seven examples of (A)PDO change transitions below illustrate the most common transitions.

7.3.1.1.1.1

Increasing the Voltage

The interaction of the *System Policy*, *Device Policy*, and power supply that **shall** be followed when changing from one (A)PDO to another while increasing the voltage is shown in [Figure 7.22, "Transition Diagram for Increasing the Voltage"](#). The sequence that **shall** be followed is described in [Table 7.1, "Sequence Description for Increasing the Voltage"](#). The timing parameters that **shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Note: In [Figure 7.22, "Transition Diagram for Increasing the Voltage"](#), the Sink has previously sent a *Request Message* to the Source.

Figure 7.22 Transition Diagram for Increasing the Voltage

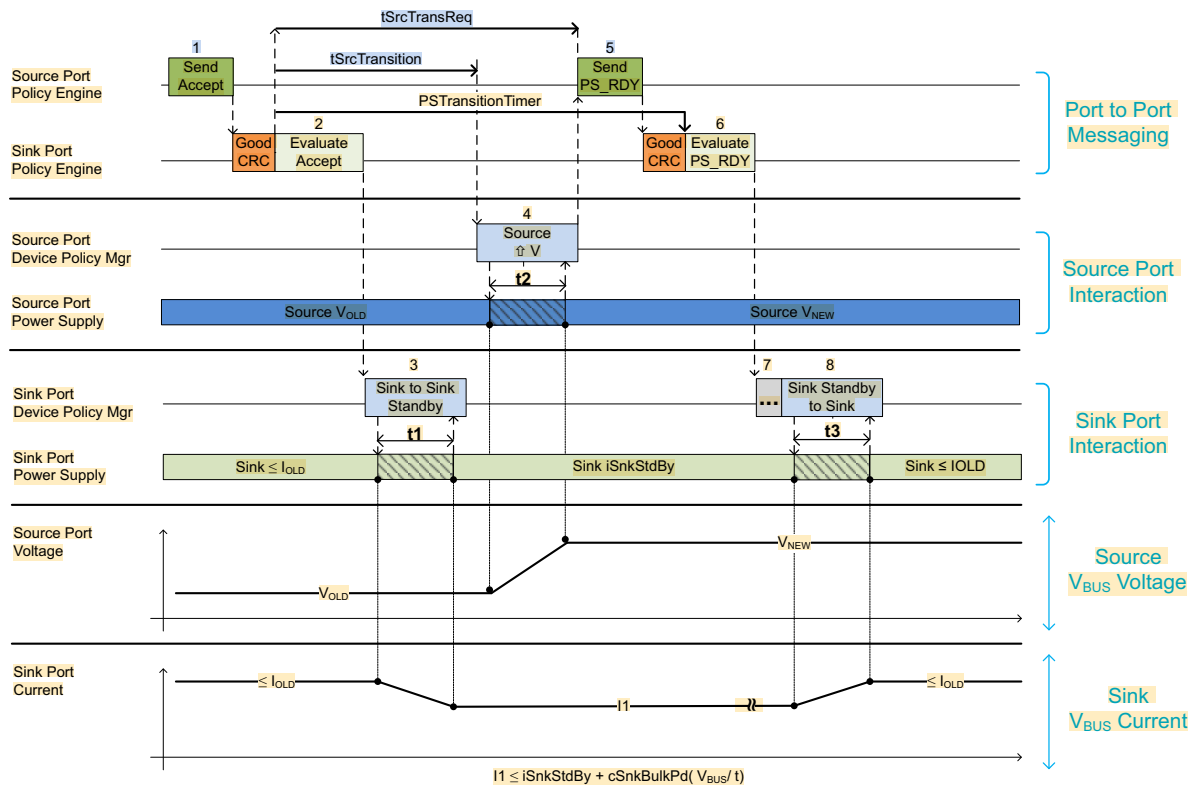


Table 7.1 Sequence Description for Increasing the Voltage

Step	Source Port	Sink Port
1	<i>Policy Engine sends the Accept Message to the Sink.</i>	<i>Policy Engine receives the Accept Message.</i>
2	<i>Protocol Layer receives the GoodCRC Message from the Sink. The Policy Engine tells the Device Policy Manager to instruct the power supply to modify its output power.</i>	<i>Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then starts the PSTransitionTimer and evaluates the Accept Message.</i>
3		<i>Policy Engine tells the Device Policy Manager to instruct the power supply to reduce current drawn to iSnkStdby within tSnkStdby (t1); t1 Shall complete before tSrcTransition. The Sink Shall Not violate transient load behavior defined in Section 7.2.6, "Transient Load Behavior" while transitioning to and operating at the new power level.</i>
4	<i>tSrcTransition after the GoodCRC Message was received, the power supply starts to change its output power capability. The power supply Shall be ready to operate at the new power level within tSrcReady (t2). The power supply informs the Device Policy Manager that it is ready to operate at the new power level. The power supply status is passed to the Policy Engine.</i>	
5	<i>The Policy Engine sends the PS_RDY Message to the Sink starting within tSrcTransReq of the end of the GoodCRC Message following the Accept Message.</i>	<i>The Policy Engine receives the PS_RDY Message from the Source.</i>
6	<i>Protocol Layer receives the GoodCRC Message from the Sink.</i>	<i>Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then stops the PSTransitionTimer, evaluates the PS_RDY Message from the Source and tells the Device Policy Manager that the Source is operating at the new (A)PDO. If the PS_RDY Message is not received before PSTransitionTimer times out the Sink sends Hard Reset Signaling.</i>
7		<i>The Sink May begin operating at the new power level any time after evaluation of the PS_RDY Message. This time duration is indeterminate.</i>
8		<i>The Sink Shall Not violate the transient load behavior defined in Section 7.2.6, "Transient Load Behavior" while transitioning to and operating at the new power level. The time duration (t3) depends on the magnitude of the load change.</i>

The interaction of the *System Policy*, *Device Policy*, and power supply that **shall** be followed when changing from one (A)PDO to another while increasing the voltage and current is shown in [Figure 7.23, "Transition Diagram for Increasing the Voltage and Current"](#). The sequence that **shall** be followed is described in [Table 7.2, "Sequence Diagram for Increasing the Voltage and Current"](#). The timing parameters that **shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Note: In [Figure 7.23, "Transition Diagram for Increasing the Voltage and Current"](#), the Sink has previously sent a *Request Message* to the Source.

Figure 7.23 Transition Diagram for Increasing the Voltage and Current

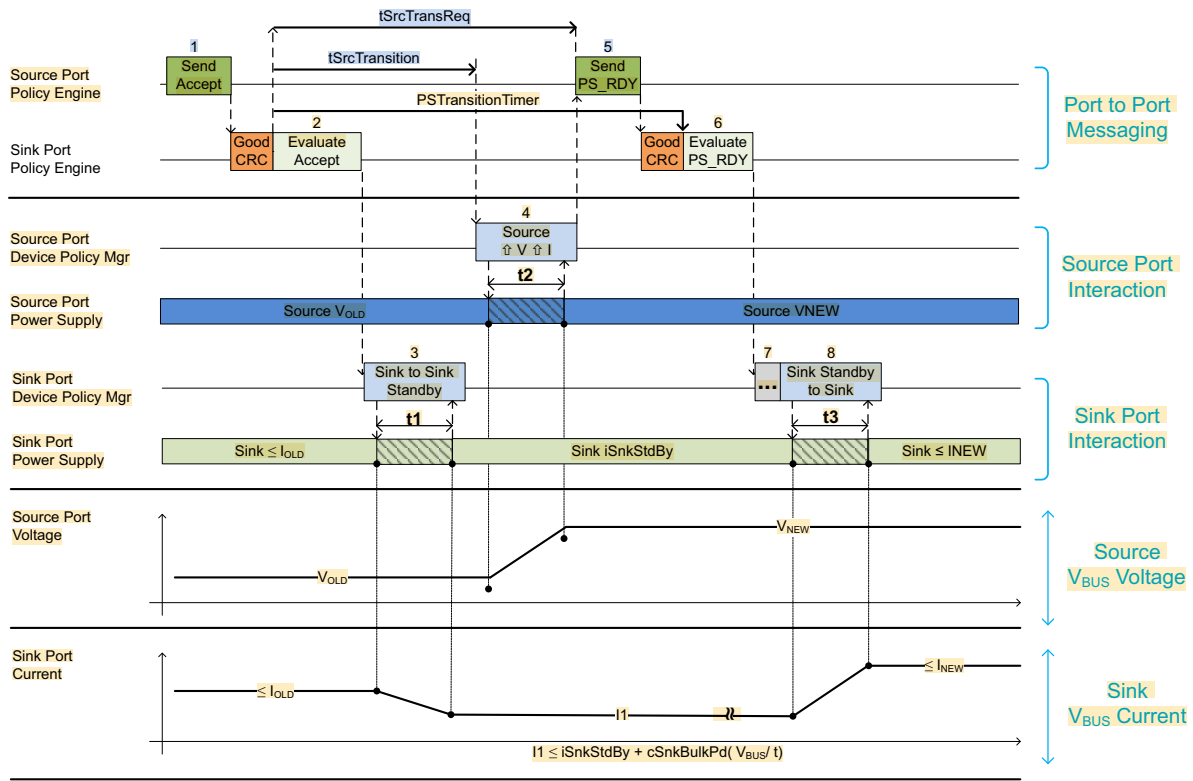


Table 7.2 Sequence Diagram for Increasing the Voltage and Current

Step	Source Port	Sink Port
1	Policy Engine sends the Accept Message to the Sink.	Policy Engine receives the Accept .
2	Protocol Layer receives the GoodCRC Message from the Sink. The Policy Engine tells the Device Policy Manager to instruct the power supply to modify its output power.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then starts the PSTransitionTimer and evaluates the Accept Message.
3		Policy Engine tells the Device Policy Manager to instruct the power supply to reduce current drawn to iSnkStdby within tSnkStdby (t1); t1 Shall complete before tSrcTransition . The Sink Shall Not violate transient load behavior defined in Section 7.2.6, "Transient Load Behavior" while transitioning to and operating at the new power level.
4	tSrcTransition after the GoodCRC Message was received, the power supply starts to change its output power capability. The power supply Shall be ready to operate at the new power level within tSrcReady (t2). The power supply informs the Device Policy Manager that it is ready to operate at the new power level. The power supply status is passed to the Policy Engine.	
5	The Policy Engine sends the PS_RDY Message to the Sink starting within tSrcTransReq of the end of the GoodCRC Message following the Accept Message.	The Policy Engine receives the PS_RDY Message from the Source.
6	Protocol Layer receives the GoodCRC Message from the Sink.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then stops the PSTransitionTimer , evaluates the PS_RDY Message from the Source and tells the Device Policy Manager that the Source is operating at the new (A)PDO. If the PS_RDY Message is not received before PSTransitionTimer times out, the Sink sends Hard Reset Signaling .
7		The Sink May begin operating at the new power level any time after evaluation of the PS_RDY Message. This time duration is indeterminate.
8		The Sink Shall Not violate the transient load behavior defined in Section 7.2.6, "Transient Load Behavior" while transitioning to and operating at the new power level. The time duration (t3) depends on the magnitude of the load change.

The interaction of the *System Policy*, *Device Policy*, and power supply that **Shall** be followed when changing from one (A)PDO to another while increasing the voltage and decreasing the current is shown in [Figure 7.24, "Transition Diagram for Increasing the Voltage and Decreasing the Current"](#). The sequence that **Shall** be followed is described in [Table 7.3, "Sequence Description for Increasing the Voltage and Decreasing the Current"](#). The timing parameters that **Shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Note: In [Figure 7.24, "Transition Diagram for Increasing the Voltage and Decreasing the Current"](#), the Sink has previously sent a *Request Message* to the *Source*.

Figure 7.24 Transition Diagram for Increasing the Voltage and Decreasing the Current

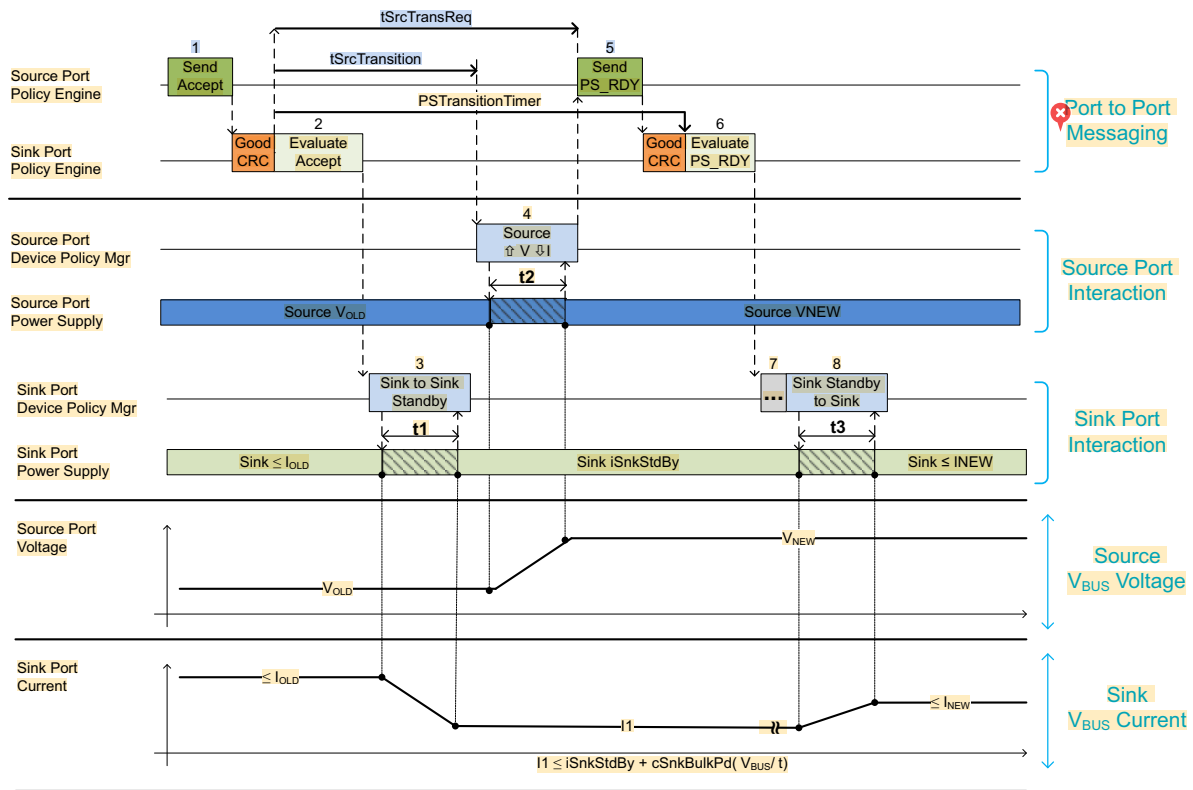


Table 7.3 Sequence Description for Increasing the Voltage and Decreasing the Current

Step	Source Port	Sink Port
1	Policy Engine sends the Accept Message to the Sink.	Policy Engine receives the Accept Message.
2	Protocol Layer receives the GoodCRC Message from the Sink. The Policy Engine tells the Device Policy Manager to instruct the power supply to modify its output power.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then starts the PSTransitionTimer and evaluates the Accept Message.
3		Policy Engine tells the Device Policy Manager to instruct the power supply to reduce current drawn to iSnkStdby within tSnkStdby (t1); t1 Shall complete before tSrcTransition . The Sink Shall Not violate transient load behavior defined in Section 7.2.6, "Transient Load Behavior" while transitioning to and operating at the new power level.
4	tSrcTransition after the GoodCRC Message was received, the power supply starts to change its output power capability. The power supply Shall be ready to operate at the new power level within tSrcReady (t2). The power supply informs the Device Policy Manager that it is ready to operate at the new power level. The power supply status is passed to the Policy Engine.	
5	The Policy Engine sends the PS_RDY Message to the Sink starting within tSrcTransReq of the end of the GoodCRC Message following the Accept Message.	The Policy Engine receives the PS_RDY Message from the Source.
6	Protocol Layer receives the GoodCRC Message from the Sink.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then stops the PSTransitionTimer , evaluates the PS_RDY Message from the Source and tells the Device Policy Manager that the Source is operating at the new (A)PDO. If the PS_RDY Message is not received before PSTransitionTimer times out the Sink sends Hard Reset Signaling .
7		The Sink May begin operating at the new power level any time after evaluation of the PS_RDY Message. This time duration is indeterminate.
8		The Sink Shall Not violate the transient load behavior defined in Section 7.2.6, "Transient Load Behavior" while transitioning to and operating at the new power level. The time duration (t3) depends on the magnitude of the load change.

The interaction of the *System Policy*, *Device Policy*, and power supply that **Shall** be followed when changing from one (A)PDO to another while decreasing the voltage and increasing the current is shown in [Figure 7.25, "Transition Diagram for Decreasing the Voltage and Increasing the Current"](#). The sequence that **Shall** be followed is described in [Table 7.4, "Sequence Description for Decreasing the Voltage and Increasing the Current"](#). The timing parameters that **Shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Note: In [Figure 7.25, "Transition Diagram for Decreasing the Voltage and Increasing the Current"](#), the Sink has previously sent a *Request Message* to the *Source*.

Figure 7.25 Transition Diagram for Decreasing the Voltage and Increasing the Current

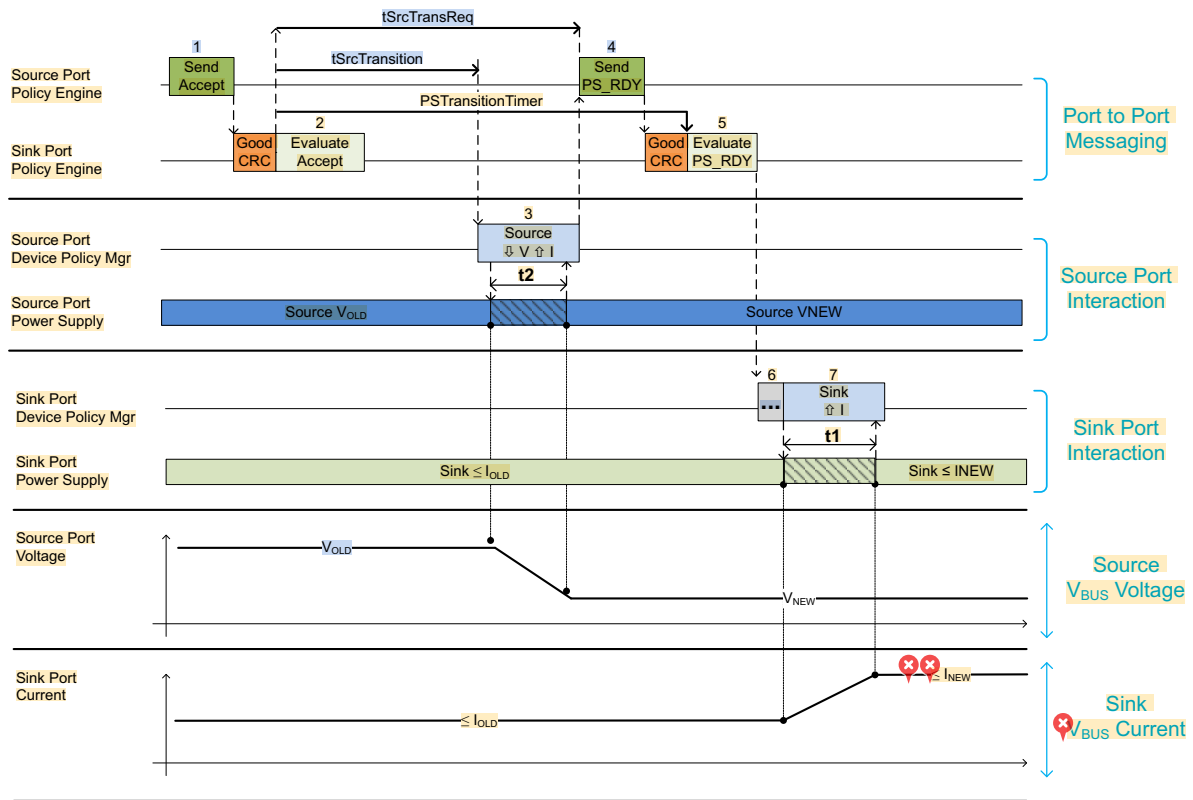


Table 7.4 Sequence Description for Decreasing the Voltage and Increasing the Current

Step	Source Port	Sink Port
1	Policy Engine sends the Accept Message to the Sink.	Policy Engine receives the Accept Message.
2	Protocol Layer receives the GoodCRC Message from the Sink. The Policy Engine tells the Device Policy Manager to instruct the power supply to modify its output power.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then starts the PSTransitionTimer and evaluates the Accept Message.
3	tSrcTransition after the GoodCRC Message was received, the power supply starts to change its output power capability. The power supply Shall be ready to operate at the new power level within tSrcReady (t1). The power supply informs the Device Policy Manager that it is ready to operate at the new power level. The power supply status is passed to the Policy Engine.	
4	The Policy Engine sends the PS_RDY Message to the Sink starting within tSrcTransReq of the end of the GoodCRC Message following the Accept Message.	The Policy Engine receives the PS_RDY Message from the Source.
5	Protocol Layer receives the GoodCRC Message from the Sink.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then evaluates the PS_RDY Message from the Source and tells the Device Policy Manager it is okay to operate at the new power level.
6		The Sink May begin operating at the new power level any time after evaluation of the PS_RDY Message. This time duration is indeterminate.

The interaction of the *System Policy*, *Device Policy*, and power supply that **shall** be followed when changing from one (A)PDO to another while decreasing the voltage is shown in [Figure 7.26, "Transition Diagram for Decreasing the Voltage"](#). The sequence that **shall** be followed is described in [Table 7.5, "Sequence Description for Decreasing the Voltage"](#). The timing parameters that **shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Note: In [Figure 7.26, "Transition Diagram for Decreasing the Voltage"](#), the Sink has previously sent a *Request Message* to the Source.

Figure 7.26 Transition Diagram for Decreasing the Voltage

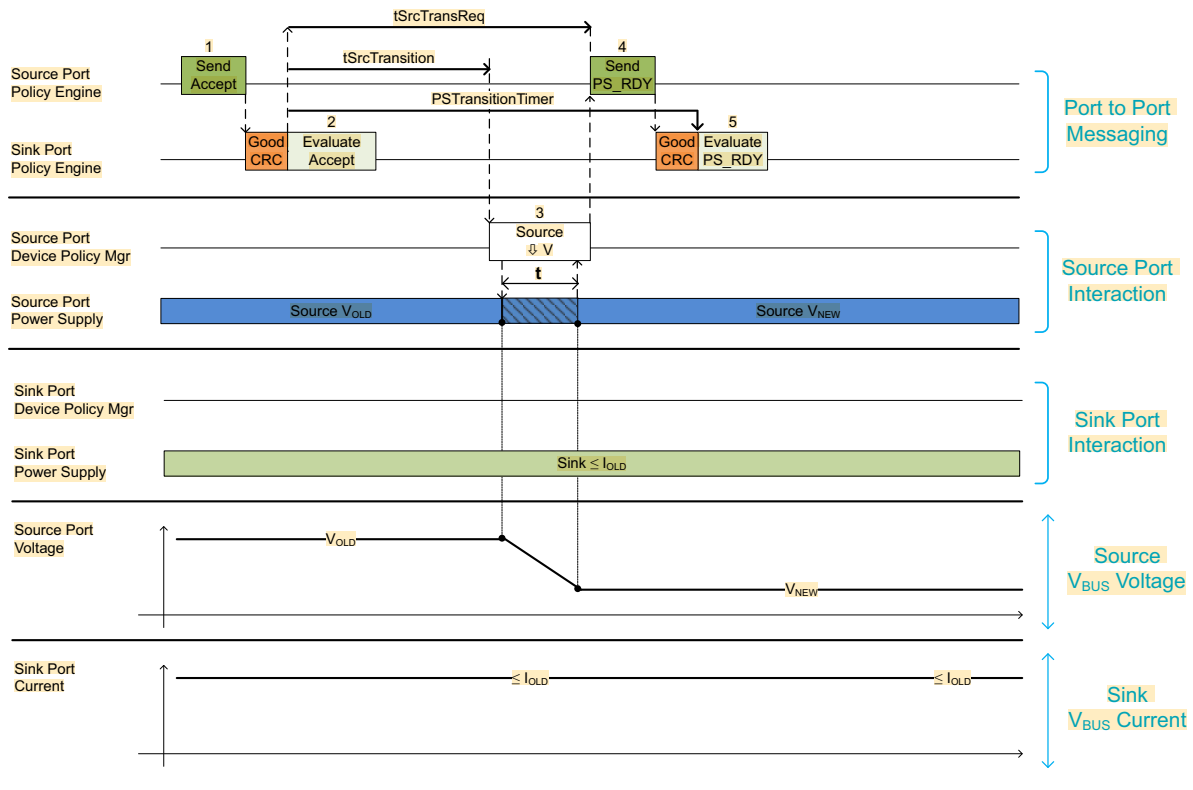


Table 7.5 Sequence Description for Decreasing the Voltage

Step	Source Port	Sink Port
1	Policy Engine sends the Accept Message to the Sink.	Policy Engine receives the Accept Message.
2	Protocol Layer receives the GoodCRC Message from the Sink. The Policy Engine tells the Device Policy Manager to instruct the power supply to modify its output power.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then starts the PSTransitionTimer and evaluates the Accept Message.
3	tSrcTransition after the GoodCRC Message was received, the power supply starts to change its output power capability. The power supply Shall be ready to operate at the new power level within tSrcReady (t1). The power supply informs the Device Policy Manager that it is ready to operate at the new power level. The power supply status is passed to the Policy Engine.	
4	The Policy Engine sends the PS_RDY Message to the Sink within tSrcTransReq of the end of the GoodCRC Message following the Accept Message.	The Policy Engine receives the PS_RDY Message from the Source.
5	Protocol Layer receives the GoodCRC Message from the Sink.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then stops the PSTransitionTimer , evaluates the PS_RDY Message from the Source and tells the Device Policy Manager that the Source is operating at the new (A)PDO. If the PS_RDY Message is not received before PSTransitionTimer times out the Sink sends Hard Reset Signaling .

The interaction of the *System Policy*, *Device Policy*, and power supply that **Shall** be followed when changing from one (A)PDO to another while decreasing the voltage and current is shown in [Figure 7.28, "Transition Diagram for no change in Current or Voltage"](#). The sequence that **Shall** be followed is described in [Table 7.6, "Sequence Description for Decreasing the Voltage and the Current"](#). The timing parameters that **Shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Note: In [Figure 7.27, "Transition Diagram for Decreasing the Voltage and the Current"](#), the Sink has previously sent a *Request Message* to the Source .

Figure 7.27 Transition Diagram for Decreasing the Voltage and the Current

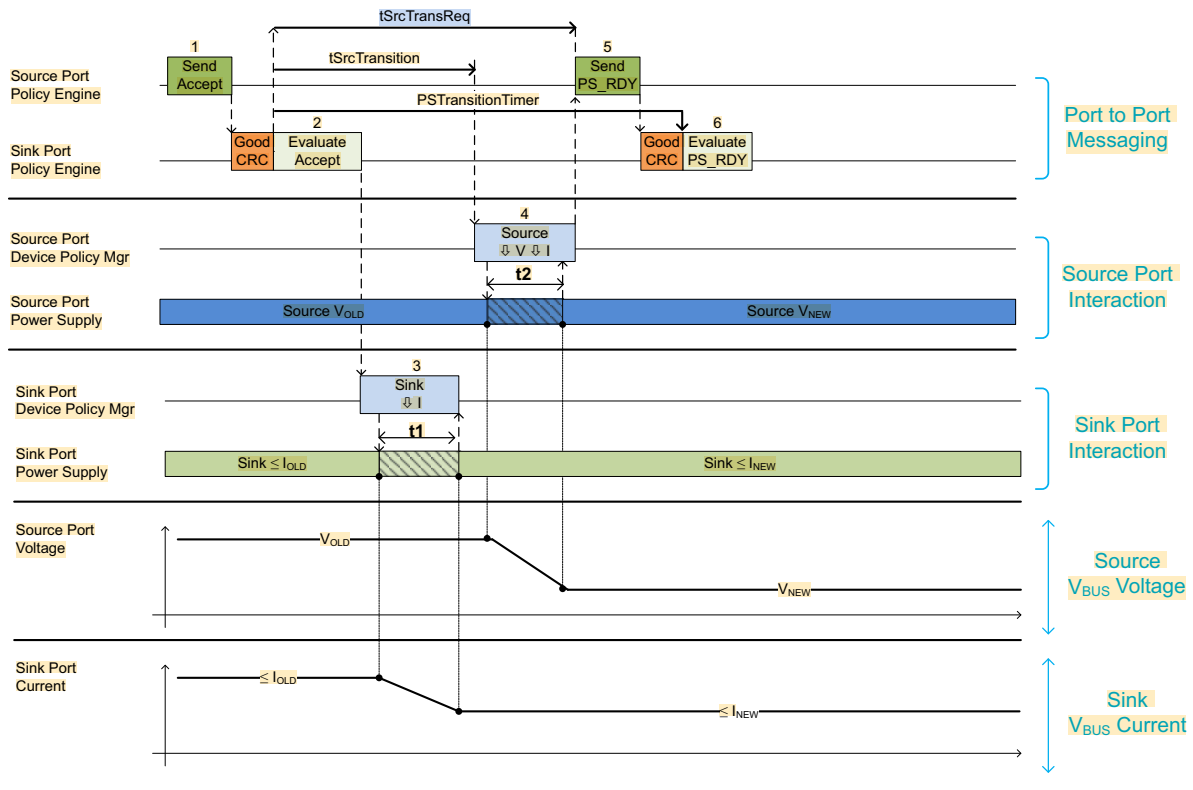


Table 7.6 Sequence Description for Decreasing the Voltage and the Current

Step	Source Port	Sink Port
1	Policy Engine sends the Accept Message to the Sink.	Policy Engine receives the Accept Message.
2	Protocol Layer receives the GoodCRC Message from the Sink. The Policy Engine tells the Device Policy Manager to instruct the power supply to modify its output power.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then starts the PSTransitionTimer and evaluates the Accept Message.
3		The Sink Shall be able to operate with lower current within tSnkNewPower (t1); t1 Shall complete before tSrcTransition . The Sink Shall Not violate transient load behavior defined in Section 7.2.6, "Transient Load Behavior" while transitioning to and operating at the new power level.
4	tSrcTransition after the GoodCRC Message was received, the power supply starts to change its output power capability. The power supply Shall be ready to operate at the new power level within tSrcReady (t2). The power supply informs the Device Policy Manager that it is ready to operate at the new power level. The power supply status is passed to the Policy Engine.	
5	The Policy Engine sends the PS_RDY Message to the Sink starting within tSrcTransReq of the end of the GoodCRC Message following the Accept Message.	The Policy Engine receives the PS_RDY Message from the Source.
6	Protocol Layer receives the GoodCRC Message from the Sink.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then stops the PSTransitionTimer , evaluates the PS_RDY Message from the Source and tells the Device Policy Manager that the Source is operating at the new (A)PDO. If the PS_RDY Message is not received before PSTransitionTimer times out the Sink sends Hard Reset Signaling .
7		The Sink Shall Not violate the transient load behavior defined in Section 7.2.6, "Transient Load Behavior" while transitioning to and operating at the new power level.



The interaction of the *System Policy*, *Device Policy*, and power supply that **Shall** be followed when changing from one (A)PDO to another while the *Sink* requests the same voltage and Current as it is currently operating at is shown in [Figure 7.28, "Transition Diagram for no change in Current or Voltage"](#). The sequence that **Shall** be followed is described in [Table 7.7, "Sequence Description for no change in Current or Voltage"](#). The timing parameters that **Shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Note: In [Figure 7.28, "Transition Diagram for no change in Current or Voltage"](#), the *Sink* has previously sent a *Request Message* to the *Source*.

Figure 7.28 Transition Diagram for no change in Current or Voltage

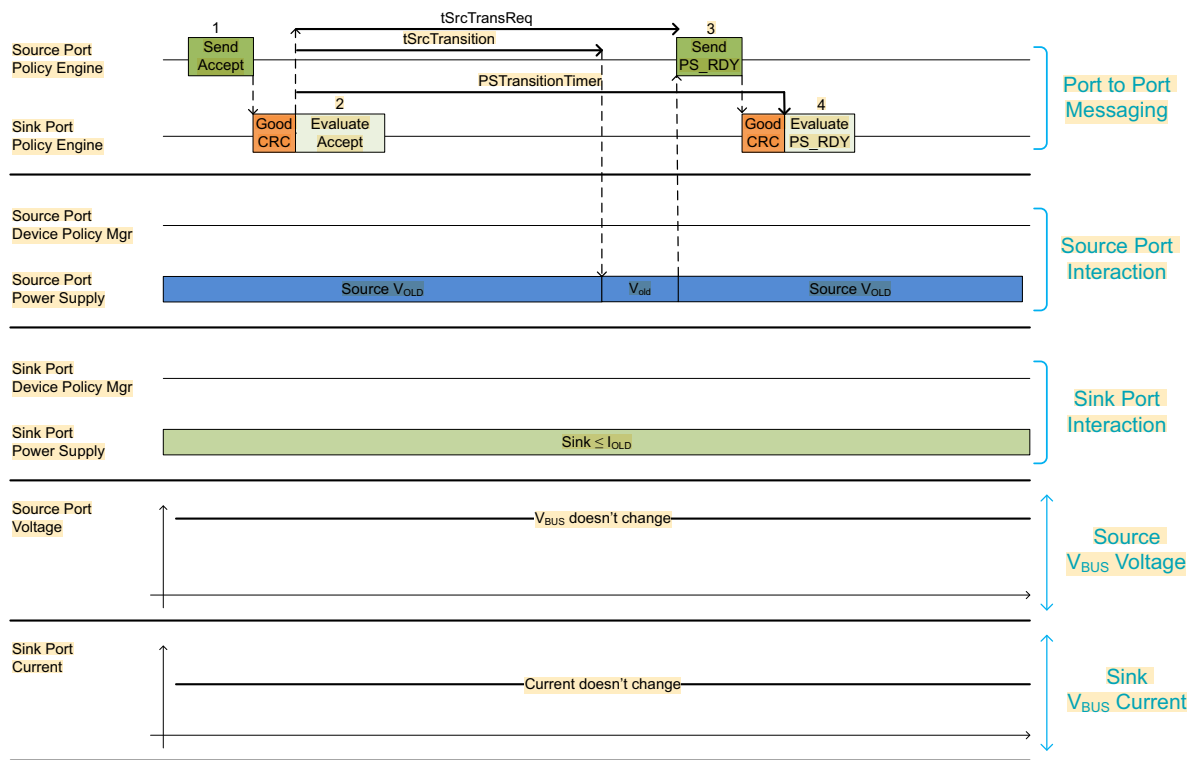


Table 7.7 Sequence Description for no change in Current or Voltage

Step	Source Port	Sink Port
1	Policy Engine sends the Accept Message to the Sink.	Policy Engine receives the Accept Message.
2	Protocol Layer receives the GoodCRC Message from the Sink.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then starts the PStTransitionTimer and evaluates the Accept Message.
3	The Policy Engine waits $t_{SrcTransition}$ then sends the PS_RDY Message to the Sink starting within $t_{SrcTransReq}$ of the end of the GoodCRC Message following the Accept Message.	Policy Engine receives the PS_RDY Message.
4	Policy Engine receives the GoodCRC Message from the Sink. Note: The decision that no power transition is required could be made either by the Device Policy Manager or the power supply depending on implementation.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine evaluates the PS_RDY Message.



7.3.1.2

Transitions within the same Fixed, Battery or Variable PDO or between Different (A)PDOs

7.3.1.2.1

Increasing the Current Only

The interaction of the *System Policy*, *Device Policy*, and power supply that **Shall** be followed when increasing the current without changing the voltage is shown in [Figure 7.29, "Transition Diagram for Increasing the Current"](#). The sequence that **Shall** be followed is described in [Table 7.8, "Sequence Description for Increasing the Current"](#). The timing parameters that **Shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Note: In [Figure 7.29, "Transition Diagram for Increasing the Current"](#), the Sink has previously sent a *Request Message* to the Source.

Figure 7.29 Transition Diagram for Increasing the Current

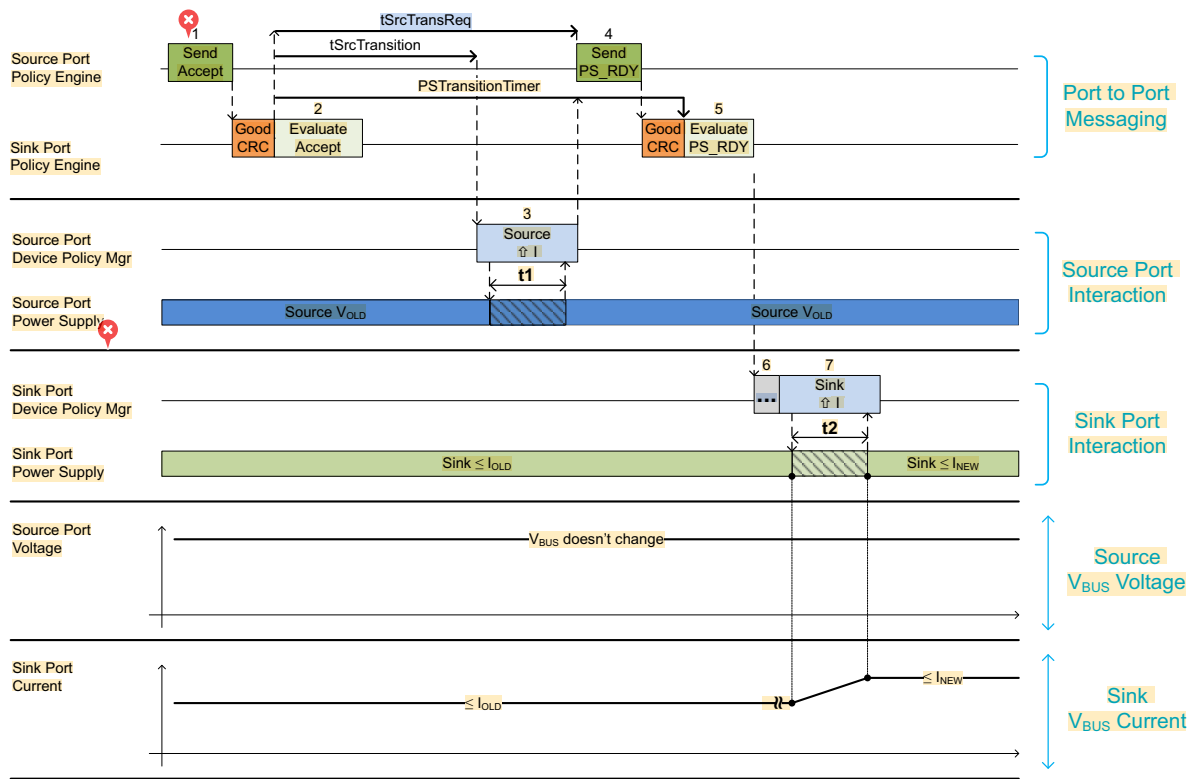


Table 7.8 Sequence Description for Increasing the Current

Step	Source Port	Sink Port
1	Policy Engine sends the Accept Message to the Sink.	Policy Engine receives the Accept Message.
2	Protocol Layer receives the GoodCRC Message from the Sink. The Policy Engine tells the Device Policy Manager to instruct the power supply to modify its output power.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then starts the PSTransitionTimer and evaluates the Accept Message.
3	tSrcTransition after the GoodCRC Message was received, the power supply starts to change its output power capability. The power supply Shall be ready to operate at the new power level within tSrcReady (t1). The power supply informs the Device Policy Manager that it is ready to operate at the new power level. The power supply status is passed to the Policy Engine.	
4	The Policy Engine sends the PS_RDY Message to the Sink starting within tSrcTransReq of the end of the GoodCRC Message following the Accept Message.	The Policy Engine receives the PS_RDY Message from the Source.
5	Protocol Layer receives the GoodCRC Message from the Sink.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then stops the PSTransitionTimer , evaluates the PS_RDY Message from the Source and tells the Device Policy Manager that the Source is operating at the new (A)PDO. If the PS_RDY Message is not received before PSTransitionTimer times out the Sink sends Hard Reset Signaling .
6		The Sink May begin operating at the new power level any time after evaluation of the PS_RDY Message. This time duration is indeterminate.
7		The Sink Shall Not violate the transient load behavior defined in Section 7.2.6, "Transient Load Behavior" while transitioning to and operating at the new power level. The time duration (t2) depends on the magnitude of the load change.

7.3.1.2.2 Decreasing the Current Only

The interaction of the *System Policy*, *Device Policy*, and power supply that **Shall** be followed when decreasing the current without changing the voltage is shown in [Figure 7.30, "Transition Diagram for Decreasing the Current"](#). The sequence that **Shall** be followed is described in [Table 7.9, "Sequence Description for Decreasing the Current"](#). The timing parameters that **Shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Note: In [Figure 7.30, "Transition Diagram for Decreasing the Current"](#), the Sink has previously sent a *Request Message* to the Source.

Figure 7.30 Transition Diagram for Decreasing the Current

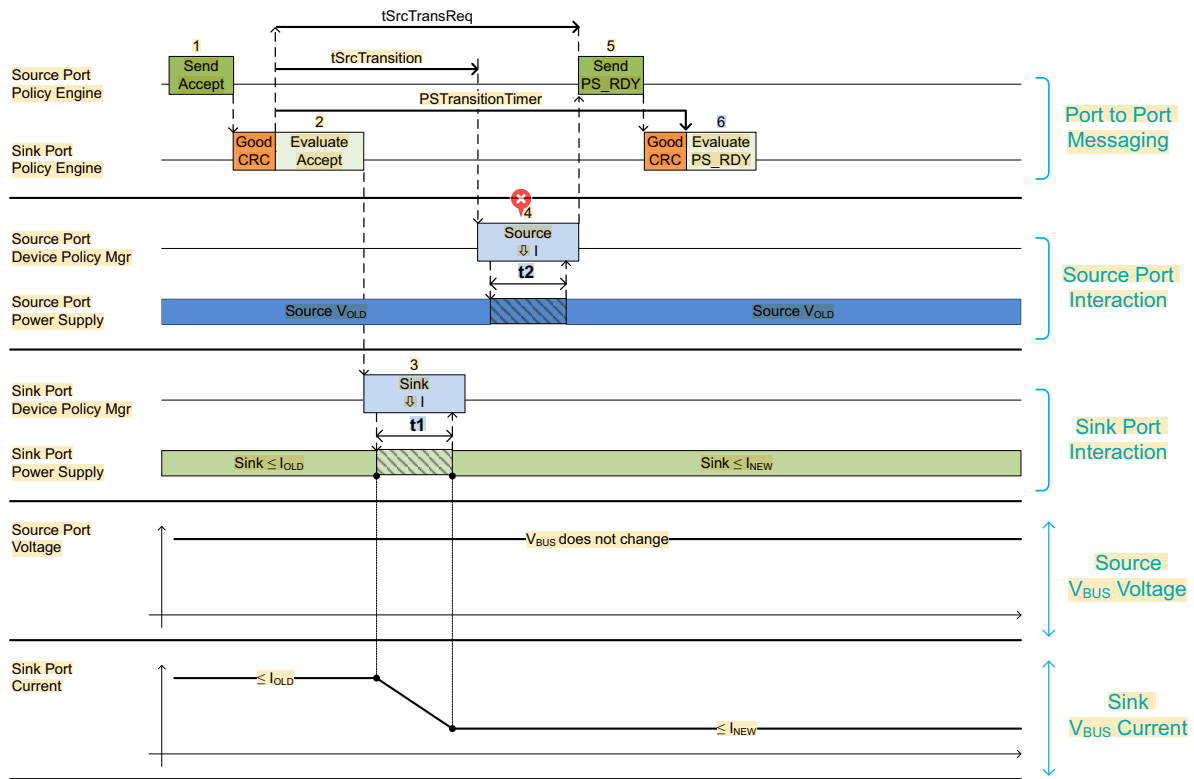


Table 7.9 Sequence Description for Decreasing the Current

Step	Source Port	Sink Port
1	Policy Engine sends the Accept Message to the Sink.	Policy Engine receives the Accept Message.
2	Protocol Layer receives the GoodCRC Message from the Sink. The Policy Engine tells the Device Policy Manager to instruct the power supply to modify its output power.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then starts the PSTransitionTimer and evaluates the Accept Message. Policy Engine tells the Device Policy Manager to instruct the power supply to reduce power consumption.
3		The Sink Shall Not violate the transient load behavior defined in Section 7.2.6, "Transient Load Behavior" while transitioning to and operating at the new power level. The Sink Shall be able to operate with lower current within tSnkNewPower (t1); t1 Shall complete before tSrcTransition .
4	tSrcTransition after the GoodCRC Message was received, the power supply starts to change its output power capability. The power supply Shall be ready to operate at the new power level within tSrcReady (t2). The power supply informs the Device Policy Manager that it is ready to operate at the new power level. The power supply status is passed to the Policy Engine.	
5	The Policy Engine sends the PS_RDY Message to the Sink starting within tSrcTransReq of the end of the GoodCRC Message following the Accept Message.	The Policy Engine receives the PS_RDY Message from the Source.
6	Protocol Layer receives the GoodCRC Message from the Sink.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then stops the PSTransitionTimer , evaluates the PS_RDY Message from the Source and tells the Device Policy Manager that the Source is operating at the new (A)PDO. If the PS_RDY Message is not received before PSTransitionTimer times out the Sink sends Hard Reset Signaling .

7.3.1.3 Changing Voltage or Current within the same PPS APDO

7.3.1.3.1 Increasing the Programmable Power Supply (PPS) Voltage

The interaction of the *System Policy*, *Device Policy*, and power supply that **Shall** be followed when increasing the voltage is shown in [Figure 7.31, "Transition Diagram for Increasing the Programmable Power Supply Voltage"](#). The sequence that **Shall** be followed is described in [Table 7.10, "Sequence Description for Increasing the Programmable Power Supply Voltage"](#). The timing parameters that **Shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Note: In [Figure 7.31, "Transition Diagram for Increasing the Programmable Power Supply Voltage"](#), the Sink has previously sent a *Request Message* to the Source.

Figure 7.31 Transition Diagram for Increasing the Programmable Power Supply Voltage

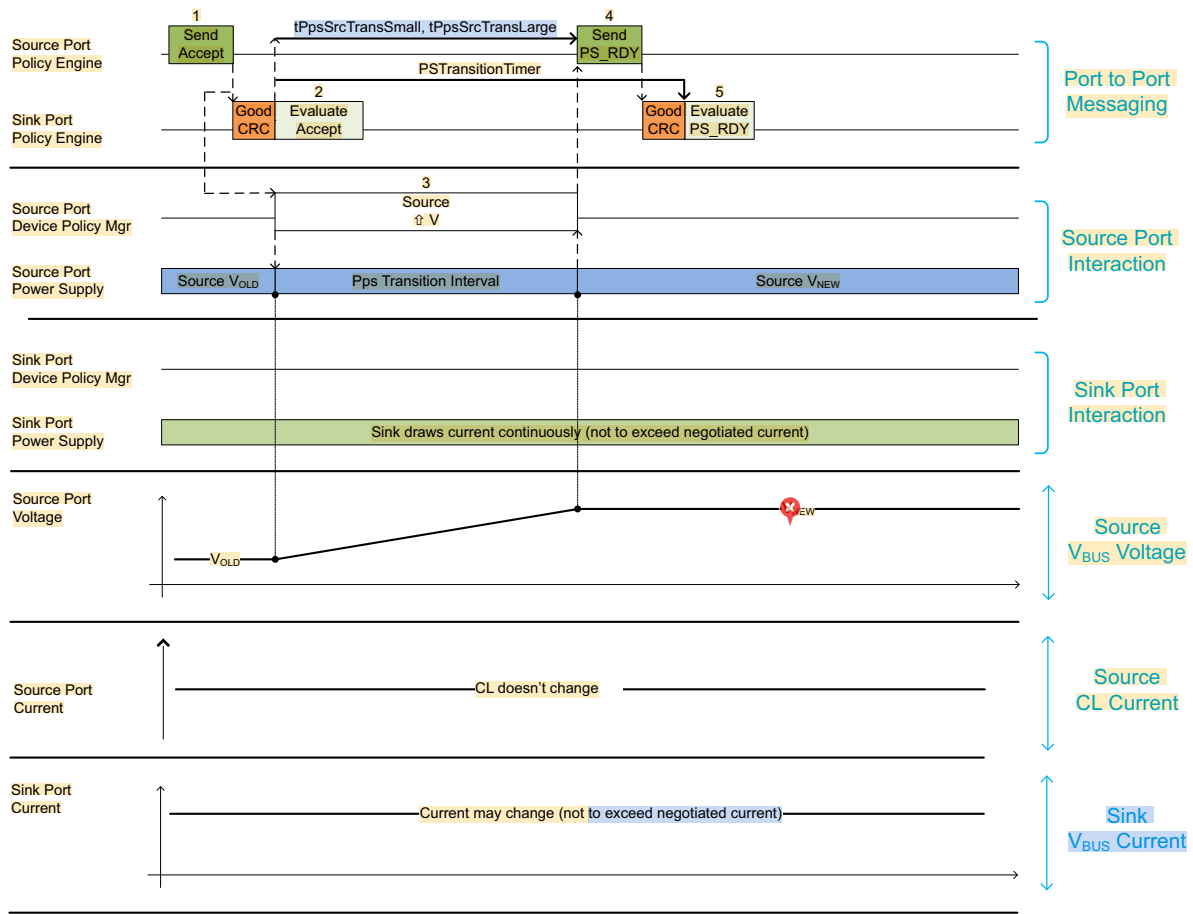


Table 7.10 Sequence Description for Increasing the Programmable Power Supply Voltage

Step	Source Port	Sink Port
1	Policy Engine sends the Accept Message to the Sink.	Policy Engine receives the Accept Message.
2	Protocol Layer receives the GoodCRC Message from the Sink. The Policy Engine tells the Device Policy Manager to instruct the power supply to increase its output voltage.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine. Policy Engine then starts the PSTransitionTimer and evaluates the Accept Message.
3	After sending the Accept Message, the Programmable Power Supply starts to increase its output voltage. The Programmable Power Supply new voltage set-point shall be reached by tPpsSrcTransLarge for steps larger than vPpsSmallStep or else by tPpsSrcTransSmall . The power supply informs the Device Policy Manager that it has reached the new set-point and whether V_{BUS} is at the corresponding new level, or if the supply is operating in CL mode. The power supply status is passed to the Policy Engine.	
4	The Policy Engine sends the PS_RDY Message to the Sink starting within tPpsSrcTransSmall or tPpsSrcTransLarge of the end of the GoodCRC Message following the Accept Message.	The Policy Engine receives the PS_RDY Message from the Source.
5	Protocol Layer receives the GoodCRC Message from the Sink.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then stops the PSTransitionTimer , evaluates the PS_RDY Message from the Source and tells the Device Policy Manager that the Source is operating at the new voltage set point (corresponding to vPpsNew). If the PS_RDY Message is not received before PSTransitionTimer times out the Sink sends Hard Reset Signaling .

7.3.1.3.2 Decreasing the Programmable Power Supply (PPS) Voltage

The interaction of the *System Policy*, *Device Policy*, and power supply that **Shall** be followed when decreasing the voltage is shown in [Figure 7.32, "Transition Diagram for Decreasing the Programmable Power Supply Voltage"](#). The sequence that **Shall** be followed is described in [Table 7.11, "Sequence Description for Decreasing the Programmable Power Supply Voltage"](#). The timing parameters that **Shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Note: In [Figure 7.32, "Transition Diagram for Decreasing the Programmable Power Supply Voltage"](#), the Sink has previously sent a *Request Message* to the Source.

Figure 7.32 Transition Diagram for Decreasing the Programmable Power Supply Voltage

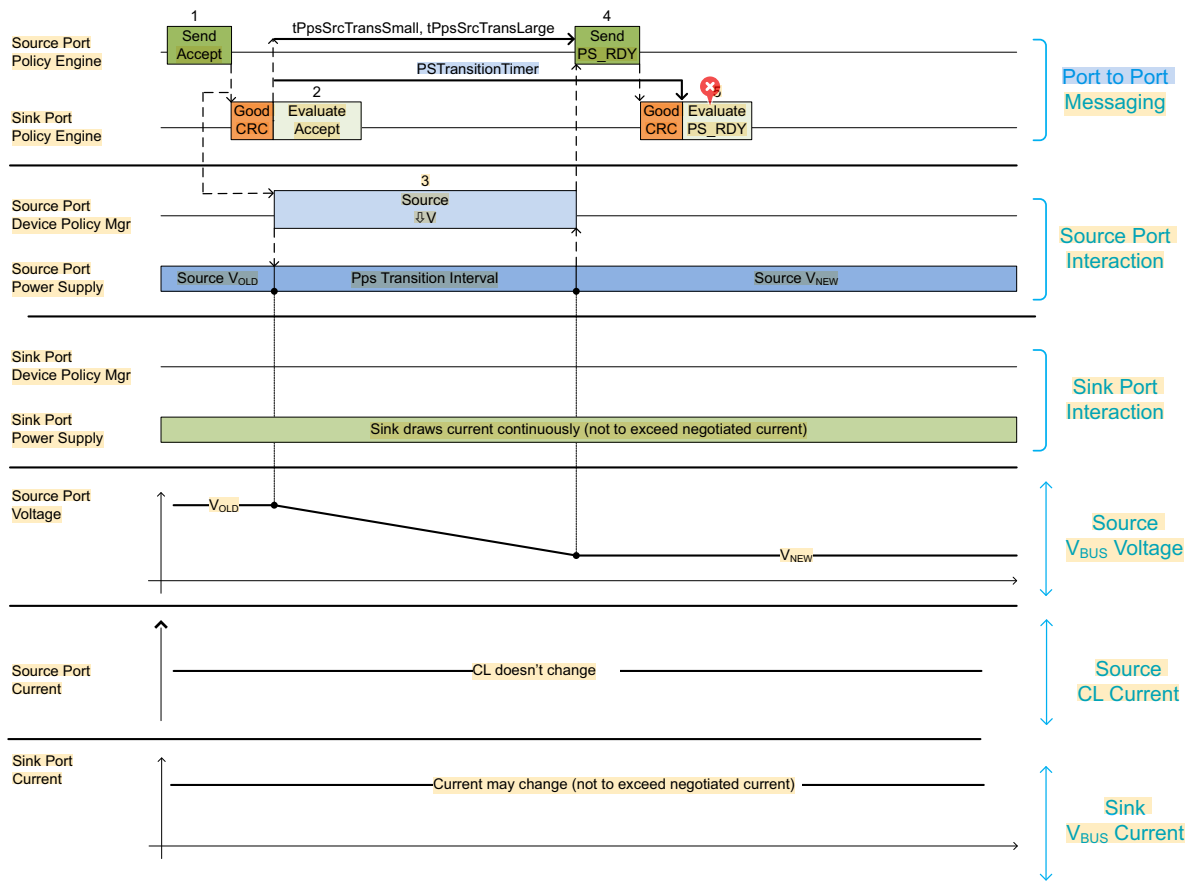


Table 7.11 Sequence Description for Decreasing the Programmable Power Supply Voltage

Step	Source Port	Sink Port
1	Policy Engine sends the Accept Message to the Sink.	Policy Engine receives the Accept Message.
2	Protocol Layer receives the GoodCRC Message from the Sink. The Policy Engine tells the Device Policy Manager to instruct the power supply to decrease its output voltage.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine. Policy Engine then starts the PSTransitionTimer and evaluates the Accept Message.
3	After sending the Accept Message, the Programmable Power Supply starts to decrease its output voltage. The Programmable Power Supply new voltage set-point (corresponding to vPpsNew) Shall be reached by tPpsSrcTransLarge for steps larger than vPpsSmallStep or else by tPpsSrcTransSmall . The power supply informs the Device Policy Manager that it has reached the new level. The power supply status is passed to the Policy Engine.	
4	The Policy Engine sends the PS_RDY Message to the Sink starting within tPpsSrcTransSmall or tPpsSrcTransLarge of the end of the GoodCRC Message following the Accept Message.	The Policy Engine receives the PS_RDY Message from the Source.
5	Protocol Layer receives the GoodCRC Message from the Sink.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then stops the PSTransitionTimer , evaluates the PS_RDY Message from the Source and tells the Device Policy Manager that the Source is operating at the new voltage set point (corresponding to vPpsNew). If the PS_RDY Message is not received before PSTransitionTimer times out the Sink sends Hard Reset Signaling .

7.3.1.3.3 Increasing the Programmable Power Supply (PPS) Current

The interaction of the *System Policy*, *Device Policy*, and power supply that **Shall** be followed when increasing the current limit in the same *APDO*, not exceeding the maximum for that *APDO* and without changing the requested voltage is shown in [Figure 7.33, "Transition Diagram for increasing the Current in PPS mode"](#). The sequence that **Shall** be followed is described in [Table 7.12, "Sequence Description for increasing the Current in PPS mode"](#). The timing parameters that **Shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Note: In [Figure 7.33, "Transition Diagram for increasing the Current in PPS mode"](#), the *Sink* has previously sent a **Request** Message to the *Source*.

The *Sink* **May** draw current equal to the increasing *Current Limit* of the *Source* before it has received the **PS_RDY** Message for the new *Request*.

Figure 7.33 Transition Diagram for increasing the Current in PPS mode

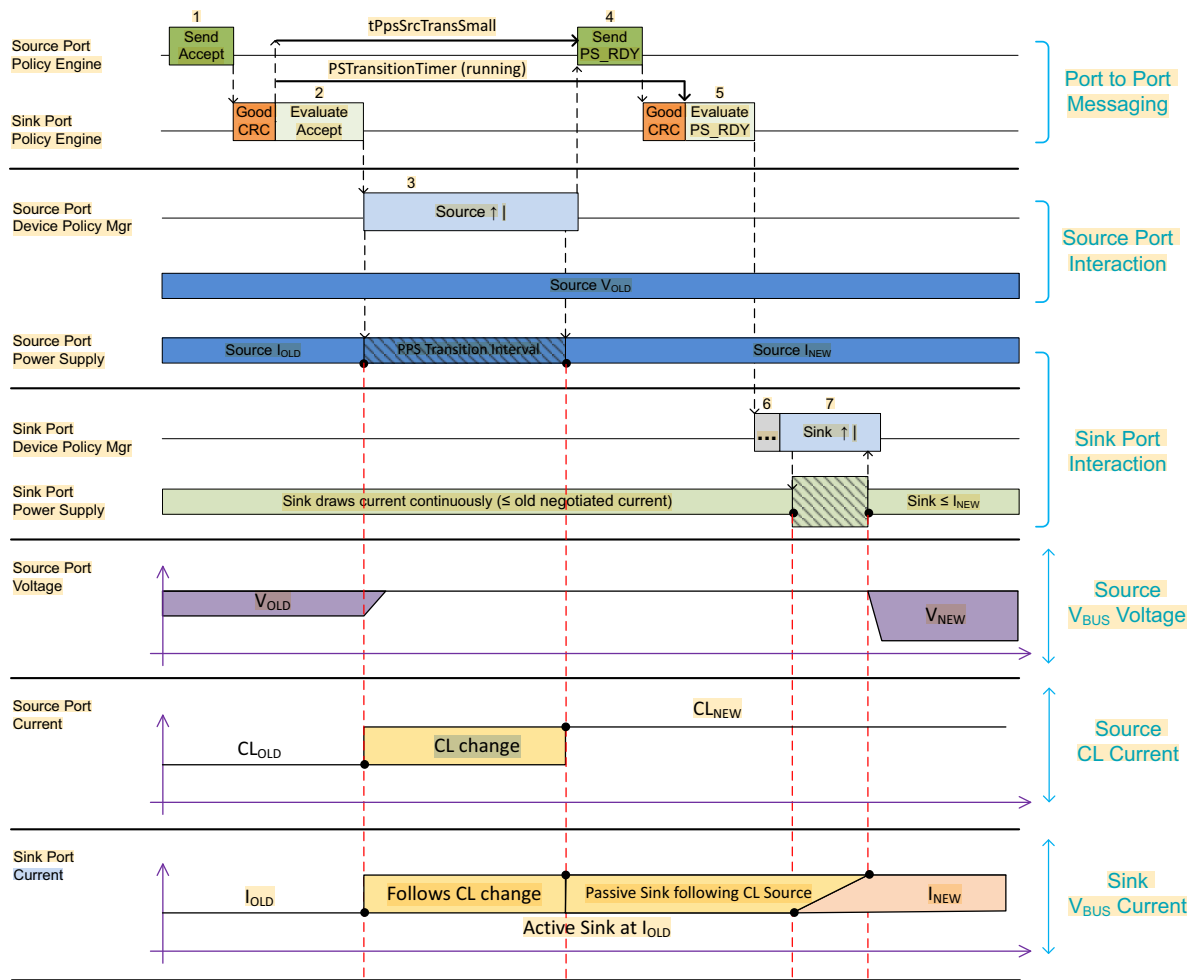


Table 7.12 Sequence Description for increasing the Current in PPS mode

Step	Source Port	Sink Port
1	Policy Engine sends the Accept Message to the Sink.	Policy Engine receives the Accept Message.
2	Protocol Layer receives the GoodCRC Message from the Sink. The Policy Engine tells the Device Policy Manager to instruct the power supply to increase its set-point for the current limit.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then starts the PSTransitionTimer and evaluates the Accept Message.
3	The Power Supply increases its Current Limit set-point to the new requested value.	The Sink draws current according to the increased Current Limit of the Source.
4	The Policy Engine waits tPpsSrcTransSmall then sends the PS_RDY Message to the Sink starting within tPpsCLProgramSettle of the end of the GoodCRC Message following the Accept Message.	Policy Engine receives the PS_RDY Message.
5	Policy Engine receives the GoodCRC Message from the Sink.	Protocol Layer sends the GoodCRC Message to the Source.
6		Policy Engine then stops the PSTransitionTimer , evaluates the PS_RDY Message and tells the Device Policy Manager it can increase the current up to the requested value without the Source going into CL mode. If the PS_RDY Message is not received before PSTransitionTimer times out the Sink sends Hard Reset Signaling .
7		The Sink increases its current.

7.3.1.3.4 Decreasing the Programmable Power Supply (PPS) Current

The interaction of the *System Policy*, *Device Policy*, and power supply that **Shall** be followed when decreasing the current limit in the same *APDO*, not exceeding the minimum for that *APDO* and without changing the requested voltage is shown in [Figure 7.34, "Transition Diagram for decreasing the Current in PPS mode"](#). The sequence that **Shall** be followed is described in [Table 7.13, "Sequence Description for decreasing the Current in PPS mode"](#). The timing parameters that **Shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Note: In [Figure 7.34, "Transition Diagram for decreasing the Current in PPS mode"](#), the *Sink* has previously sent a *Request Message* to the *Source*.

Figure 7.34 Transition Diagram for decreasing the Current in PPS mode

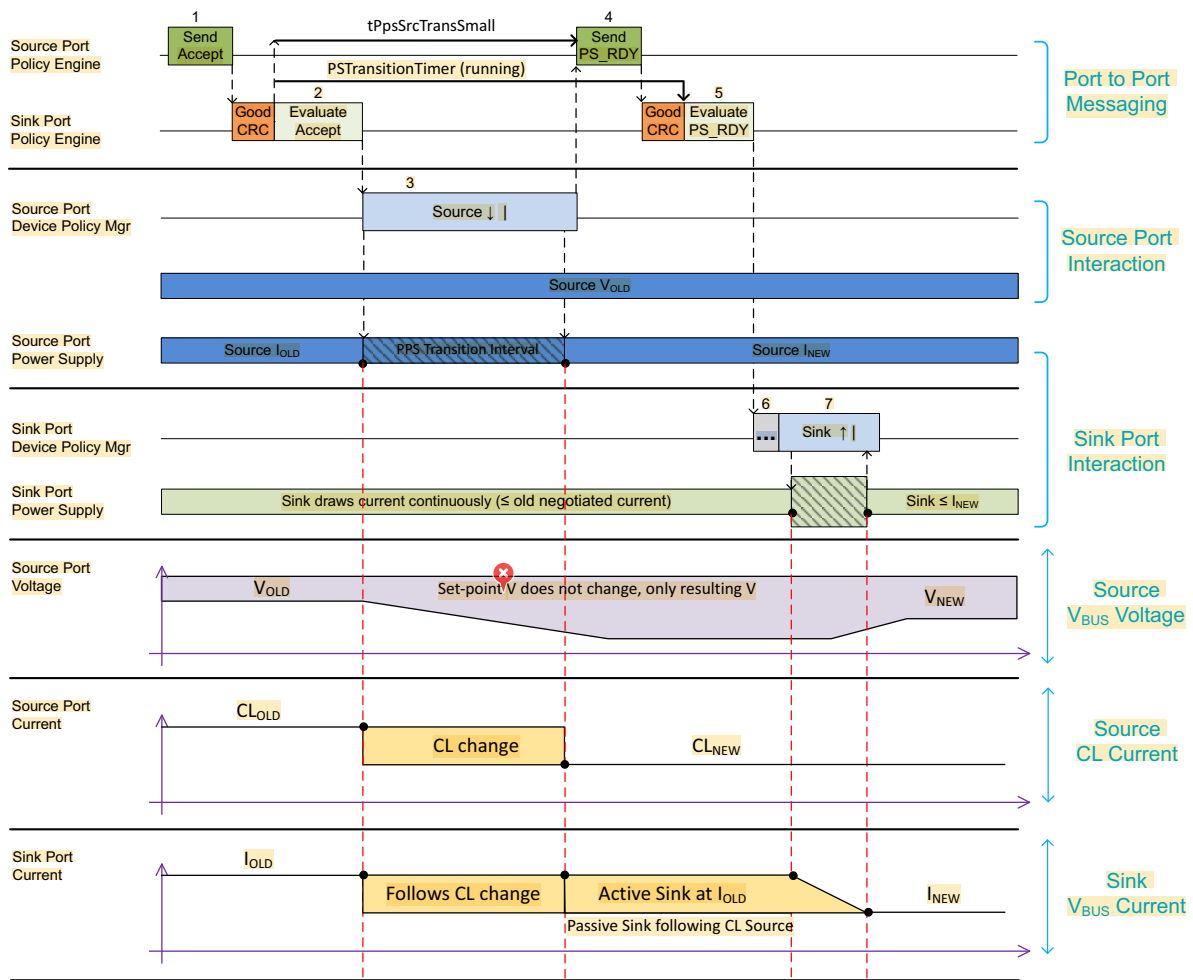


Table 7.13 Sequence Description for decreasing the Current in PPS mode

Step	Source Port	Sink Port
1	Policy Engine sends the Accept Message to the Sink.	Policy Engine receives the Accept Message.
2	Protocol Layer receives the GoodCRC Message from the Sink. The Policy Engine tells the Device Policy Manager to instruct the power supply to decrease its set-point for the current limit.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then evaluates the Accept Message and instructs the Sink to reduce its current to below the new Negotiated current level and starts the PSTransitionTimer .
3	The Power Supply decreases its Current Limit set-point to the new Negotiated value.	The Sink reduces its current to less than the new Negotiated current to prevent the Source from going into Current Limit .
4	The Policy Engine sends the PS_RDY Message to the Sink starting within tPpsSrcTransSmall of the end of the GoodCRC Message following the Accept Message.	
5	Policy Engine receives the GoodCRC Message from the Sink.	Policy Engine receives the PS_RDY Message.
6		Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then stops the PSTransitionTimer and evaluates the PS_RDY Message. If the PS_RDY Message is not received before PSTransitionTimer times out the Sink sends Hard Reset Signaling .
7		The Sink is allowed to draw I_{NEW} but must be aware the voltage on V_{BUS} can drop doing so.

The interaction of the *System Policy*, *Device Policy*, and power supply that **Shall** be followed when the *Sink* requests the same voltage and current levels as the present *Negotiated* levels for voltage and current is shown in [Figure 7.35, "Transition Diagram for no change in Current or Voltage in PPS mode"](#). The sequence that **Shall** be followed is described in [Table 7.14, "Sequence Description for no change in Current or Voltage in PPS mode"](#). The timing parameters that **Shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Note: In [Figure 7.35, "Transition Diagram for no change in Current or Voltage in PPS mode"](#), the *Sink* has previously sent a *Request Message* to the *Source*.

Figure 7.35 Transition Diagram for no change in Current or Voltage in PPS mode

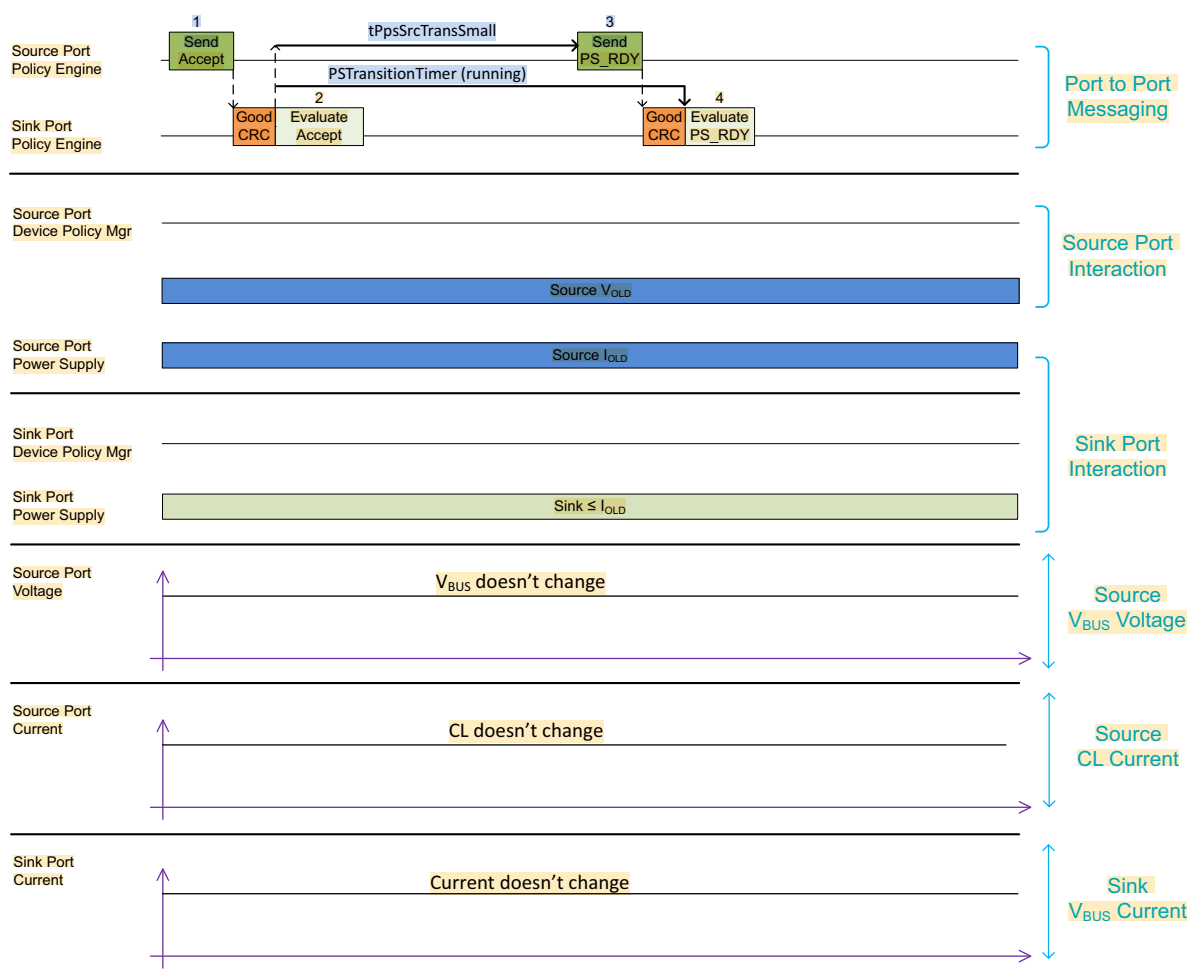


Table 7.14 Sequence Description for no change in Current or Voltage in PPS mode

Step	Source Port	Sink Port
1	Policy Engine sends the Accept Message to the Sink.	Policy Engine receives the Accept Message.
2	Protocol Layer receives the GoodCRC Message from the Sink.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then evaluates the Accept Message and starts the PSTransitionTimer .
3	The Policy Engine then sends the PS_RDY Message to the Sink starting within tPpsSrcTransSmall of the end of the GoodCRC Message following the Accept Message.	Policy Engine receives the PS_RDY Message.
4	Policy Engine receives the GoodCRC Message from the Sink. Note: The decision that no power transition is required could be made either by the Device Policy Manager or the power supply depending on implementation.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then stops the PSTransitionTimer and evaluates the PS_RDY Message from the Source. The Sink is already operating at the new power level, so no further action is required. If the PS_RDY Message is not received before PSTransitionTimer times out the Sink sends Hard Reset Signaling .

The interaction of the *System Policy*, *Device Policy*, and power supply that **shall** be followed when increasing the voltage is shown in [Figure 7.36, "Transition Diagram for Increasing the Adjustable Power Supply Voltage"](#). The sequence that **shall** be followed is described in [Table 7.15, "Sequence Description for Increasing the Adjustable Voltage Supply Voltage"](#). The timing parameters that **shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Note: In [Figure 7.36, "Transition Diagram for Increasing the Adjustable Power Supply Voltage"](#), the Sink has previously sent a *Request Message* to the Source.

Figure 7.36 Transition Diagram for Increasing the Adjustable Power Supply Voltage

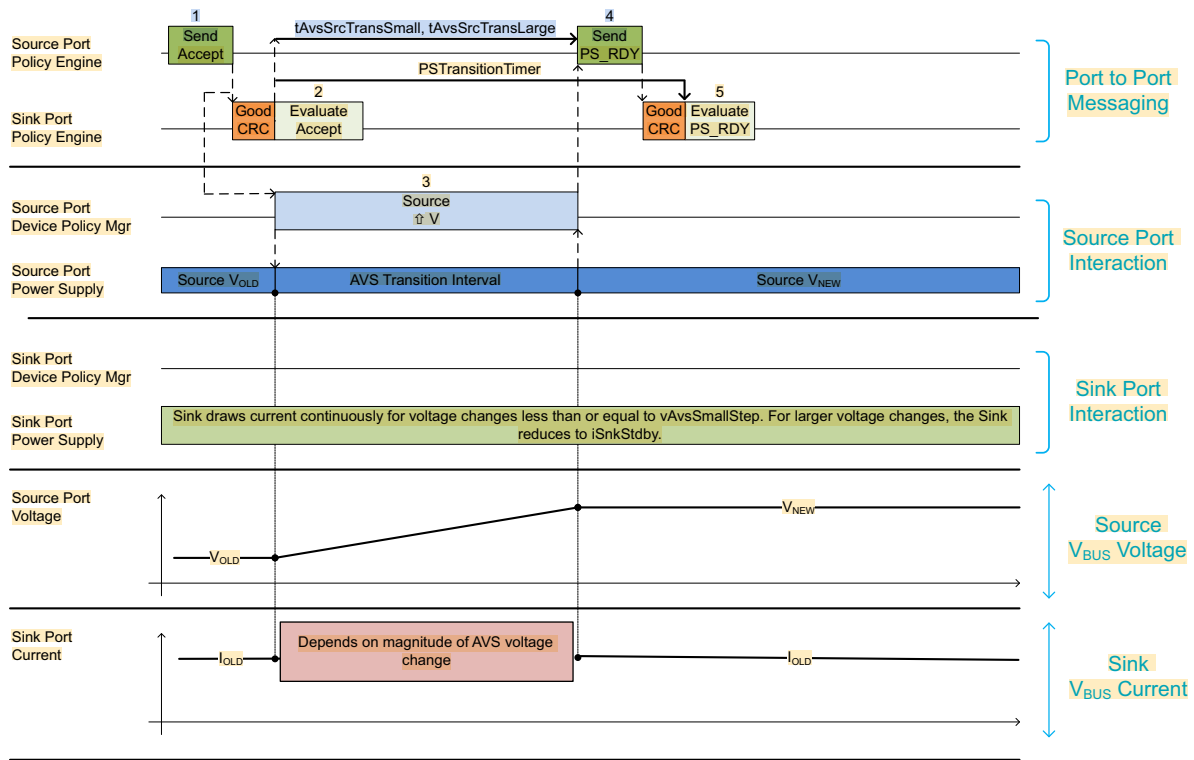


Table 7.15 Sequence Description for Increasing the Adjustable Voltage Supply Voltage

Step	Source Port	Sink Port
1	Policy Engine sends the Accept Message to the Sink.	Policy Engine receives the Accept Message.
2	Protocol Layer receives the GoodCRC Message from the Sink. The Policy Engine tells the Device Policy Manager to instruct the power supply to increase its output voltage.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine. Policy Engine then starts the PSTransitionTimer and evaluates the Accept Message. If the voltage increase is larger than vAvsSmallStep, the Sink Shall reduce its current draw to iSnkStdby within tSnkStdby. The reduction to iSnkStdby is not required if the voltage increase is less than or equal to vAvsSmallStep.
3	After sending the Accept Message, the AVS starts to increase its output voltage. The AVS new voltage set-point Shall be reached by tAvsSrcTransLarge for steps larger than vAvsSmallStep or else by tAvsSrcTransSmall . The power supply informs the Device Policy Manager that it has reached the new level. The power supply status is passed to the Policy Engine.	
4	The Policy Engine sends  PS_RDY Message to the Sink starting within tAvsSrcTransSmall or tAvsSrcTransLarge of the end of the GoodCRC Message following the Accept Message.	The Policy Engine receives the PS_RDY Message from the Source.
5	Protocol Layer receives the GoodCRC Message from the Sink.	Protocol Layer sends the GoodCRC Message to the Source.  Policy Engine then stops the PSTransitionTimer, evaluates the PS_RDY Message from the Source and tells the Device Policy Manager that the Source is operating at the new voltage set point. The Sink May begin operating at the new power level any time after evaluation of the PS_RDY Message. If the PS_RDY Message is not received before PSTransitionTimer times out the Sink sends Hard Reset Signaling.

7.3.1.4.2 Decreasing the Adjustable Voltage Supply (AVS) Voltage

The interaction of the *System Policy*, *Device Policy*, and power supply that **Shall** be followed when decreasing the voltage is shown in [Figure 7.37, "Transition Diagram for Decreasing the Adjustable Voltage Supply Voltage"](#). The sequence that **Shall** be followed is described in [Table 7.16, "Sequence Description for Decreasing the Adjustable Voltage Supply Voltage"](#). The timing parameters that **Shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Note: In [Figure 7.37, "Transition Diagram for Decreasing the Adjustable Voltage Supply Voltage"](#), the Sink has previously sent a *Request Message* to the Source.

Figure 7.37 Transition Diagram for Decreasing the Adjustable Voltage Supply Voltage

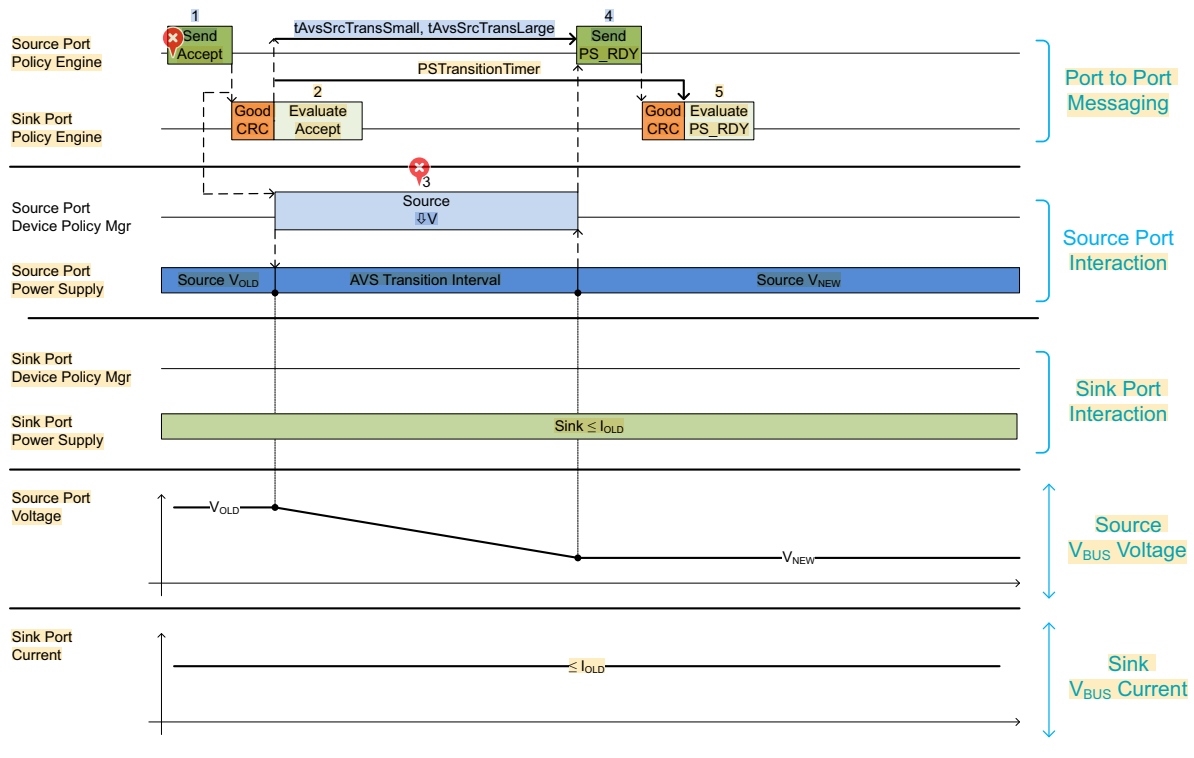


Table 7.16 Sequence Description for Decreasing the Adjustable Voltage Supply Voltage

Step	Source Port	Sink Port
1	Policy Engine sends the Accept Message to the Sink.	Policy Engine receives the Accept Message.
2	Protocol Layer receives the GoodCRC Message from the Sink. The Policy Engine tells the Device Policy Manager to instruct the power supply to decrease its output voltage.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then starts the PSTransitionTimer and evaluates the Accept Message.
3	After sending the Accept Message, the AVS starts to decrease its output voltage. The AVS new voltage set-point Shall be reached by tAvsSrcTransLarge for steps larger than vAvsSmallStep or else by tAvsSrcTransSmall . The power supply informs the Device Policy Manager that it has reached the new level. The power supply status is passed to the Policy Engine.	
4	The Policy Engine sends the PS_RDY Message to the Sink starting within tAvsSrcTransSmall or tAvsSrcTransLarge of the end of the GoodCRC Message following the Accept Message.	The Policy Engine receives the PS_RDY Message from the Source.
5	Protocol Layer receives the GoodCRC Message from the Sink.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then stops the PSTransitionTimer , evaluates the PS_RDY Message from the Source and tells the Device Policy Manager that the Source is operating at the new voltage set point (corresponding to vAvsNew). If the PS_RDY Message is not received before PSTransitionTimer times out the Sink sends Hard Reset Signaling .

7.3.1.4.3 Same Request Adjustable Voltage Supply (AVS) Voltage

The interaction of the *System Policy*, *Device Policy*, and power supply that **shall** be followed when the *Sink* requests the same voltage and current levels as the present *Negotiated* levels for voltage and current as shown in [Figure 7.38, "Transition Diagram for no change in Current or Voltage in AVS mode"](#). The sequence that **shall** be followed is described in [Table 7.17, "Sequence Description for no change in Current or Voltage in AVS mode"](#). The timing parameters that **shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Note: In [Figure 7.38, "Transition Diagram for no change in Current or Voltage in AVS mode"](#), the *Sink* has previously sent a *Request Message* to the *Source*.

Figure 7.38 Transition Diagram for no change in Current or Voltage in AVS mode

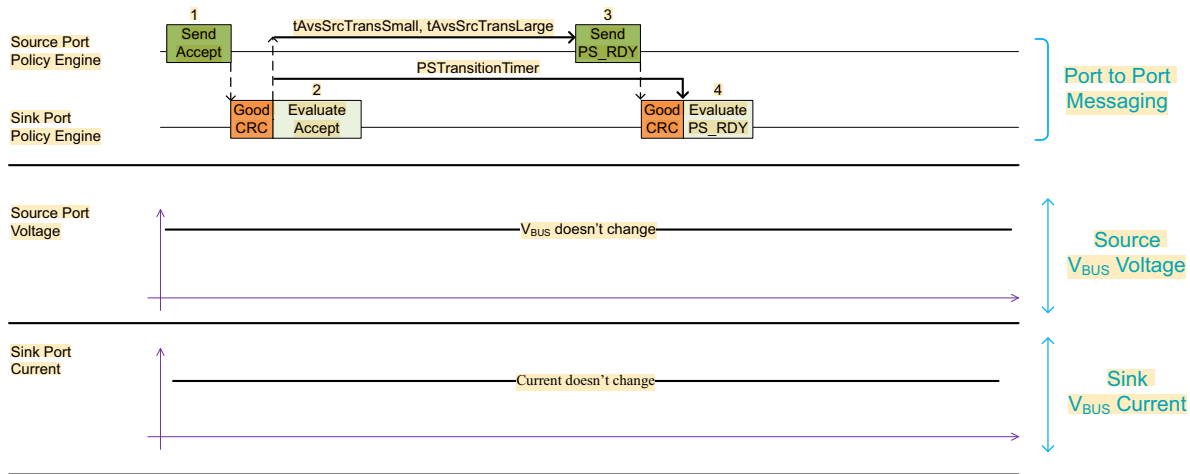


Table 7.17 Sequence Description for no change in Current or Voltage in AVS mode

Step	Source Port	Sink Port
1	Policy Engine sends the Accept Message to the Sink.	Policy Engine receives the Accept Message.
2	Protocol Layer receives the GoodCRC Message from the Sink.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then starts the PSTransitionTimer and evaluates the Accept Message.
3	The Policy Engine sends the PS_RDY Message to the Sink starting within $t_{AvsSrcTransSmall}$ of the end of the GoodCRC Message following the Accept Message.	The Policy Engine receives the PS_RDY Message from the Source.
4	Protocol Layer receives the GoodCRC Message from the Sink. Note: The decision that no power transition is required could be made either by the <i>Device Policy Manager</i> or the power supply depending on implementation.	Protocol Layer sends the GoodCRC Message to the Source. Policy Engine then stops the PSTransitionTimer , evaluates the PS_RDY Message from the Source. The Sink is already operating at the new power level, so no further action is required. If the PS_RDY Message is not received before PSTransitionTimer times out the Sink sends Hard Reset Signaling .

The interaction of the *System Policy*, *Device Policy*, and power supply that **shall** be followed during a *Sink* requested *Power Role Swap* is shown in [Figure 7.39, "Transition Diagram for a Sink Requested Power Role Swap"](#). The sequence that **shall** be followed is described in [Table 7.18, "Sequence Description for a Sink Requested Power Role Swap"](#). The timing parameters that **shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Note: In [Figure 7.39, "Transition Diagram for a Sink Requested Power Role Swap"](#), the *Sink* has previously sent a **PR_Swap** Message to the *Source*.

Figure 7.39 Transition Diagram for a Sink Requested Power Role Swap

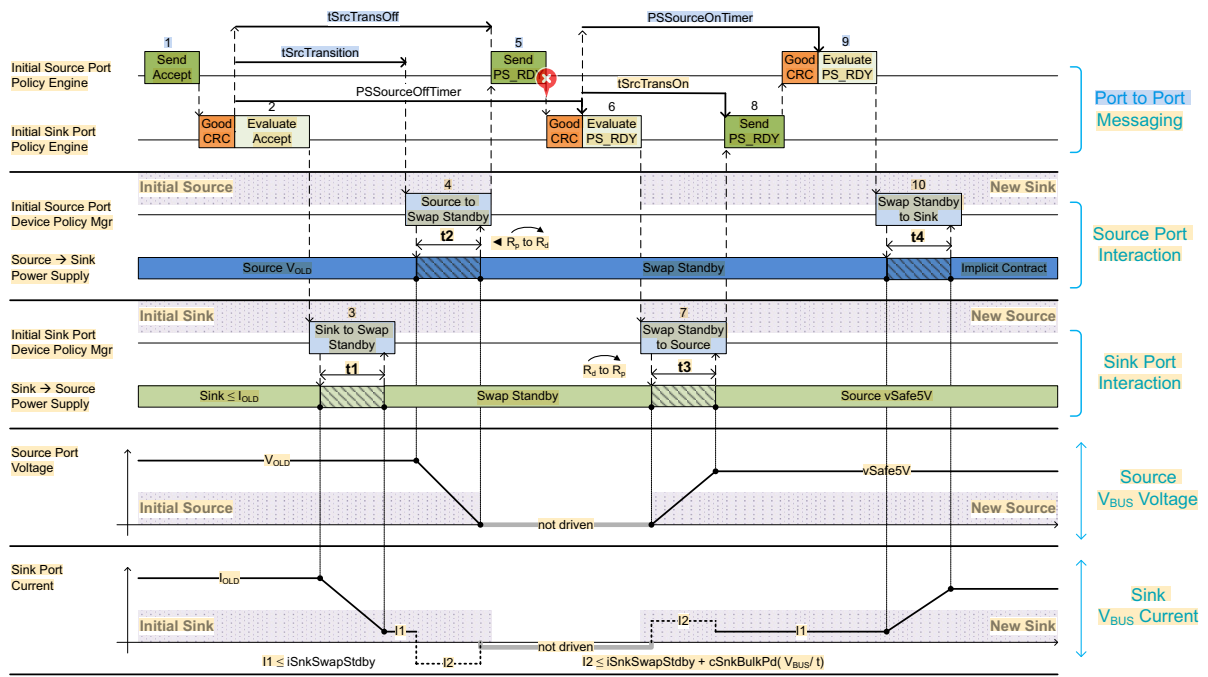


Table 7.18 Sequence Description for a Sink Requested Power Role Swap

Step	Initial Source Port → New Sink Port	Initial Sink Port → New Source Port
1	Policy Engine sends the Accept Message to the Initial Sink.	Policy Engine receives the Accept .
2	Protocol Layer receives the GoodCRC Message from the Sink. The Policy Engine tells the Device Policy Manager to instruct the power supply to modify its output power.	Protocol Layer sends the GoodCRC Message to the Initial Source. Policy Engine then starts the PSSourceOffTimer and evaluates the Accept Message.
3		Policy Engine tells the Device Policy Manager to instruct the power supply to transition to Swap Standby within tSnkStdby (t1); t1 shall complete before tSrcTransition min. When in Sink Standby the Initial Sink shall not draw more than iSnkSwapStdby (I1). The Sink shall not violate transient load behavior defined in Section 7.2.6, "Transient Load Behavior" while transitioning to and operating at the new power level.
4	tSrcTransition after the GoodCRC Message was received, the power supply starts to change its output power capability to Swap Standby (see Section 7.1.10, "Swap Standby for Sources"). The power supply shall complete the transition to Swap Standby within tSrcSwapStdby (t2). The power supply informs the Device Policy Manager that it is ready to operate as the New Sink. The CC termination is changed from R_p to R_d (see [USB Type-C 2.4]). The power supply status is passed to the Policy Engine.	
5	The power supply is ready, and the Policy Engine sends the PS_RDY Message to the device that will become the New Source, starting within tSrcTransOff of the end of the GoodCRC Message following the Accept Message.	
6	Protocol Layer receives the GoodCRC Message from the device that will become the New Source. Policy Engine starts the PSSourceOnTimer . Upon sending the PS_RDY Message and receiving the GoodCRC Message the Initial Source is ready to be the New Sink.	The Protocol Layer sends the GoodCRC Message to the New Sink. Policy Engine stops the PSSourceOffTimer and tells the Device Policy Manager to instruct the power supply to operate as the New Source. If the PS_RDY Message is not received before PSTransitionTimer times out the Sink sends Hard Reset Signaling .
7		The CC termination is changed from R_d to R_p (see [USB Type-C 2.4]). The power supply as the New Source transitions from Swap Standby to sourcing default vSafe5V within tNewSrc (t3). The power supply informs the Device Policy Manager that it is operating as the New Source.
8	Policy Engine receives the PS_RDY Message from the Source.	Device Policy Manager informs the Policy Engine the power supply is ready, and the Policy Engine sends the PS_RDY Message to the New Sink, starting within tSrcTransOn of the end of the GoodCRC Message following the Accept Message.

Table 7.18 Sequence Description for a Sink Requested Power Role Swap (Continued)

Step	Initial Source Port → New Sink Port	Initial Sink Port → New Source Port
9	<i>Protocol Layer</i> sends the GoodCRC Message to the <i>New Source</i> and then stops the PSSourceOnTimer. <i>Policy Engine</i> evaluates the PS_RDY Message from the <i>New Source</i> and tells the <i>Device Policy Manager</i> to instruct the power supply to draw current as the <i>New Sink</i>.	<i>Protocol Layer</i> receives the GoodCRC Message from the <i>New Sink</i>.
10	The power supply as the <i>New Sink</i> transitions from <i>Swap Standby</i> and begins to drawing the current allowed by the <i>Implicit Contract</i>. The power supply informs the <i>Device Policy Manager</i> that it is operating as the <i>New Sink</i>. At this point subsequent <i>Negotiations</i> between the <i>New Source</i> and the <i>New Sink</i> May proceed as normal. The <i>Sink</i> Shall Not violate the transient load behavior defined in Section 7.2.6, "Transient Load Behavior" while transitioning to and operating at the new power level. The time duration (t4) depends on the magnitude of the load change (iLoadStepRate).	

The interaction of the *System Policy*, *Device Policy*, and power supply that **shall** be followed during a *Source requested Power Role Swap* is shown in [Figure 7.40, "Transition Diagram for a Source Requested Power Role Swap"](#). The sequence that **shall** be followed is described in [Table 7.19, "Sequence Description for a Source Requested Power Role Swap"](#). The timing parameters that **shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Note: In [Figure 7.40, "Transition Diagram for a Source Requested Power Role Swap"](#), the *Source* has previously sent a **PR_Swap** Message to the *Sink*.

Figure 7.40 Transition Diagram for a Source Requested Power Role Swap

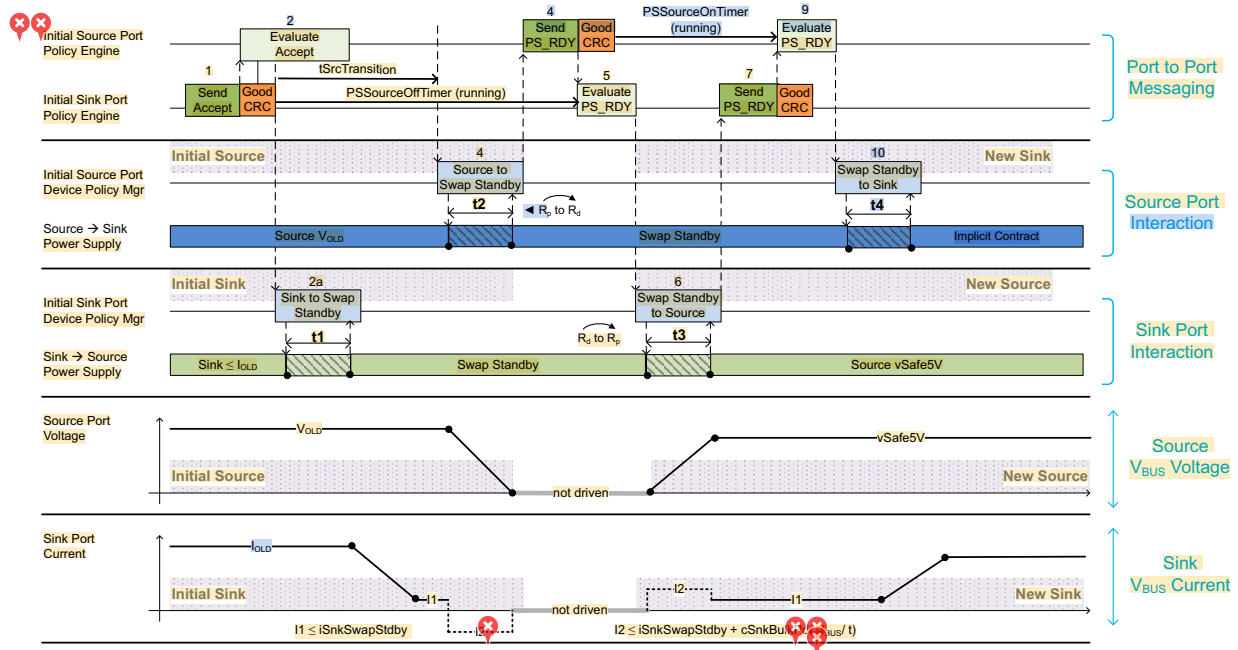


Table 7.19 Sequence Description for a Source Requested Power Role Swap

Step	Initial Source Port → New Sink Port	Initial Sink Port → New Source Port
1	Policy Engine receives the Accept Message.	Policy Engine sends the Accept Message to the Initial Source.
2	Protocol Layer sends the GoodCRC Message to the Sink. The Policy Engine tells the Device Policy Manager to instruct the power supply to modify its output power.	Protocol Layer receives the GoodCRC Message from the Initial Source. Policy Engine starts the PSSourceOffTimer .
2a		The Policy Engine tells the Device Policy Manager to instruct the power supply to transition to Swap Standby . The power supply Shall complete the transition to Swap Standby within tSnkStdby (t1); t1 Shall complete before tSrcTransition . The Sink Shall Not violate the transient load behavior defined in Section 7.2.6, "Transient Load Behavior" while transitioning to and operating at the new power level. When in Sink Standby the Initial Sink Shall Not draw more than iSnkSwapStdby (I1).
3	tSrcTransition after the GoodCRC Message was sent the power supply starts to change its output power capability to Swap Standby (see Section 7.1.10, "Swap Standby for Sources"). The power supply Shall complete the transition to Swap Standby within tSrcSwapStdby (t2). The power supply informs the Device Policy Manager that it is ready to operate as the New Sink . The CC termination is changed from R_p to R_d (see [USB Type-C 2.4]). The power supply status is passed to the Policy Engine.	
4	The Policy Engine sends the PS_RDY Message to the device that will become the New Source , starting within tSrcTransOff of the end of the GoodCRC Message following the Accept Message.	Policy Engine receives the PS_RDY .
5	Protocol Layer receives the GoodCRC Message from the soon to be New Source . Policy Engine starts the PSSourceOnTimer . At this point the Initial Source is ready to be the New Sink .	Protocol Layer sends the GoodCRC Message to the New Sink. Policy Engine then stops the PSSourceOffTimer and tells the Device Policy Manager to instruct the power supply to operate as the New Source . If the PS_RDY Message is not received before the PSSourceOffTimer times out the Sink starts sending Hard Reset Signaling .
6		The CC termination is changed from R_d to R_p (see [USB Type-C 2.4]). The power supply as the New Source transitions from Swap Standby to sourcing default vSafe5V within tNewSrc (t3). The power supply informs the Device Policy Manager that it is operating as the New Source .
7	Policy Engine receives the PS_RDY Message.	Device Policy Manager informs the Policy Engine the power supply is ready, and the Policy Engine sends the PS_RDY Message to the New Sink , starting within tSrcTransOn of the end of the GoodCRC Message following the Accept Message.

Table 7.19 Sequence Description for a Source Requested Power Role Swap (Continued)

Step	Initial Source Port → New Sink Port	Initial Sink Port → New Source Port
8	<i>Protocol Layer</i> sends the GoodCRC Message to the <i>New Source</i> and then stops the PSSourceOnTimer. <i>Policy Engine</i> evaluates the PS_RDY Message from the <i>New Source</i> and tells the <i>Device Policy Manager</i> to instruct the power supply to draw current as the <i>New Sink</i>.	<i>Protocol Layer</i> receives the GoodCRC Message from the <i>New Sink</i>.
9	The power supply as the <i>New Sink</i> transitions from <i>Swap Standby</i> to drawing the power allowed by the <i>Implicit Contract</i>. The power supply informs the <i>Device Policy Manager</i> that it is operating as the <i>New Sink</i>. At this point subsequent Negotiations between the <i>New Source</i> and the <i>New Sink</i> May proceed as normal. The <i>New Sink</i> Shall Not violate the transient load behavior defined in Section 7.2.6, "Transient Load Behavior" while transitioning to and operating at the new power level. The time duration (t4) depends on the magnitude of the load change (iLoadStepRate).	

7.3.3

Transitions Caused by Hard Reset

7.3.3.1

Source Initiated Hard Reset

The interaction of the *System Policy*, *Device Policy*, and power supply that **shall** be followed during a *Source Initiated Hard Reset* is shown in [Figure 7.41, "Transition Diagram for a Source Initiated Hard Reset"](#). The sequence that **shall** be followed is described in [Table 7.20, "Sequence Description for a Source Initiated Hard Reset"](#). The timing parameters that **shall** be applied are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Figure 7.41 Transition Diagram for a Source Initiated Hard Reset

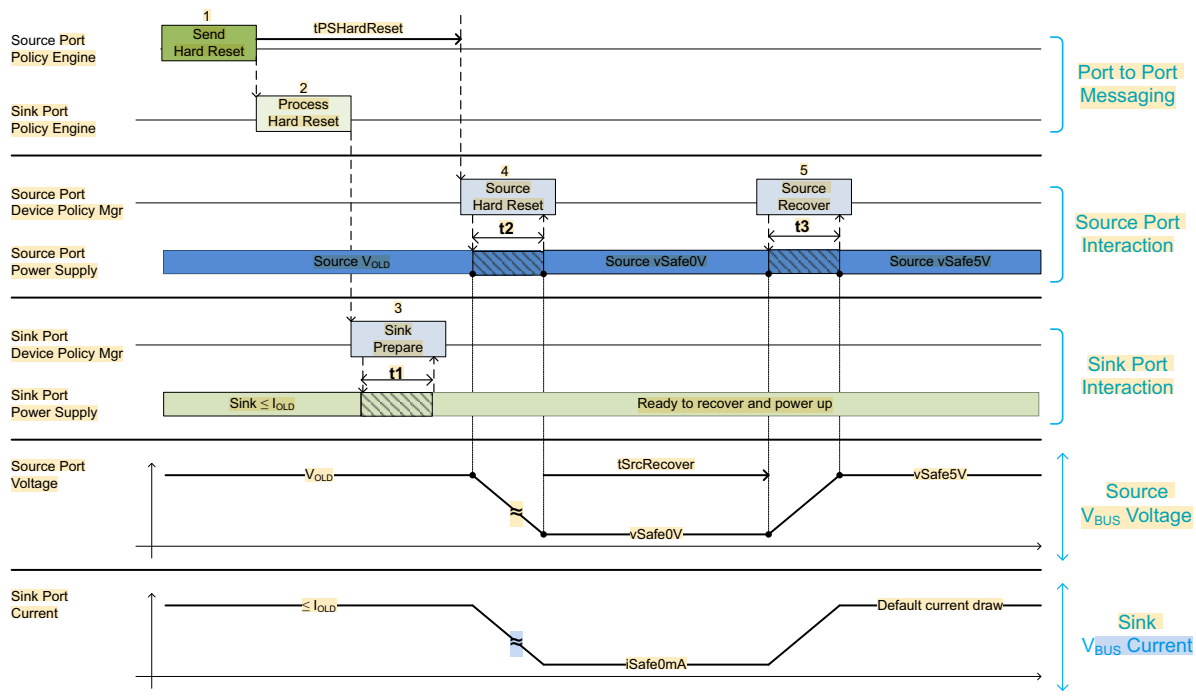


Table 7.20 Sequence Description for a Source Initiated Hard Reset

Step	Source Port	Sink Port
1	Policy Engine sends Hard Reset Signaling to the Sink.	Sink receives Hard Reset Signaling.
2		Policy Engine is informed of the Hard Reset . Policy Engine tells the Device Policy Manager to instruct the power supply to prepare for a Hard Reset .
3		The Sink prepares for the Hard Reset within $t_{SnkHardResetPrepare}$ and passes an indication to the Device Policy Manager. The Sink Shall Not draw more than $iSafe0mA$ when V_{BUS} is driven to $vSafe0V$.
4	Policy Engine waits $t_{PSHardReset}$ after sending Hard Reset Signaling and then tells the Device Policy Manager to instruct the power supply to perform a Hard Reset . The transition to $vSafe0V$ Shall occur within t_{Safe0V} (t_2).	
5	After $t_{SrcRecover}$ the Source applies power to V_{BUS} in an attempt to re-establish communication with the Sink and resume USB Default Operation . The transition to $vSafe5V$ Shall occur within $t_{SrcTurnOn}$ (t_3).	The Sink Shall Not violate the transient load behavior defined in Section 7.2.6, "Transient Load Behavior" while transitioning to and operating at the new power level.

7.3.3.2 Sink Initiated Hard Reset

The interaction of the *System Policy*, *Device Policy*, and power supply that **Shall** be followed during a *Sink Initiated Hard Reset* is shown in [Figure 7.42, "Transition Diagram for a Sink Initiated Hard Reset"](#). The sequence that **Shall** be followed is described in [Table 7.21, "Sequence Description for a Sink Initiated Hard Reset"](#). The timing parameters that **Shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Figure 7.42 Transition Diagram for a Sink Initiated Hard Reset

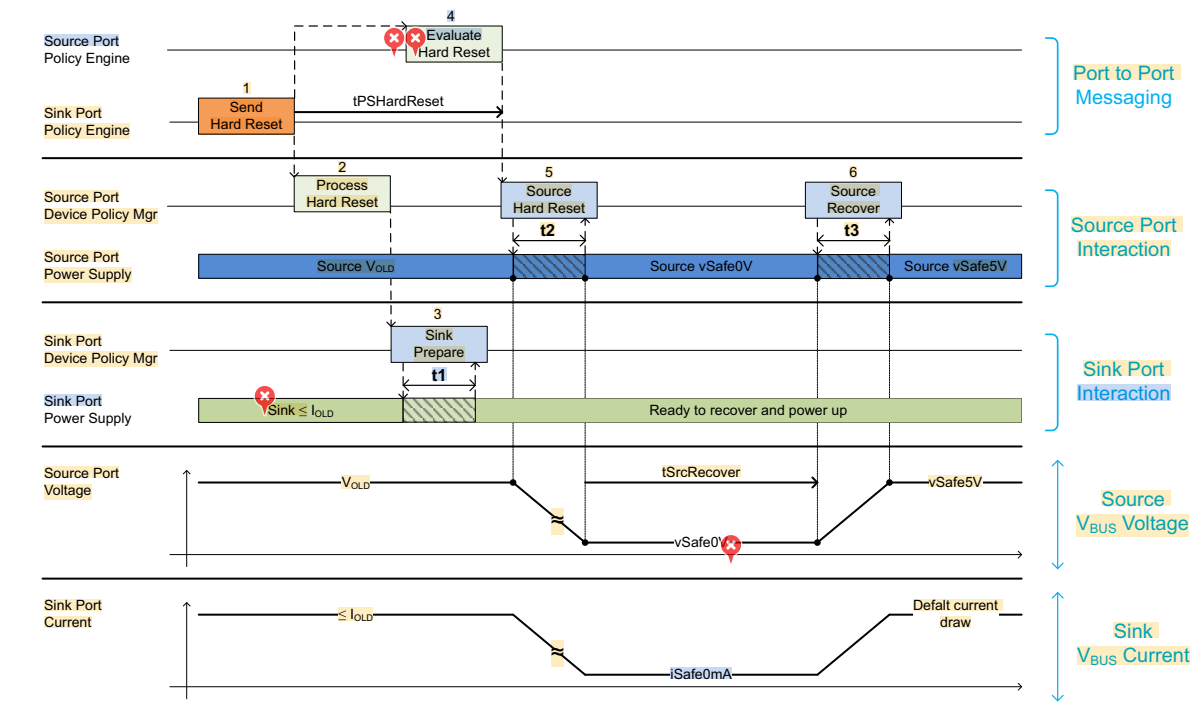


Table 7.21 Sequence Description for a Sink Initiated Hard Reset

Step	Source Port	Sink Port
1		Policy Engine sends Hard Reset Signaling to the Source.
2		Policy Engine tells the Device Policy Manager to instruct the power supply to prepare for a Hard Reset.
3		The Sink prepares for the Hard Reset within tSinkHardResetPrepare (t1) and passes an indication to the Device Policy Manager. The Sink Shall Not draw more than iSafe0mA when V_{BUS} is driven to vSafe0V .
4	Policy Engine is informed of the Hard Reset.	
5	Policy Engine waits tPSHardReset after receiving Hard Reset Signaling and then tells the Device Policy Manager to instruct the power supply to perform a Hard Reset. The transition to vSafe0V Shall occur within tSafe0V (t2).	
6	After tSrcRecover the Source applies power to V_{BUS} in an attempt to re-establish communication with the Sink and resume USB Default Operation. The transition to vSafe5V Shall occur within tSrcTurnOn (t3).	The Sink Shall Not violate the transient load behavior defined in Section 7.2.6, "Transient Load Behavior" while transitioning to and operating at the new power level.

7.3.4 Transitions Caused by Fast Role Swap

7.3.4.1 Fast Role Swap



The interaction of the *System Policy*, *Device Policy*, and power supply that **shall** be followed during a *Fast Role Swap* is shown in [Figure 7.43, "Transition Diagram for Fast Role Swap"](#). The parallel sequences that **shall** be followed are described in [Table 7.22, "Sequence Description for Fast Role Swap"](#). The timing parameters that **shall** be followed are listed in [Table 7.23, "Source Electrical Parameters"](#), [Table 7.24, "Sink Electrical Parameters"](#), and [Table 7.25, "Common Source/Sink Electrical Parameters"](#). Negotiations between the *New Source* and the *New Sink* **may** occur after the *New Source* sends the final **PS_RDY** Message.

Note: In [Figure 7.43, "Transition Diagram for Fast Role Swap"](#) and [Table 7.22, "Sequence Description for Fast Role Swap"](#) numbers are used to indicate *Message* related steps and letters are used to indicate other events.

Figure 7.43 Transition Diagram for Fast Role Swap

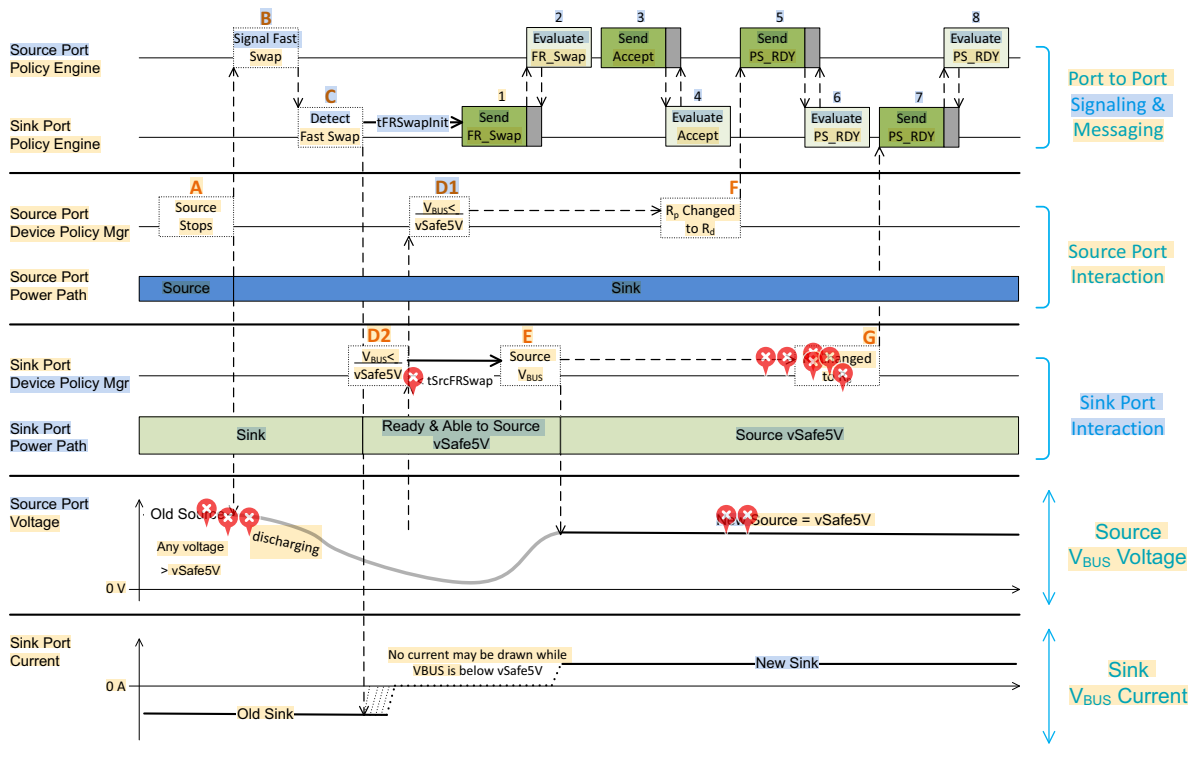


Table 7.22 Sequence Description for Fast Role Swap

Step	Initial Source Port → New Sink Port	Initial Sink Port → New Source Port
Fast Role Swap Request and Power Transition		
A	The Source connected to the Hub UFP (see Figure 7.16, "VBUS Power during Fast Role Swap") stops sourcing VBUS.	
B	Policy Engine sends the Fast Role Swap Request to the Initial Sink on the CC wire. When $V_{BUS} < v_{Safe5V}$ (min), it tells the Device Policy Manager not to draw more than $i_{SnkStdby}$ until the $t_{SnkFRSwap}$ timer has elapsed.	
C		Policy Engine detects the Fast Role Swap Request on the CC wire from the Initial Source and Shall send the FR_Swap Message back to the Initial Source (that is no longer powering VBUS) within time $t_{FRSwapInit}$.
D1	The Policy Engine monitors for $V_{BUS} \leq v_{Safe5V}$ so that a PS_RDY Message can be sent to the New Source at Step 5 of the messaging sequence.	
D2		The Policy Engine monitors for $V_{BUS} \leq v_{Safe5V}$ so the Initial Sink can assume the Power Role of New Source and begin to source VBUS.
E		When $V_{BUS} = v_{Safe5V}$ the New Source May provide power to VBUS. When $V_{BUS} < v_{Safe5V}$ the New Source Shall provide power to VBUS within $t_{SrcFRSwap}$. Once the New Source is providing power, the PS_RDY Message can be sent to the New Sink at Step 7 of the messaging sequence.
F	The CC termination is changed from R_p to R_d (see [USB Type-C 2.4]) before the New Sink sends the PS_RDY Message at Step 5 to the New Source.	
G		The CC termination is changed from R_d to R_p (see [USB Type-C 2.4]) before the New Source sends the PS_RDY Message at Step 7 to the New Sink.
Fast Role Swap Message Sequence		
1	Policy Engine receives the FR_Swap Message from the Initial Sink that is transitioning to be the New Source.	Policy Engine sends the FR_Swap Message to the Initial Source (that is no longer powering VBUS) after detecting the Fast Role Swap Request at Step C.
2	Protocol Layer sends the GoodCRC Message to the Initial Sink. Policy Engine then evaluates the FR_Swap Message.	Protocol Layer receives the GoodCRC Message from the Initial Source 
3	Policy Engine sends an Accept Message to the Initial Sink that is transitioning to be the New Source.	Policy Engine receives the Accept Message from the Initial Source that is transitioning to be the New Sink.
4	Protocol Layer receives the GoodCRC Message from the Initial Sink that is transitioning to be the New Source.	 Protocol Layer sends the GoodCRC Message to the Initial Source that is transitioning to be the New Sink 

Table 7.22 Sequence Description for Fast Role Swap (Continued)

Step	Initial Source Port → New Sink Port	Initial Sink Port → New Source Port
5	Policy Engine sends a PS_RDY Message to the Initial Sink that is transitioning to be the New Source. The Policy Engine Shall start the PS_RDY Message at least tFRSwap5V after it has sent the Accept Message, and when Step D1 has also been completed.	Policy Engine receives the PS_RDY Message from the New Sink.
6	Protocol Layer receives the GoodCRC Message from the New Source.	Protocol Layer sends the GoodCRC Message from the Initial Sink that has completed the transition to New Source. Policy Engine then evaluates the PS_RDY Message.
7	Policy Engine receives the PS_RDY Message from the New Source.	Policy Engine sends a PS_RDY Message to the New Sink. The Policy Engine Shall wait for Step E before sending the PS_RDY Message, and Shall send the PS_RDY Message within tFRSwapComplete of receiving the PS_RDY Message from the Initial Source Port.

7.4 Electrical Parameters

7.4.1 Source Electrical Parameters

The Source Electrical Parameters that **shall** be followed are specified in [Table 7.23, "Source Electrical Parameters"](#).

Table 7.23 Source Electrical Parameters

Parameter	Description	MIN	TYP	MAX	UNITS	Reference
cSrcBulk	Source bulk capacitance when a Port is powered from a dedicated supply. ¹	10			μF	Section 7.1.2
cSrcBulkShared	Source bulk capacitance when a Port is powered from a shared supply. ¹	120			μF	Section 7.1.2
DNL (Differential Non-Linearity)	Deviation between ideal analog values corresponding to adjacent input digital values	-1	0	+1	LSB	Section 7.1.4.2.1
iPpsCLMin	SPR PPS Minimum Current Limit setting.	1			A	Section 7.1.4.2.2
iPpsCLNew	Current Limit accuracy					Section 7.1.4.2.2
	1A ≤ Operating Current ≤ 3A	150		150	mA	
	Operating current > 3A	5		5	%	
iPpsCLStep	SPR PPS Current Limit programming step size (1 LSB).		50		mA	Section 7.1.4.2.2
iPpsCLLoadReleaseRate	Maximum load decrease slew rate during Current Limit set-point changes.	-150			mA/μs	Section 7.1.4.2.2
iPpsCLLoadStepRate	Maximum load increase slew rate during Current Limit set-point changes.			150	mA/μs	Section 7.1.4.2.2
iPpsCLTransient	Allowed output current overshoot when a load increase occurs while in CL mode.			New load + 100	mA	Section 7.1.4.2.2
	Allowed output current undershoot when a load decrease occurs while in CL mode.	New load - 100			mA	

1) The Source **shall** charge and discharge the total bulk capacitance to meet the transition time requirements.

Table 7.23 Source Electrical Parameters (Continued)

Parameter	Description	MIN	TYP	MAX	UNITS	Reference
<i>iPpsCVCLTransient</i>	CV to CL transient current bounds assuming the Operating Voltage reduction of Section 7.2.3.1, "Programmable Power Supply Sink Standby" .	<i>iPpsCLNew</i> - 100		New load + 500	mA	Section 7.1.4.2.2
<i>tAvsTransient</i>	The maximum time for the AVS to be between <i>vAvsNew</i> and <i>vAvsValid</i> in response to a load transient.			5	ms	Section 7.1.8.4
<i>tAvsSrcTransLarge</i>	The time the AVS set-point shall transition between requested voltages for steps larger than <i>vAvsSmallStep</i> .	0		700	ms	Section 7.1.4.3.1
<i>tAvsSrcTransSmall</i>	The time the AVS set-point shall transition between requested voltages for steps smaller than <i>vAvsSmallStep</i> .	0		50	ms	Section 7.1.4.3.1
<i>tNewSnk</i>	Time allowed for an <i>Initial Source</i> in Swap Standby to transition <i>New Sink</i> operation.			15	ms	Section 7.1.10 Figure 7.39 Figure 7.40
<i>tPpsCLCVTransient</i>	CL to CV transient voltage settling time.			275	ms	Section 7.1.4.2.2
<i>tPpsCLProgramSettle</i>	SPR PPS Current Limit programming settling time.			250	ms	Section 7.1.4.2.2
<i>tPpsCLSettle</i>	CL load transient current settling time.			250	ms	Section 7.1.4.2.2
<i>tPpsCVCLTransient</i>	CV to CL transient settling time.			250	ms	Section 7.1.8.3
<i>tPpsSrcTransLarge</i>	The time the Programmable Power Supply's set-point shall transition between requested voltages for steps larger than <i>vPpsSmallStep</i> .	0		275	ms	Section 7.3.1.3

1) The Source **shall** charge and discharge the total bulk capacitance to meet the transition time requirements.

Table 7.23 Source Electrical Parameters (Continued)

Parameter	Description	MIN	TYP	MAX	UNITS	Reference
<i>tPpsSrcTransSmall</i>	The time the Programmable Power Supply's set-point shall transition between requested voltages for steps less than or equal to <i>vPpsSmallStep</i> .	0		25	ms	Section 7.3.1.3
<i>tPpsTransient</i>	The maximum time for the Programmable Power Supply to be between <i>vPpsNew</i> and <i>vPpsValid</i> in response to a load transient when target load is greater than or equal to 60mA.			5	ms	Section 7.1.8.3
	The maximum time for the Programmable Power Supply to be between <i>vPpsNew</i> and <i>vPpsValid</i> in response to a load transient when target load is less than 60mA.			150	ms	Section 7.1.8.3
<i>tSrcFRSwap</i>	Time from the <i>Initial Sink</i> detecting that <i>VBUS</i> has dropped below <i>vSafe5V</i> until the <i>Initial Sink</i> /new <i>Source</i> is able to supply <i>USB Type-C</i> Current (see [USB Type-C 2.4])			150	μs	Section 7.1.13
1)	The <i>Source</i> shall charge and discharge the total bulk capacitance to meet the transition time requirements.					

Table 7.23 Source Electrical Parameters (Continued)

Parameter		Description	MIN	TYP	MAX	UNITS	Reference
tSrcReady	SPR Mode	Time from positive/negative transition start (t0) to when the Source is ready to provide the newly Negotiated power level. Applies only to SPR Mode voltage transitions.			285	ms	Figure 7.2 Figure 7.3
	EPR Mode	Time from positive/negative transition start (t0) to when the Source is ready to provide the newly Negotiated power level. Applies to EPR Mode voltage transitions and any voltage transition that either begins or ends in EPR Mode .			720		
tSrcRecover	SPR Mode	Time allotted for the Source to recover.	0.66		1.0	s	Section 7.1.5
	EPR Mode		1.085		1.425		
tSrcSettle	SPR Mode	Time from positive/negative transition start (t0) to when the transitioning voltage is within the range vSrcNew . Applies only to SPR Mode voltage transitions.			275	ms	Figure 7.2
	EPR Mode	Time from positive/negative transition start (t0) to when the transitioning voltage is within the range vAvsNew . Applies to EPR Mode voltage transitions and any voltage transition that either begins or ends in EPR Mode .			700		
tSrcSwapStdby		The maximum time for the Source to transition to Swap Standby .			650	ms	Section 7.1.10 Figure 7.17 Figure 7.18
1) The Source Shall charge and discharge the total bulk capacitance to meet the transition time requirements.							

Table 7.23 Source Electrical Parameters (Continued)

Parameter		Description	MIN	TYP	MAX	UNITS	Reference
<i>tSrcTransient</i>		The maximum time for the <i>Source</i> output voltage to be between <i>vSrcNew</i> and <i>vSrcValid</i> in response to a load transient when target load is greater or equal to than 60mA.			5	ms	Section 7.1.8
		* The maximum time for the <i>Source</i> output voltage to be between <i>vSrcNew</i> and <i>vSrcValid</i> in response to a load transient when target load is less than 60mA.			150	ms	Section 7.1.8
<i>tSrcTransition</i>		The time the <i>Source</i> <i>Shall</i> wait before transitioning the power supply to ensure that the <i>Sink</i> has sufficient time to prepare (does not apply to transitions within the same PPS or AVS APDO).	* 25		35	ms	Section 7.3
<i>tSrcTransOff</i>	<i>SPR Mode</i>	Time from the last bit of the <i>GoodCRC</i> Message acknowledging the <i>Accept</i> Message in response to the <i>PR_Swap</i> Message until the <i>PS_RDY</i> Message must be started. Applies only to <i>SPR Mode</i> voltage transitions.			690	ms	Section 7.3.2
<i>tSrcTransOn</i>		Time from the last bit of the <i>GoodCRC</i> Message acknowledging the <i>PS_RDY</i> Message sent by the new <i>Source</i> , in response to the <i>PR_Swap</i> Message until the <i>PS_RDY</i> Message must be started.			280	ms	Section 7.3.2
1) The <i>Source</i> <i>Shall</i> charge and discharge the total bulk capacitance to meet the transition time requirements.							

Table 7.23 Source Electrical Parameters (Continued)

Parameter	Description	MIN	TYP	MAX	UNITS	Reference
tSrcTransReq	SPR Mode Time from the last bit of the GoodCRC Message acknowledging the Accept Message in response to the Request Message until the PS_RDY Message must be started. Applies only to SPR Mode voltage transitions.			325	ms	Section 7.3
	EPR Mode Time from the last bit of the GoodCRC Message acknowledging the Accept Message in response to the Request Message until the PS_RDY Message must be started. Applies to EPR Mode voltage transitions and any voltage transition that either begins or ends in EPR Mode.			760	ms	Section 7.3
tSrcTurnOn	Transition time from vSafe0V to vSafe5V.			275	ms	Figure 7.10 Table 7.20 Table 7.21
vAvsMaxVoltage	Maximum Voltage Field in the AVS APDO .	APDO Max Voltage *0.95		APDO Max Voltage *1.05	V	Section 7.1.4.3.1
vAvsMinVoltage	Minimum Voltage Field in the AVS APDO .	APDO Min Voltage *0.95		APDO Min Voltage *1.05	V	Section 7.1.4.3.1
vAvsNew	Adjustable RDO Output Voltage measured at the Source receptacle.	RDO Output Voltage *0.95	RDO Output Voltage	RDO Output Voltage *1.05	V	Section 7.1.8.4
vAvsSlewNeg	AVS maximum slew rate for negative voltage changes.			-30	mV/μs	Section 7.1.8.4
1) The Source Shall charge and discharge the total bulk capacitance to meet the transition time requirements.						

Table 7.23 Source Electrical Parameters (Continued)

Parameter	Description	MIN	TYP	MAX	UNITS	Reference
vAvsSlewPos	AVS maximum slew rate for positive voltage changes.			30	mV/μs	Section 7.1.8.4
vAvsSmallStep	AVS step size defined as a small step relative to the previous vAvsNew.	-1.0		1.0	V	Section 7.1.4.3.1
vAvsStep	AVS voltage programming step size.		100		mV	Section 7.1.8.4
vAvsValid	The range in addition to vAvsNew which the AVS output is considered Valid during and after a transition as well as in response to a transient load condition.	-0.5		0.5	V	Section 7.1.8.4
vPpsCLCVTransient	CL to CV load transient voltage bounds.	Operating Voltage * 0.95 – 0.1V		Operating Voltage * 1.05 + 0.1V	V	Section 7.1.4.2.2
vPpsMaxVoltage	Maximum Voltage Field in the Programmable Power Supply APDO.	APDO Max Voltage *0.95		APDO Max Voltage * 1.05	V	Section 7.1.4.2.1
vPpsMinVoltage	Minimum Voltage Field in the Programmable Power Supply APDO.	APDO Min Voltage *0.95		APDO Min Voltage * 1.05	V	Section 7.1.4.2.1
vPpsNew	Programmable RDO Output Voltage measured at the Source receptacle.	RDO Output Voltage *0.95	RDO Output Voltage	RDO Output Voltage *1.05	V	Section 7.1.8.3
vPpsShutdown	The voltage at which the SPR PPS shuts down when operating in CL.	APDO Minimum Voltage * 0.85		APDO Minimum Voltage * 0.95	V	Section 7.1.4.2.2
vPpsSlewNeg	Programmable Power Supply maximum slew rate for negative voltage changes			-30	mV/μs	Section 7.1.8.3
vPpsSlewPos	Programmable Power Supply maximum slew rate for positive voltage changes			30	mV/μs	Section 7.1.8.3

1) The Source **Shall** charge and discharge the total bulk capacitance to meet the transition time requirements.

Table 7.23 Source Electrical Parameters (Continued)

Parameter	Description	MIN	TYP	MAX	UNITS	Reference
<i>vPpsSmallStep</i>	PPS Step size defined as a small step relative to the previous <i>vPpsNew</i> .	-500		500	mV	Section 7.1.4.2.2
<i>vPpsStep</i>	PPS voltage programming step size (1 LSB).		20		mV	Section 7.1.8.3
<i>vPpsValid</i>	The range in addition to <i>vPpsNew</i> which the Programmable Power Supply output is considered Valid in response to a load step.	-0.1		0.1	V	Section 7.1.8.3
<i>vSmallStep</i>	<i>V_{BUS}</i> step size increase defined as a small step relative to the previous <i>V_{BUS}</i> when Requesting a different (A)PDO.			500	mV	Section 7.1.4.3.1
<i>vSrcNeg</i>	Most negative voltage allowed during transition.			-0.3	V	Figure 7.10
<i>vSrcNew</i>	Fixed Supply output measured at the Source receptacle.	PDO Voltage *0.95	PDO Voltage	PDO Voltage *1.05	V	Table 7.2
	Variable Supply output measured at the Source receptacle.	PDO Minimum Voltage		PDO Maximum Voltage	V	
	Battery Supply output measured at the Source receptacle.	PDO Minimum Voltage		PDO Maximum Voltage	V	
<i>vSrcPeak</i>	The range that a Fixed Supply or EPR AVS in Peak Current operation is allowed when overload conditions occur.	PDO Voltage *0.90		PDO Voltage *1.05	V	Table 6.10 Table 6.16 Figure 7.14
<i>vSrcSlewNeg</i>	Maximum slew rate allowed for negative voltage transitions. Limits current based on a 3 A connector rating and maximum Sink bulk capacitance of 100 µF.			-30	mV/µs	Section 7.1.4.2 Table 7.2

1) The Source **Shall** charge and discharge the total bulk capacitance to meet the transition time requirements.

Table 7.23 Source Electrical Parameters (Continued)

Parameter	Description	MIN	TYP	MAX	UNITS	Reference
vSrcSlewPos	Maximum slew rate allowed for positive voltage transitions. Limits current based on a 3 A connector rating and maximum Sink bulk capacitance of 100 μ F.			30	mV/ μ s	Section 7.1.4 Figure 7.2
vSrcValid	The range in addition to vSrcNew which a newly <i>Negotiated</i> voltage is considered Valid during and after a transition as well as in response to a transient load condition. This range also applies to vSafe5V .	-0.5		0.5	V	Figure 7.2 Figure 7.3 Section 7.1.8
1) The Source Shall charge and discharge the total bulk capacitance to meet the transition time requirements.						

7.4.2 Sink Electrical Parameters

The *Sink* Electrical Parameters that **shall** be followed are specified in [Table 7.24, "Sink Electrical Parameters"](#).

Table 7.24 Sink Electrical Parameters

Parameter	Description	MIN	TYP	MAX	UNITS	Reference
<i>cSnkBulk</i>	<i>Sink</i> bulk capacitance on <i>V_{BUS}</i> at <i>Attach</i> and during <i>FRS</i> after the <i>Initial Source</i> stops sourcing and prior to establishing the <i>First Explicit Contract</i> (see Appendix E, "FRS System Level Example" for an example). ¹			See [USB 3.2]		Section 7.2.2 [USB 3.2]
<i>cSnkBulkPd</i>	Bulk capacitance on <i>V_{BUS}</i> a <i>Sink</i> is allowed after a successful <i>Negotiation</i> . ¹			100	μF	Section 7.2.2
<i>iLoadReleaseRate</i>	Load release di/dt.	-150			mA/ μs	Section 7.2.6
<i>iLoadStepRate</i>	Load step di/dt.			150	mA/ μs	Section 7.2.6
<i>iNewFrSsSink</i>	Maximum current the <i>New Sink</i> can draw during a <i>Fast Role Swap</i> until the <i>New Source</i> applies <i>R_p</i> . Matches the required <i>Fast Role Swap required USB Type-C Current</i> Current field of the <i>Fixed Supply PDO</i> of the <i>Initial Source's Sink Capabilities</i> Message.			Default USB current or 1.5 or 3.0	A	Section 7.1.13
<i>iOvershoot</i>	Positive or negative overshoot when a load change occurs less than or equal to <i>iLoadStepRate</i> ; relative to the settled value after the load change.	-230		230	mA	Section 7.2.6
<i>iPpsCLLoadStep</i>	Maximum Current set-point change while operating in <i>CL</i> mode.	-500		500	mA	Section 7.2.3.1
<i>iSafe0mA</i>	Maximum current a <i>Sink</i> is allowed to draw when <i>V_{BUS}</i> is driven to <i>vSafe0V</i> .			1.0	mA	Figure 7.29 Figure 7.30
<i>iSnkStdby</i>	Maximum current during voltage transition.			500	mA	Section 7.2.3
<i>iSnkSwapStdby</i>	Maximum current a <i>Sink</i> can draw during <i>Swap Standby</i> . Ideally this current is very near to 0 mA largely influenced by <i>Port</i> leakage current.			2.5	mA	Section 7.2.7

1) If more bypass capacitance than *cSnkBulk* max or *cSnkBulkPd* max is required in the device, then the device **shall** incorporate some form of *V_{BUS}* surge current limiting as described in [\[USB 3.2\]](#) Section 11.4.4.1.

Table 7.24 Sink Electrical Parameters (Continued)

Parameter	Description	MIN	TYP	MAX	UNITS	Reference
pHubSusp	Suspend power consumption for a <i>Hub</i> . 25mW + 25mW per downstream <i>Port</i> for up to 4 ports.			125	mW	Section 7.2.3
pSnkSusp	Suspend power consumption for a peripheral device. 			25	mW	Section 7.2.3
tNewSrc	Maximum time allowed for an <i>Initial Sink</i> in <i>Swap Standby</i> to transition to <i>New Source</i> operation.			275	ms	Section 7.2.7 Table 7.18  Table 7.19
tSnkFRSwap	Time during a <i>Fast Role Swap</i> when the <i>New Sink</i> can draw no more than <i>iSnkStdby</i> .			200	μs	Section 7.1.13
tSnkHardResetPrepare	Time allotted for the <i>Sink</i> power electronics to prepare for a <i>Hard Reset</i> .			15	ms	Table 7.12
tSnkNewPower	Maximum transition time between power levels.			15	ms	Section 7.2.3
tSnkRecover	Time for the <i>Sink</i> to resume <i>USB Default Operation</i> .			150	ms	Table 7.20
tSnkStdby	Time to transition to <i>Sink Standby</i> from <i>Sink</i> .			15	ms	Section 7.2.3
tSnkSwapStdby	Maximum time for the <i>Sink</i> to transition to <i>Swap Standby</i> .			15	ms	Section 7.2.7
vEprMax	Highest voltage an <i>EPR Sink</i> is expected to tolerate			55	V	Section 7.2.9.2
vSprMax	Highest voltage an <i>SPR Sink</i> is expected to tolerate			24	V	Section 7.2.9.2
1)	If more bypass capacitance than <i>cSnkBulk</i> max or <i>cSnkBulkPd</i> max is required in the device, then the device Shall incorporate some form of <i>VBUS</i> surge current limiting as described in [USB 3.2] Section 11.4.4.1.					

7.4.3 Common Electrical Parameters

Electrical Parameters that are common to both the *Source* and the *Sink* that **Shall** be followed are specified in [Table 7.25, "Common Source/Sink Electrical Parameters"](#).

Table 7.25 Common Source/Sink Electrical Parameters

Parameter	Description	MIN	TYP	MAX	UNITS	Reference
<i>tSafe0V</i>	Time to reach <i>vSafe0V</i> max.			650	ms	Section 7.1.5 Figure 7.10 Table 7.20 Table 7.21
<i>tSafe5V</i>	Time to reach <i>vSafe5V</i> max.			275	ms	Section 7.1.5 Figure 7.10 Table 7.20 Table 7.21
<i>tVCONNReapplied</i>	When the <i>UFP</i> is the <i>VCONN Source</i> : time from the last bit of the <i>GoodCRC</i> acknowledging the <i>PS_RDY</i> Message before reapplying <i>VCONN</i> . When the <i>DFP</i> is the <i>VCONN Source</i> : time from when <i>VCONN</i> drops below <i>vRaReconnect</i> .	10		20	ms	Figure 7.19 Figure 7.20
<i>tVCONNValid</i>	Time from <i>tVCONNReapplied</i> until <i>VCONN</i> is within <i>vVconnValid</i> (see [USB Type-C 2.4]). ¹	0		5	ms	Figure 7.19 Figure 7.20
<i>tVCONNZero</i>	Time from the last bit of the <i>GoodCRC</i> acknowledging the <i>Accept</i> Message in response to the <i>Data_Reset</i> Message until <i>VCONN</i> is below <i>vRaReconnect</i> (see [USB Type-C 2.4]). 			125	ms	Figure 7.19 Figure 7.20
<i>vSafe0V</i>	Safe operating voltage at "zero volts".	0		0.8	V	Section 7.1.5
<i>vSafe5V</i>	Safe operating voltage at 5V. See [USB 2.0] and [USB 3.2] for allowable <i>VBUS</i> voltage range.	4.75		5.5	V	Section 7.1.5
1) <i>tVCONNStable</i> (See [USB Type-C 2.4]) still applies.						